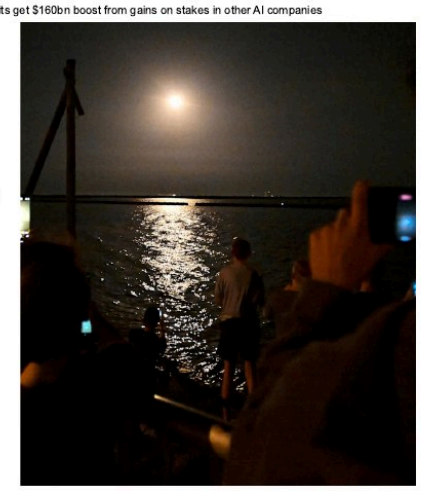
Big Tech 投資利得新聞｜PDF 第 1/4 頁

投資其他 AI 公司的帳面利得，墊高了大型科技公司的獲利



Technology sector

Big Tech profits get $160bn boost from gains on stakes in other AI companies

Analysts say large paper windfalls on investments in OpenAI, Anthropic and SpaceX have muddied the tech sector's earnings metrics

'ech groups' equity gains have been turbocharged this year by the public float of SpaceX, in which Google-parent Alphabet Nvidia both own stakes © Miguel J. Rodriguez Carrillo/Getty Images

Ramsay Hodgson and **Emily Herbert** in London

Published AUG 31 2026

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Big tech giants booked a more than $160bn windfall last quarter from investments in other AI companies, flattering their earnings and raising concerns that paper gains are overstating the strength of the AI boom.

Alphabet, Amazon, Nvidia and Microsoft all reported substantial boosts to their pre-tax profits in their most recent results from the 'other income' line in their accounts, which largely comes from valuation gains on equity stakes in other leading AI companies.

These one-off valuation boosts, derived in large part from enthusiasm around AI, risk distorting the financial picture at a time when investors are closely scrutinising tech earnings to gauge the health of the AI ecosystem.

本頁中文重點

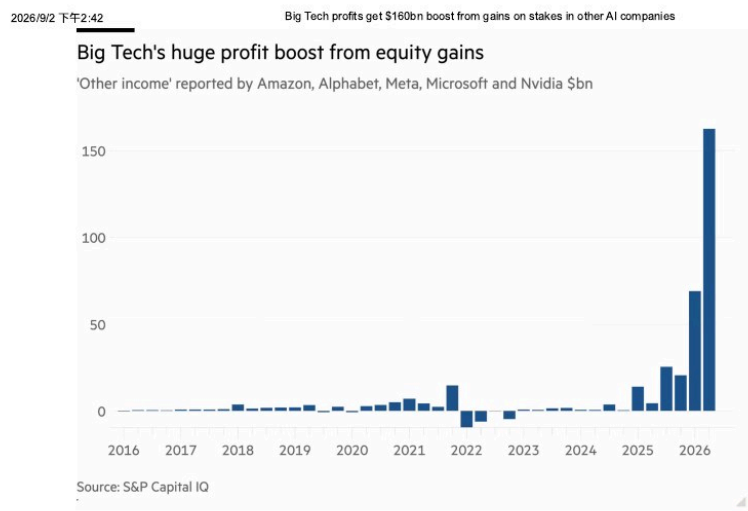
- 文章出處：Financial Times，〈Big Tech profits get $160bn boost from gains on stakes in other AI companies〉，記者 Ramsay Hodgson、Emily Herbert（倫敦），發布日 2026/08/31。頁首的 2026/09/02 是列印時間。
- 導言：分析師認為，對 OpenAI、Anthropic 與 SpaceX 的投資產生大筆帳面利得，已經攪亂了科技業的獲利指標。
- 上一季大型科技公司從投資其他 AI 公司認列超過 1,600 億美元的進帳，使財報變好看，也引發「帳面利得誇大了 AI 榮景實力」的疑慮。
- Alphabet、Amazon、Nvidia 與 Microsoft 最近一期財報的稅前獲利都被「其他收入（other income）」這一行大幅墊高，而這一行主要來自持有其他領先 AI 公司股權的評價利得。
- 這些一次性的評價利得大多源自市場對 AI 的熱度，在投資人正緊盯科技業獲利、藉此判斷 AI 生態系是否健康的時候，可能扭曲財務全貌。

老師課堂說明

- 這一篇的逐頁討論沒有出現在本週的錄音裡（錄音從講義開始），以下是老師在講課途中回頭引用這篇文章的部分。
- 老師在講「現金流怎麼估」時回頭提到這篇：這些公司的稅前盈餘（EBT）看起來很高，但那不是品質好的現金流，因為 EBT 會被其他收入推上去。
- 老師接著說，如果改用 EBIT，就不會受其他收入影響，所以「選哪一個會計科目當估值的起點」是關鍵決定，而不是隨便拿一個獲利數字來折現。**即：先問清楚這個獲利數字是從哪裡來的，再決定要不要拿它去折現。**

Big Tech 投資利得新聞｜PDF 第 2/4 頁

『其他收入』的規模與來源



"A couple of years ago investors started to ask about the circularity of [Big Tech] revenues," said Ben Snider, chief US equities strategist at Goldman Sachs.

The profit boost from AI investments now "raises the question of whether the growth these companies are reporting is based on underlying demand or if it is misleading in some way," he added.

Big Tech groups' investments in OpenAI and Anthropic have increasingly shown up in their financial reporting in recent quarters, as accounting rules dictate that changes in the value of equity investments at the end of each quarter appear as part of the company's profit or loss.

But their equity gains have been turbocharged this year by the public float of SpaceX, in which Google-parent Alphabet and Nvidia both own stakes. The rocket company absorbed Elon Musk's xAI before going public, making the promise of AI data centres in orbit a key part of its pitch. The listing delivered a huge uplift for existing investors.

The 'other income' contribution to profits last quarter was more than double the roughly $69bn boost in the previous three-month period. The expected public offerings of Anthropic and OpenAI would see the phenomenon continue in the coming year.

本頁中文重點

- 圖表「Big Tech's huge profit boost from equity gains」：Amazon、Alphabet、Meta、Microsoft 與 Nvidia 合計認列的其他收入（十億美元）。2016–2024 年多數季度在正負個位數之間，2025 年起快速放大，最近一季衝到約 160。資料來源 S&P Capital IQ。
- 高盛首席美股策略師 Ben Snider：「兩年前投資人開始追問大型科技公司營收的循環性問題。」他說現在這些投資帶來的獲利增幅，「讓人要問這些公司報出來的成長是來自真實需求，還是在某種程度上具有誤導性」。
- 會計規則規定，股權投資在每季季末的價值變動要計入當期損益，所以這些公司對 OpenAI 與 Anthropic 的投資，近幾季越來越明顯地出現在財報上。**即：手上的股票漲了就要記成當期的獲利，所以這幾季的財報才會被投資的帳面漲幅撐起來。**
- 今年把帳面利得推高的是 SpaceX 公開上市，Alphabet 與 Nvidia 都持有它的股權。SpaceX 在上市前先吸收了 Elon Musk 的 xAI，把「在軌道上建 AI 資料中心」當成招股重點，上市為既有投資人帶來很大的價值提升。
- 上一季其他收入對獲利的貢獻，是前一季約 690 億美元的兩倍以上。Anthropic 與 OpenAI 若依預期公開發行，這個現象明年還會延續。

會計分錄 **季末持股漲價：沒有收到現金，損益表卻多了一筆「其他收入」**

單位：十億美元

會計科目	借方	貸方
季末依公允價值重新評價（五家公司合計約 160）		
借：股權投資（透過損益按公允價值衡量）	160	
貸：其他收入—投資評價利益		160

分錄裡沒有現金。這筆利益讓稅前淨利變大，但在現金流量表的營業活動要扣回去，所以估值時不能把它當成本業的現金流。

Big Tech 投資利得新聞｜PDF 第 3/4 頁

稅前獲利創高，但來源是持股評價而非本業

In the most recent round of earnings reports, pre-tax profits reached a record at several of the Big Tech "hyperscalers". But the main source of these increases at Alphabet and Amazon was 'other income' from gains on SpaceX and Anthropic, rather than new business lines or fresh cash generation.



'Other income' more than doubled to $97.9bn at Google parent Alphabet, and more than tripled to $53.4bn at Amazon for the three months to June 30, compared with the previous three months.

Nvidia, which has a large portfolio of equity holdings, disclosed ownership of nearly 123mn shares in SpaceX at the end of June. In the three months to the end of July, it reported $7.7bn in 'other income'. This was less than the previous quarter, when it booked big gains on its shares in Intel.

Analysts note that the underlying earnings from the US tech sector, which plays a pivotal role in overall stock market performance, are still strong. But the lumpy appearance of AI investment gains makes it harder to get a clear picture.

These gains raised "earnings quality questions", said Manish Kabra, head of US equity strategy at Société Générale, which had contributed to the market trimming the companies' price-to-earnings multiples from around 25 times to 20 times.

本頁中文重點

- 最近一輪財報中，幾家大型科技「超大規模業者（hyperscalers）」的稅前獲利創下紀錄。但 Alphabet 與 Amazon 的增幅主要來自 SpaceX 與 Anthropic 的其他收入，不是新的業務線，也不是新產生的現金。
- 圖表「Hyperscaler earnings risk being distorted by stakes in other companies」：Alphabet、Amazon、Meta、Microsoft、Nvidia 合計的每季其他收入、稅前盈餘與自由現金流（十億美元）。最近一季稅前盈餘衝到約 350，其中深色的其他收入約 160，而自由現金流只有約 30。註記說明 Nvidia 的財季已對齊曆年季度以便比較。資料來源 S&P Capital IQ 與 FT 計算。
- 具體數字：到 6 月 30 日止的三個月，Alphabet 的其他收入較前三個月增加超過一倍到 979 億美元，Amazon 增加超過兩倍到 534 億美元。
- Nvidia 持有大量股權部位，6 月底揭露持有近 1.23 億股 SpaceX。到 7 月底為止的三個月它認列 77 億美元其他收入，低於前一季（前一季因英特爾持股認列了大額利得）。
- 分析師指出美國科技業的本業獲利仍然強勁，但 AI 投資利得忽高忽低地出現，使人更難看清真實情況。
- 法國興業銀行美股策略主管 Manish Kabra 說這些利得引發「盈餘品質的問題」，市場也因此把這些公司的本益比從約 25 倍調降到 20 倍。**即：同樣的獲利，如果大家覺得它靠不住，願意付的倍數就會往下掉。**

Big Tech 投資利得新聞｜PDF 第 4/4 頁

為什麼這類利得不能當成可重複的獲利能力

Kasper Elmgreen, fixed income and equities chief investment officer at Nordea Asset Management, said first-half earnings "materially overstated repeatable earnings power" and that the circularity of these AI gains was "a point of attention".

The companies only revalue their holdings in private companies following a fresh funding round, while publicly traded holdings are adjusted each quarter.

This can create a distorted impression of earnings growth for investors and analysts looking at high-level financial metrics that do not adjust for these factors company by company.

"For top-down investors, yes there's a concern on the investments that have gone on within the tech space," Kabra said.

Louise Dudley, a global equities portfolio manager at Federated Hermes, said that the "size of the adjustments" was "difficult to ignore" and made analysis more complex and would "undoubtedly introduce additional risk".

But she added that the companies "deserve some credit for investing in and partnering with highly successful businesses".

The large one-off equity revisions so far this year had the potential to create nasty downward surprises for future earning periods. "The earnings contribution you get from this sets up for potential negative earnings growth next year," said Scott Chronert, US equity strategist at Citi.

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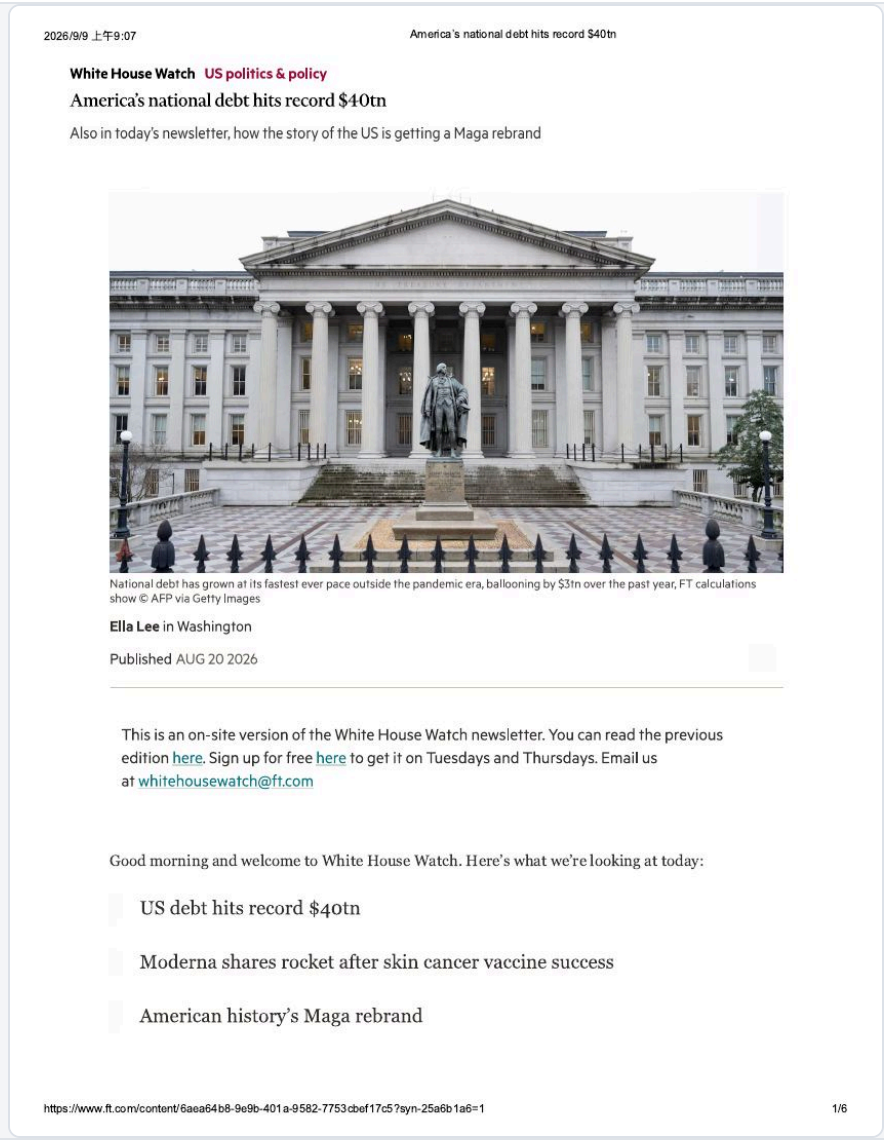
US companies
Technology sector
Corporate earnings and results
Equity earnings
Artificial intelligence

本頁中文重點

- Nordea Asset Management 固定收益與股票投資長 Kasper Elmgreen 說，上半年的盈餘「明顯高估了可重複的獲利能力」，而這些 AI 利得的循環性是「值得注意的一點」。
- 重要的認列差異：這些公司只有在被投資的私人公司完成新一輪募資後才重估持股，公開上市的持股則是每季調整。
- 這會讓只看高層次財務指標、沒有逐家還原調整的投資人與分析師，對獲利成長產生失真的印象。
- Federated Hermes 全球股票基金經理人 Louise Dudley 說調整的「規模大到難以忽視」，使分析更複雜，也「無疑會帶來額外風險」。但她也認為這些公司「投資並與非常成功的企業合作，值得一些肯定」。
- 花旗美股策略師 Scott Chronert 指出，今年這些大額一次性股權重估，可能替未來的獲利期間埋下大幅下修的風險：「你今年拿到的這塊獲利貢獻，會替明年的負成長鋪路。」
- 本頁末為版權聲明與主題標籤：US companies、Technology sector、Corporate earnings and results、Equity earnings、Artificial intelligence。

美國國債新聞｜PDF 第 1/6 頁

美國國債突破 40 兆美元：風險資產與無風險資產的界線



2026/9/9 上午9:07 America's national debt hits record $40tn

White House Watch US politics & policy

America's national debt hits record $40tn

Also in today's newsletter, how the story of the US is getting a Maga rebrand

National debt has grown at its fastest ever pace outside the pandemic era, ballooning by $3tn over the past year, FT calculations show © AFP via Getty Images

Ella Lee in Washington

Published AUG 20 2026

This is an on-site version of the White House Watch newsletter. You can read the previous edition here. Sign up for free here to get it on Tuesdays and Thursdays. Email us at whitehousewatch@ft.com

Good morning and welcome to White House Watch. Here's what we're looking at today:

- US debt hits record $40tn
- Moderna shares rocket after skin cancer vaccine success
- American history's Maga rebrand

https://www.ft.com/content/6aea64b8-9e9b-401a-9582-7753cbef17c5?syn-25a6b1a6=1 1/6

文章重點

- 美國聯邦總債務升至 40 兆美元，其中由公眾持有的債務超過 32 兆美元，規模已接近美國一整年的經濟產出。
- 財政部一方面大量發債，一方面宣布擴大長天期公債買回，目的在改善流動性與市場運作，但也突顯債券供給與殖利率壓力。
- 債務規模上升會提高市場對償債能力、通膨與期限風險的要求。債券價格因供給增加而承壓，殖利率與政府再融資成本因而上升。

老師課堂補充

- 老師先把金融資產依違約風險分成 risky 與 risk-free。股票、公司債與加密資產具有風險。黃金雖不涉及發行人違約，但交易成本與稅負使其不等同於標準的無風險資產。
- 市場通常以一年內到期的美國國庫券（Treasury bills）作為近似無風險資產。這並非表示完全沒有利率、通膨或價格波動，而是主要著眼於到期償付的違約風險極低。
- 全球投資組合普遍以美國公債作為基準，因此信任是核心。當債務與通膨同時升高，投資人會重新評估美國公債能否繼續扮演無風險資產。**即：美債之所以能當成無風險的那把尺，是因為大家相信它一定會還。這份信任一鬆動，整把尺就得重新量。**
- 大量新發債會增加供給，通常壓低債券價格並推升殖利率。政府未來付息與再融資負擔也會跟著提高。

課堂提問與解答

問 為什麼美國國庫券常被當成無風險資產？

答 因為分析重點是到期違約風險，而美國短期國庫券長期具有高度償付能力與市場流動性。但它仍可能承受通膨、利率與價格風險，所以是『近似無風險』而不是沒有任何風險。**即：近似無風險的意思是「幾乎一定拿得回本金」，不是「價格不會跌」。**

問 債券供給增加，價格與殖利率通常如何變化？

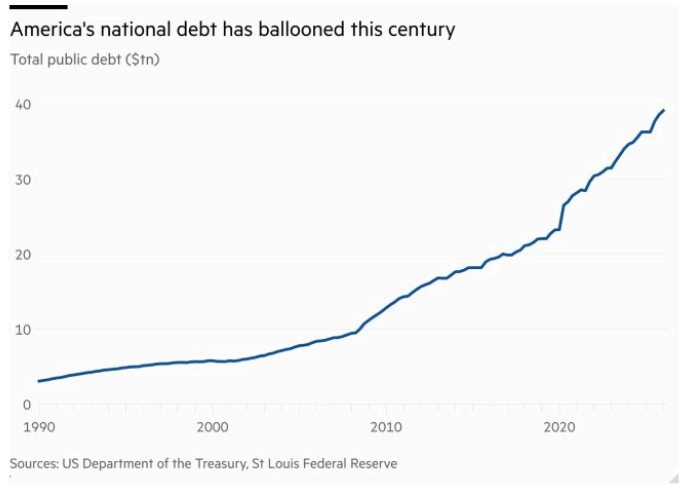
答 其他條件不變時，供給增加會壓低債券價格。因票息與到期本金固定，價格下降會使殖利率上升，政府的新借款成本也提高。

美國國債新聞｜PDF 第 2/6 頁 債務規模、買回長債與通膨疑慮

America's national debt has hit a record $40tn as borrowing rises at a historic pace, despite Donald Trump's vow to rein in government spending.

Gross federal debt hit the mark on Tuesday, according to Treasury department data released on Wednesday afternoon. It has grown at its fastest ever pace outside the pandemic era, ballooning by $3tn over the past year, FT calculations show.

Marc Goldwein, senior policy director at the Committee for a Responsible Federal Budget think-tank, called the numbers a "gigantic flashing 'check engine' light".



The record-high comes as the US Treasury said yesterday it would "at least double" purchases of long-term government debt in a bid to contain a sell-off that has sent borrowing costs rocketing.

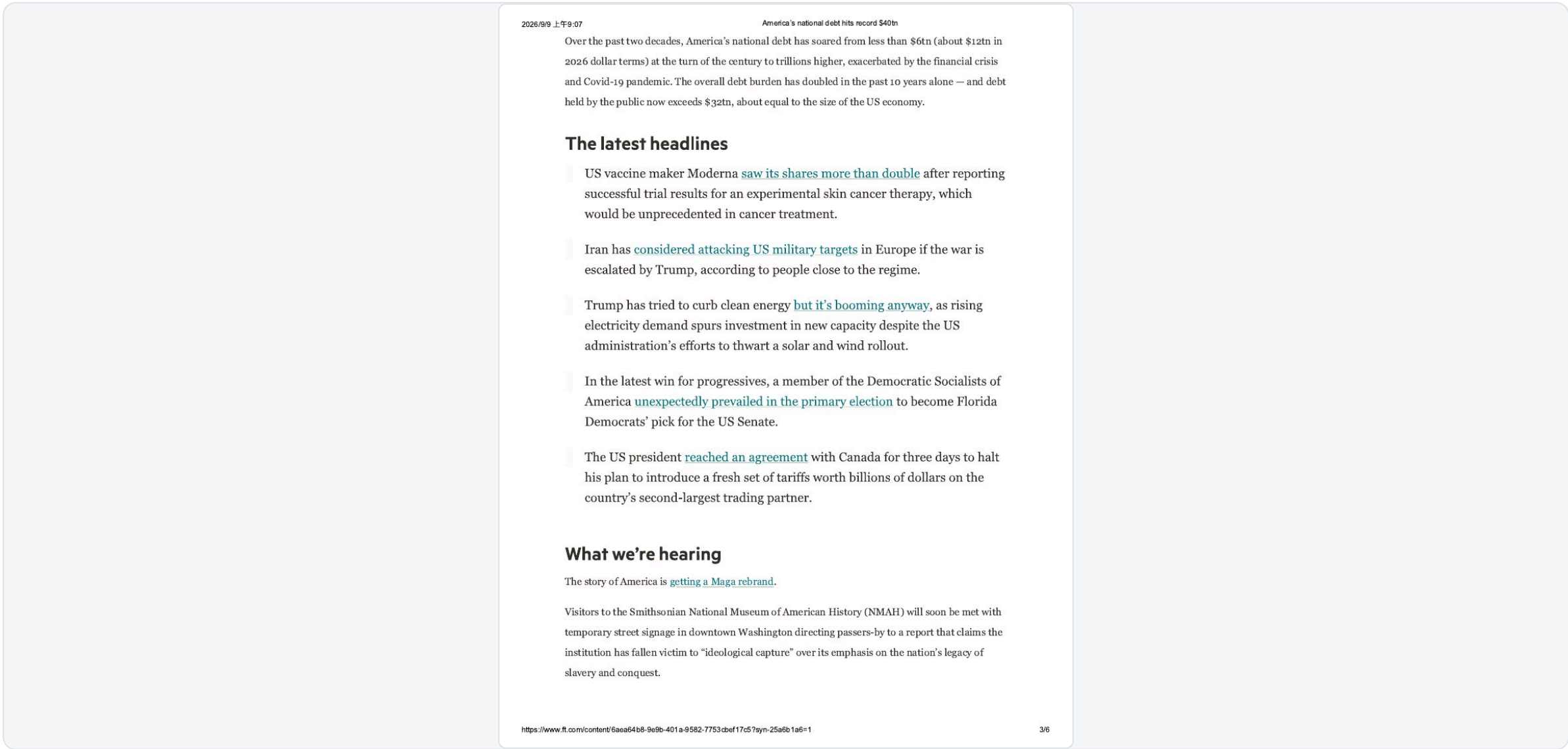
And it comes as US central bankers raised concerns in minutes of the Federal Reserve's latest policy vote that persistently high inflation will lead to rapid price rises becoming embedded in the world's largest economy, according to minutes of the Federal Reserve's latest policy vote.

Long story short: investor concern is bubbling over as inflation triggered by Trump's Iran war and Washington's mounting public-debt burden has escalated stress.

本頁中文重點

- 美國國債達到創紀錄的 40 兆美元，借款速度為歷史級別，儘管川普誓言要控制政府支出。
- 依財政部資料，聯邦總債務在週二觸及這個數字。FT 計算顯示，除疫情期間外，這是史上最快的成長速度，過去一年膨脹了 3 兆美元。
- 智庫 Committee for a Responsible Federal Budget 資深政策主任 Marc Goldwein 形容這些數字是「一個巨大的、閃爍的『引擎故障』警示燈」。
- 圖表「America's national debt has balloooned this century」：公共債務總額（兆美元），1990 年約 3，2000 年約 5.6，2008 年後快速上升，2020 年疫情時出現跳升，到 2026 年接近 40。資料來源：美國財政部、聖路易聯準銀行。
- 這個紀錄出現的同時，美國財政部表示將「至少加倍」買回長天期公債，目的是抑制一波已使借款成本急升的賣壓。
- 另一方面，聯準會最新一次決策會議紀要顯示官員擔心持續偏高的通膨會讓物價快速上漲的預期固著在美國經濟裡。
- 文章的總結：伊朗戰爭引發的通膨與華府不斷累積的公共債務負擔，使投資人的擔憂持續升溫。

美國國債新聞｜PDF 第 3/6 頁　二十年的債務軌跡，以及當期其他新聞



2026/9/9 上午9:07 America's national debt hits record $40tn

Over the past two decades, America's national debt has soared from less than $6tn (about $12tn in 2026 dollar terms) at the turn of the century to trillions higher, exacerbated by the financial crisis and Covid-19 pandemic. The overall debt burden has doubled in the past 10 years alone — and debt held by the public now exceeds $32tn, about equal to the size of the US economy.

The latest headlines

US vaccine maker Moderna saw its shares more than double after reporting successful trial results for an experimental skin cancer therapy, which would be unprecedented in cancer treatment.

Iran has considered attacking US military targets in Europe if the war is escalated by Trump, according to people close to the regime.

Trump has tried to curb clean energy but it's booming anyway, as rising electricity demand spurs investment in new capacity despite the US administration's efforts to thwart a solar and wind rollout.

In the latest win for progressives, a member of the Democratic Socialists of America unexpectedly prevailed in the primary election to become Florida Democrats' pick for the US Senate.

The US president reached an agreement with Canada for three days to halt his plan to introduce a fresh set of tariffs worth billions of dollars on the country's second-largest trading partner.

What we're hearing

The story of America is getting a Maga rebrand.

Visitors to the Smithsonian National Museum of American History (NMAH) will soon be met with temporary street signage in downtown Washington directing passers-by to a report that claims the institution has fallen victim to "ideological capture" over its emphasis on the nation's legacy of slavery and conquest.

https://www.ft.com/content/6aea64b8-9e9b-401a-9582-7753cbef17c5?syn-25a6b1a6=1 3/6

本頁中文重點

- 過去二十年，美國國債從世紀初不到 6 兆美元（約相當於 2026 年幣值的 12 兆）一路攀升，金融危機與 Covid-19 疫情都讓它惡化。光是過去十年，整體債務負擔就翻了一倍，而由公眾持有的部分現在超過 32 兆美元，大約等於美國一整年的經濟規模。
- 本頁其餘為這份 White House Watch 電子報當日的其他頭條，與本課程主題無直接關係：Moderna 皮膚癌療法試驗成功使股價翻倍、伊朗考慮攻擊美軍歐洲目標、川普試圖壓抑綠能但投資仍成長、民主社會主義者意外贏得佛州參議員初選、美加暫停三天關稅計畫。
- 另起「What we're hearing」段落，談美國史敘事的政治爭議（史密森尼美國歷史博物館）。

美國國債新聞｜PDF 第 4/6 頁

電子報其餘內容：博物館爭議（與課程主題無關）

It's the latest front in America's culture wars. The battle over how the country's 250-year history should be framed has been thrust into the spotlight during Trump's second term as the administration attempts to wind back what it sees as a "woke" upheaval spurred by the Black Lives Matter movement.

White House official Vince Haley told the FT in a statement: "Very few Americans want the Smithsonian National Museum of American History to be used, in the words of its director, as a 'prime tool of social justice' and as an instrument of ideological activism and advocacy."

He added that the institution should provide an "honest and inspiring account of American history . . . telling the story of our Founding Fathers — who they were, what they did, and what they built."



An exhibit at the museum. The Smithsonian is not directly controlled by the US government and is governed by a bipartisan board © Finn Gomez/Getty Images

Haley was behind a 162-page report released last month called "Saving America's Story", in which the White House took aim at the museum's "thinly veiled anti-Americanism". It criticised several exhibits, including ones that celebrate drag queens, spotlight the destruction and displacement of Native American tribes, and allude to Alexander Hamilton and Benjamin Franklin owning slaves.

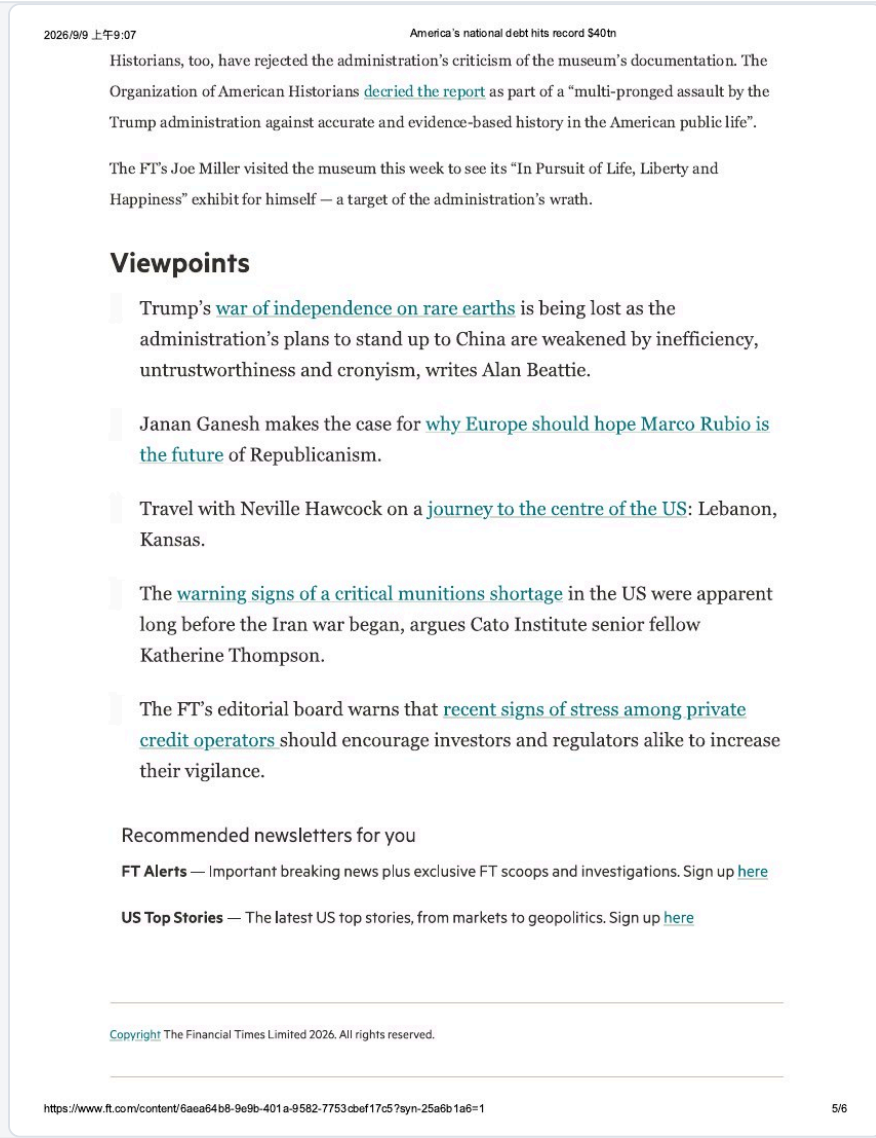
In testimony before Congress last month, NMAH director Anthea Hartig defended the museum, calling history "incredibly challenging".

本頁中文重點

- 本頁延續上一頁的文化政治議題：白宮官員 Vince Haley 批評史密森尼美國歷史博物館被用作「社會正義的主要工具」，並說該館應提供「誠實而激勵人心的美國史敘述」。
- 文中提到一份 162 頁報告〈Saving America's Story〉，批評博物館「隱約的反美傾向」。館長 Anthea Hartig 則在國會作證時為博物館辯護。
- 這一段是電子報裡的文化政治新聞，和估值沒有關係，列出來只是讓這篇 PDF 的每一頁都看得到。

美國國債新聞｜PDF 第 5/6 頁

電子報其餘內容：觀點欄（與課程主題無關）



2026/9/9 上午9:07 America's national debt hits record $40tn

Historians, too, have rejected the administration's criticism of the museum's documentation. The Organization of American Historians decried the report as part of a "multi-pronged assault by the Trump administration against accurate and evidence-based history in the American public life".

The FT's Joe Miller visited the museum this week to see its "In Pursuit of Life, Liberty and Happiness" exhibit for himself — a target of the administration's wrath.

Viewpoints

Trump's war of independence on rare earths is being lost as the administration's plans to stand up to China are weakened by inefficiency, untrustworthiness and cronyism, writes Alan Beattie.

Janan Ganesh makes the case for why Europe should hope Marco Rubio is the future of Republicanism.

Travel with Neville Hawcock on a journey to the centre of the US: Lebanon, Kansas.

The warning signs of a critical munitions shortage in the US were apparent long before the Iran war began, argues Cato Institute senior fellow Katherine Thompson.

The FT's editorial board warns that recent signs of stress among private credit operators should encourage investors and regulators alike to increase their vigilance.

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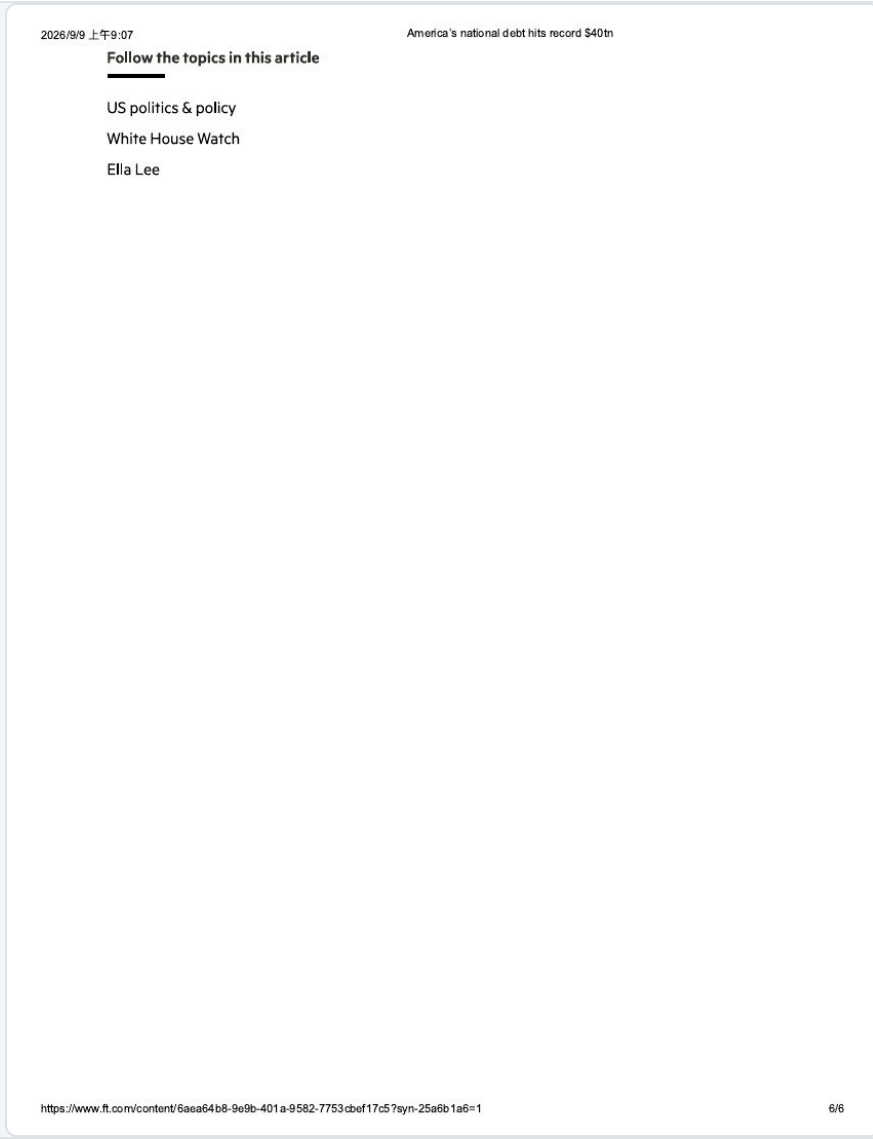
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本頁中文重點

- 歷史學者組織反對行政部門對博物館的批評。FT 記者實地參訪該展覽。
- 「Viewpoints」欄列出當期評論：稀土政策、Marco Rubio 與共和黨路線、美國中部旅行報導、彈藥短缺警訊，以及 FT 編輯部對私募信貸承作機構近期壓力的警示——最後這一則與本課程的信用風險主題有間接關聯。
- 本頁其餘為電子報推薦與版權聲明。

美國國債新聞｜PDF 第 6/6 頁

文章末頁：主題標籤

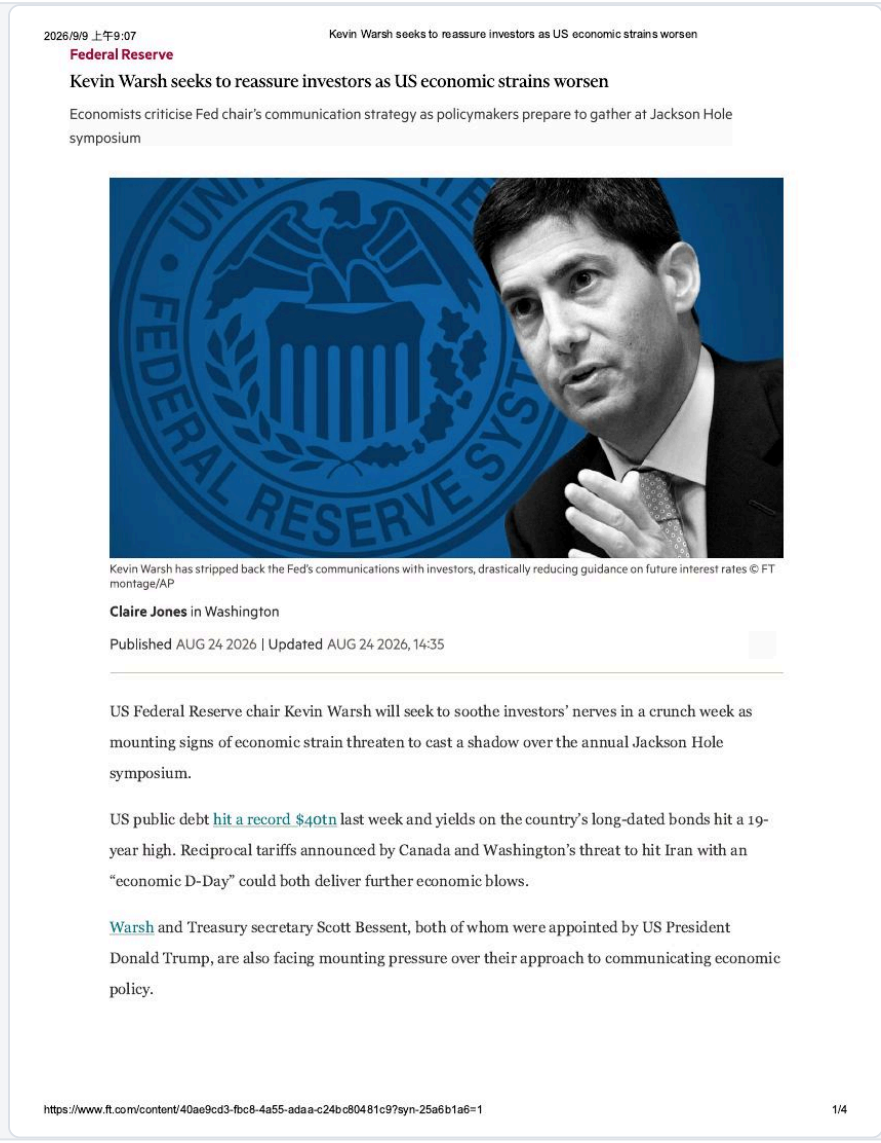


本頁中文重點

- 「Follow the topics in this article」標籤：US politics & policy、White House Watch、Ella Lee。

聯準會溝通新聞｜PDF 第 1/4 頁

Kevin Warsh 與聯準會溝通：市場信任也是政策工具



2026/9/9 上午9:07 Kevin Warsh seeks to reassure investors as US economic strains worsen

Federal Reserve

Kevin Warsh seeks to reassure investors as US economic strains worsen

Economists criticise Fed chair's communication strategy as policymakers prepare to gather at Jackson Hole symposium

Kevin Warsh has stripped back the Fed's communications with investors, drastically reducing guidance on future interest rates © FT montage/AP

Claire Jones in Washington

Published AUG 24 2026 | Updated AUG 24 2026, 14:35

US Federal Reserve chair Kevin Warsh will seek to soothe investors' nerves in a crunch week as mounting signs of economic strain threaten to cast a shadow over the annual Jackson Hole symposium.

US public debt hit a record $40tn last week and yields on the country's long-dated bonds hit a 19-year high. Reciprocal tariffs announced by Canada and Washington's threat to hit Iran with an "economic D-Day" could both deliver further economic blows.

Warsh and Treasury secretary Scott Bessent, both of whom were appointed by US President Donald Trump, are also facing mounting pressure over their approach to communicating economic policy.

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文章重點

- 文章聚焦新任 Fed 主席 Kevin Warsh 面對高債務、長天期殖利率與通膨黏著時，如何向市場建立政策可信度。
- Warsh 減少前瞻指引與公開溝通，可能讓市場重新依賴經濟資料，但在政策不確定性升高時，也可能放大期限溢酬與波動。
- 市場擔心通膨回到 2% 目標所需時間更長。若央行可信度下降，長期利率可能因投資人要求更高風險補償而上升。

老師課堂補充

- 老師說明美國政府與 Fed 在法律上相互獨立，但 Fed 主席由總統任命，因此人事與制度可信度仍會影響市場判斷。
- Fed 的雙重使命是物價穩定與就業。通膨目標約為 2%。失業率沒有固定政策目標，但市場常把約 4% 多視為觀察經濟是否轉弱的重要區間。
- 政策工具不只有調整利率。FOMC 每年固定開會，主席與官員的演說、聲明與前瞻指引也會影響投資人的預期，因而改變資產價格與借款成本。**即：聯準會不動利率也能影響市場，因為市場會先去猜它下一步要做什麼。**
- 本篇文章的重點不是單純評斷升息或降息，而是新主席在債務、關稅與通膨壓力並存時，能否以清楚溝通維持 Fed 的政策可信度。

課堂提問與解答

問 Fed 的雙重使命是什麼？

答 維持物價穩定與促進充分就業。兩者有時方向一致，有時需要取捨。

問 為什麼央行的溝通會影響長期利率？

答 長期利率反映市場對未來短期利率、通膨與風險溢酬的預期。溝通若不清楚，投資人會要求更高的不確定性補償。

聯準會溝通新聞｜PDF 第 2/4 頁

溝通減少之後，市場的反應

Some investors said Bessent's abrupt announcement of bond buybacks last week reduced the credibility of the Treasury's guidance. Warsh has stripped back the US central bank's communications with investors, drastically reducing guidance on future interest rates.

The Fed chair will address central bankers and economists at the Kansas City Fed's Jackson Hole gathering on Friday for the first time since his appointment, where he is expected to set out the framework behind his pared-back style.

In a poll of academic economists conducted for the FT by the University of Chicago's Booth School of Business, a majority said they were concerned that inflation would take longer to tame than had been the case earlier in the year.

Some also criticised the Fed chair and Bessent's approach to communications.

Nearly 60 per cent of those polled said it would take the central bank longer to get back to its 2 per cent target than when Warsh was confirmed as Fed chair by the Senate in mid-May.

Inflation has been running above the Fed's 2 per cent goal for more than five years and the conflict between Tehran and Washington has pushed up petrol and diesel costs, putting fresh upward pressure on prices.

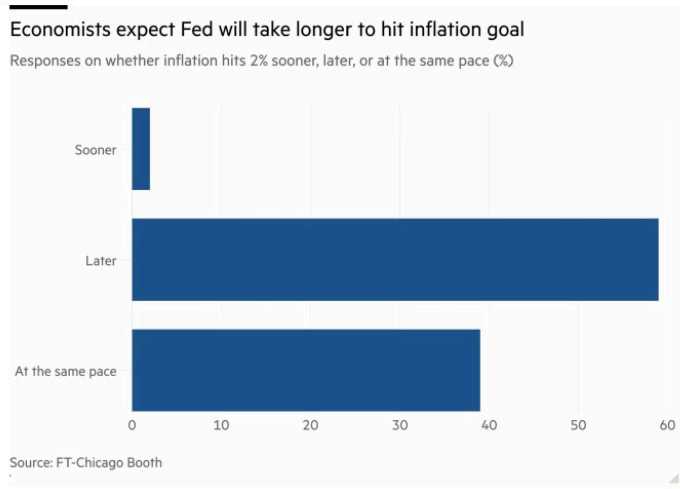
Trump's latest bout of tariffs on Canada and his administration's threats towards countries it deems to be aiding Iran could add to that.

本頁中文重點

- 有投資人認為財政部長 Scott Bessent 上週突然宣布公債買回，降低了財政部指引的可信度。Warsh 則大幅縮減聯準會對投資人的溝通，特別是未來利率路徑的指引。
- Warsh 將在堪薩斯城聯準銀行的 Jackson Hole 會議上首次以主席身分發表演說，預期會說明他這種精簡溝通風格背後的架構。
- FT 委託芝加哥大學 Booth 商學院進行的學界經濟學家調查中，多數人擔心通膨要壓下來所需的時間會比今年稍早預估的更長。部分人也批評主席與財政部長的溝通方式。
- 接近 60% 的受訪者認為，相較於 Warsh 五月中獲參議院同意任命時，央行回到 2% 目標所需的時間變得更長。
- 通膨已經連續五年多高於聯準會 2% 的目標，德黑蘭與華府的衝突又推升汽油與柴油成本，對物價形成新的上行壓力。川普最新一輪對加拿大關稅，以及對「協助伊朗」國家的威脅，可能再加一層。

聯準會溝通新聞｜PDF 第 3/4 頁

調查結果：通膨目標要更久才達成，可信度影響長率



"What is unsettling is Warsh's unwillingness to provide a clear and open assessment of the current situation and the outlook for the US economy," said Christiane Baumeister of the University of Notre Dame. "This gives rise to unnecessary speculation, which has the potential to jeopardise stability and raise concerns about the Fed's credibility."

More than 60 per cent of respondents said concerns over the US central bank's credibility had either a very substantial or a somewhat substantial impact on the rise in long-term yields since Trump nominated Warsh in late January.

Robert Barbera of Johns Hopkins University said Bessent's bond market intervention had made the "dollar look iffy". A depreciation of the US currency would raise inflationary risks by increasing the price of imports.

Eric Rosengren, a former head of the Boston Fed, said that while Warsh could remain silent on future decisions, he needed to explain current ones. "The communication strategy is driving a loss of Fed credibility," Rosengren said.

Almost three-quarters of those polled said the shift from the more open communications strategy of his predecessors — Jay Powell, Janet Yellen and Ben Bernanke — was the most important change Warsh had made since becoming chair in late May.

本頁中文重點

- 圖表「Economists expect Fed will take longer to hit inflation goal」：對「通膨回到 2% 會更快、更慢，還是速度不變」的回答比例，「更慢」約 59%、「速度不變」約 39%、「更快」約 2%。資料來源 FT-Chicago Booth。
- 聖母大學 Christiane Baumeister：「令人不安的是 Warsh 不願對當前情勢與美國經濟前景提出清楚公開的評估。這會引發不必要的揣測，可能危及穩定並引起對聯準會可信度的疑慮。」
- 超過 60% 的受訪者認為，自川普一月底提名 Warsh 以來，對央行可信度的疑慮對長期殖利率上升有「非常大」或「相當大」的影響。
- 約翰霍普金斯大學 Robert Barbera 說 Bessent 介入債市讓「美元看起來不太妙」。美元貶值會推升進口價格，從而加大通膨風險。
- 前波士頓聯準銀行總裁 Eric Rosengren 說，Warsh 可以對未來決策保持沉默，但必須解釋當前決策：「這套溝通策略正在造成聯準會可信度的流失。」
- 將近四分之三的受訪者認為，從前幾任（Powell、Yellen、Bernanke）較開放的溝通策略轉向，是 Warsh 五月底就任以來最重要的改變。

聯準會溝通新聞｜PDF 第 4/4 頁

支持方的理由，與文章末頁標籤



2026/9/9 上午9:07 Kevin Warsh seeks to reassure investors as US economic strains worsen

A large majority of respondents — 75 per cent — said Warsh's overall performance since becoming chair was in line with their expectations.

Warsh has previously said that he hoped breaking the feedback loop between the Fed and investors would mean that markets would focus less on scrutinising officials' remarks and more on economic data.

"The Fed had become more and more entwined in micromanaging markets, to the detriment of the functioning of the capital market," said Deborah Lucas, a professor at the Massachusetts Institute of Technology. "Warsh is willing to stand up against the vested interests in the status quo to try to reverse this. I commend him for it."

Jon Faust, a fellow at Johns Hopkins who worked as an adviser to former Fed chair Powell, said Warsh had allayed concerns that he would pander to pressure from Trump, who has consistently called for the US central bank to slash borrowing costs despite high inflation.

"He has pretty much overcome the sock-puppet fears and he hasn't gone in and caused turmoil inside the Fed," said Faust. "There have been some horror scenarios that have been pretty firmly ruled out."

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本頁中文重點

- 75% 的受訪者表示 Warsh 就任以來的整體表現符合他們的預期。
- Warsh 先前說過，他希望打斷聯準會與投資人之間的回饋循環，讓市場少花力氣揣摩官員談話、多看經濟數據本身。
- 麻省理工學院教授 Deborah Lucas 支持這個方向：「聯準會變得越來越深地介入微觀管理市場，對資本市場的運作有害。Warsh 願意對抗既得利益去扭轉這件事，我肯定他。」
- 曾任 Powell 顧問、現於約翰霍普金斯的 Jon Faust 說，Warsh 化解了外界擔心他會屈從川普壓力的疑慮——川普一直要求央行在高通膨下大幅調降借款成本。Faust：「他大致上克服了『傀儡』的疑慮，也沒有在聯準會內部製造動盪。」
- 本頁末為版權聲明與主題標籤：US economy、US politics & policy、US interest rates、Central banks、Monetary policy。

全球利息帳單新聞｜PDF 第 1/13 頁

全球 2 兆美元利息帳單：債務存量與借款成本的雙重壓力



The Big Read Global Economy

The world's $2tn interest bill

Many countries now spend more on debt servicing than defence, including the US, France and the UK

s stop advert in Washington highlights the rising levels of US government borrowing, which has led to an indebtedness that is ding into global markets © Jemal Countess/Getty Images

Ian Smith, **Sam Fleming** and **Emily Herbert** in London

Published YESTERDAY

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The world's governments have created a $2tn monster — a debt-servicing burden that gobbles up tax revenues and has the power to overwhelm elected leaders.

More money is now spent on servicing the national debt than on defence in the UK, France and the US; the same is true for more than a dozen states in the 38-member OECD rich countries' club.

Their problem is that borrowing costs have climbed to the highest level in almost two decades just as governments are taking on record debt.

The total owed by the US government reached a record $40tn last month. This year, OECD nations are expected to borrow $18tn between them, another all-time high. In 2025, the group's total debt-servicing bill exceeded $2tn, or 3 per cent of GDP.

文章重點

- OECD 國家 2025 年政府債務利息支出合計超過 2 兆美元，約占 GDP 的 3%。多國付息支出甚至高於國防預算。
- 疫情後的大量借款、通膨與量化寬鬆退場，使公債殖利率回升。舊債逐步到期再融資後，更高利率會持續傳導到財政支出。
- 買方結構也在改變：傳統退休基金等長期持有者減少，避險基金等要求更高報酬的投資人比重上升，使政府更難以低成本長期借款。
- 可能的調整途徑包括提高成長、增稅、削減支出或縮短債務期限。但短期化會提高再融資頻率，金融壓抑則把成本轉嫁給儲蓄者。

老師課堂補充

- 問題並非美國獨有。英國、法國及其他 OECD 國家同樣面對債務增加與付息壓力，必須同時觀察『借了多少』與『每單位債務要付多少成本』。
- 過去政府債券有大量追求安全、未必要求高報酬的機構買家。當買方減少或投資人偏好轉向高報酬，政府就需要以更高殖利率吸引資金。
- 若利息支出惡化財政前景，市場要求更高殖利率，而更高殖利率又推升利息支出，就可能形成債務—利率的負向循環。

課堂提問與解答

問 為何不能只看政府債務總額？

答 因為同一債務規模在 1% 與 5% 利率下的財政負擔完全不同。還要看債務期限、再融資速度與投資人要求的風險溢酬。

全球利息帳單新聞｜PDF 第 2/13 頁

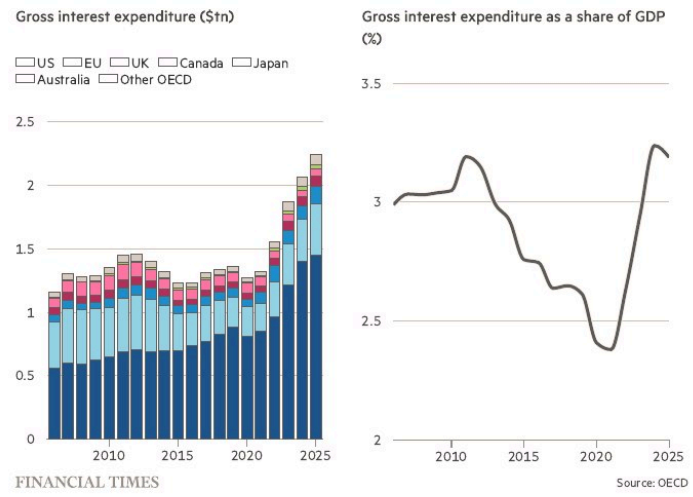
為什麼利息帳單還會繼續上升

That number is set to rise in the coming years because of the bond sell-off that has pushed up yields since the Covid-19 crisis, increasing the cost of debt for governments across the world.

"These massive sovereign debts are having to get refinanced at ever-increasing rates," says Mike Riddell, a fund manager at Fidelity International. "Global investors are already starting to get scared."

The summer rout in the global bond market has only augmented the post-Covid surge in yields triggered by inflation shocks, increased government borrowing and the end of central banks' "quantitative easing" bond-buying programmes.

Interest on government debt in the OECD now exceeds $2tn per year

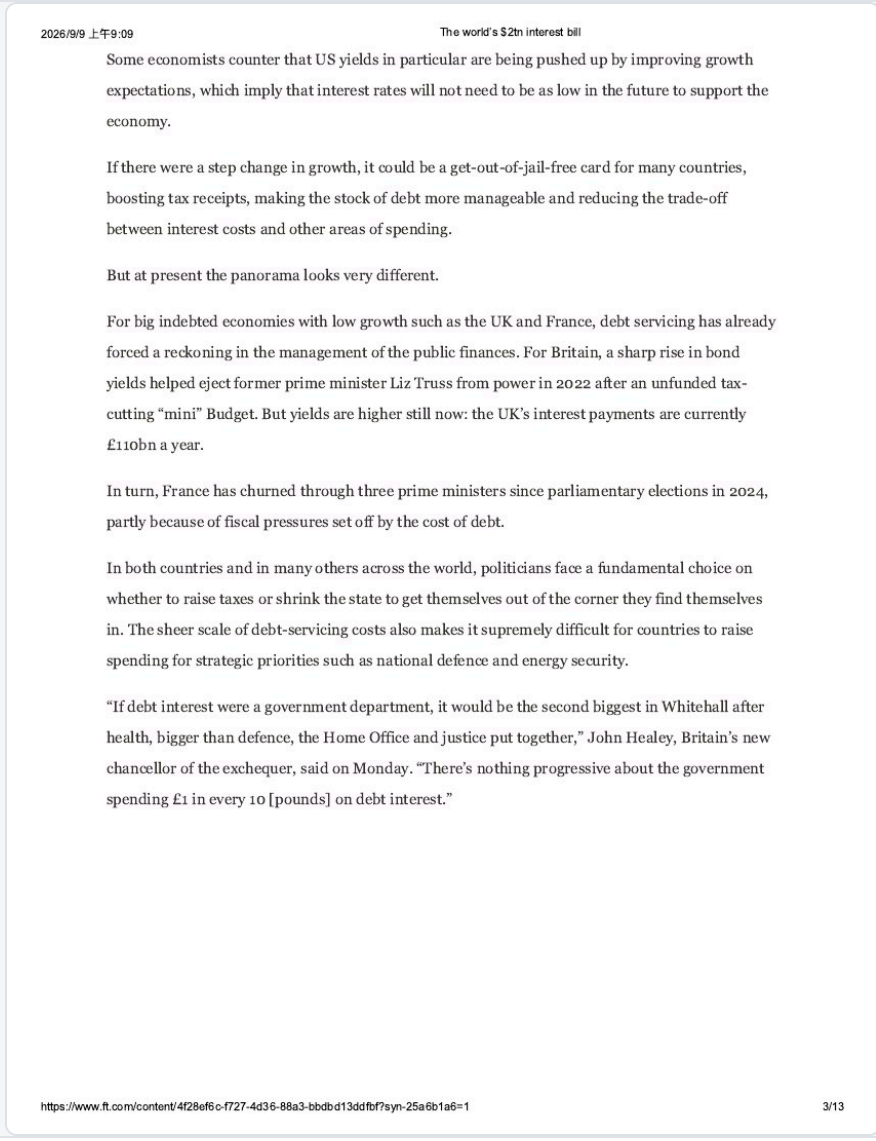


The average benchmark 10-year bond yield for G7 countries has reached 4 per cent for the first time since 2008, amid concerns about the inflationary impact of the Iran war and the sheer amount of debt being sought by both the public and private sectors — particularly tech groups.

本頁中文重點

- 這個數字未來幾年還會上升，原因是 Covid-19 危機以來的債券賣壓推高了殖利率，各國政府的債務成本因而增加。
- Fidelity International 基金經理人 Mike Riddell：「這些龐大的主權債務必須用越來越高的利率再融資。全球投資人已經開始感到害怕。」
- 今年夏天的全球債市賣壓，是在通膨衝擊、政府借款增加與央行結束量化寬鬆購債之後，再疊加上去的一波殖利率上升。
- 圖表「Interest on government debt in the OECD now exceeds $2tn per year」：左圖為毛利息支出（兆美元），依美國、歐盟、英國、加拿大、日本、澳洲與其他 OECD 分層，2010 年前後約 1.3，2021 年約 1.3，之後快速攀升到 2025 年超過 2.2。右圖為毛利息支出佔 GDP 比重（%），2010 年前後約 3.2，2021 年降到約 2.4，2025 年回升到約 3.2。資料來源 OECD。
- G7 國家十年期公債平均殖利率已達 4%，是 2008 年以來首次，背景包括伊朗戰爭的通膨影響，以及公部門與民間（特別是科技業）同時大量舉債。

全球利息帳單新聞｜PDF 第 3/13 頁 成長能否解套，以及英國與法國的處境


2026/9/9 上午9:09 The world's $2tn interest bill

Some economists counter that US yields in particular are being pushed up by improving growth expectations, which imply that interest rates will not need to be as low in the future to support the economy.

If there were a step change in growth, it could be a get-out-of-jail-free card for many countries, boosting tax receipts, making the stock of debt more manageable and reducing the trade-off between interest costs and other areas of spending.

But at present the panorama looks very different.

For big indebted economies with low growth such as the UK and France, debt servicing has already forced a reckoning in the management of the public finances. For Britain, a sharp rise in bond yields helped eject former prime minister Liz Truss from power in 2022 after an unfunded tax-cutting "mini" Budget. But yields are higher still now: the UK's interest payments are currently £110bn a year.

In turn, France has churned through three prime ministers since parliamentary elections in 2024, partly because of fiscal pressures set off by the cost of debt.

In both countries and in many others across the world, politicians face a fundamental choice on whether to raise taxes or shrink the state to get themselves out of the corner they find themselves in. The sheer scale of debt-servicing costs also makes it supremely difficult for countries to raise spending for strategic priorities such as national defence and energy security.

"If debt interest were a government department, it would be the second biggest in Whitehall after health, bigger than defence, the Home Office and justice put together," John Healey, Britain's new chancellor of the exchequer, said on Monday. "There's nothing progressive about the government spending £1 in every 10 [pounds] on debt interest."

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本頁中文重點

- 有經濟學家反駁說，美國殖利率上升部分來自成長預期改善——那意味未來不需要那麼低的利率來支撐經濟。
- 如果成長真的出現階梯式跳升，對許多國家會是脫困的關鍵：稅收增加、債務存量更可控，利息支出與其他支出之間的取捨也會緩解。
- 但目前的景象很不一樣。對英國、法國這類高債務、低成長的大型經濟體，償債負擔已經迫使公共財政management面臨清算。英國 2022 年因未提供財源的減稅「迷你預算」導致殖利率急升，前首相 Liz Truss 因此下台。而現在的殖利率比當時更高，英國一年的利息支出為 1,100 億英鎊。**即：利息支出大到一定程度，政府就沒辦法再假裝財政可以慢慢處理。**
- 法國自 2024 年國會選舉以來已經換了三位總理，部分原因就是債務成本造成的財政壓力。
- 兩國以及世界上許多國家的政治人物，面對的是加稅或縮減政府規模的根本選擇。龐大的償債成本也使各國極難為國防、能源安全這類戰略優先項目增加支出。
- 英國新任財政大臣 John Healey：「如果債務利息是一個政府部門，它會是白廳僅次於衛生的第二大部門，比國防、內政部與司法部加起來還大。政府每花 10 英鎊就有 1 英鎊付債務利息，這一點都不進步。」

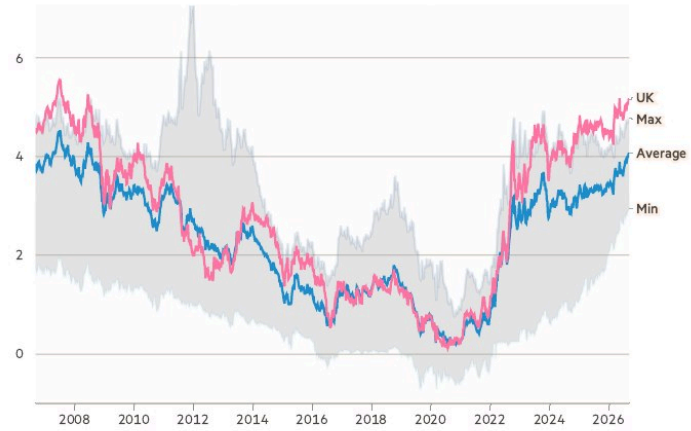
全球利息帳單新聞｜PDF 第 4/13 頁

G7 借款成本與殖利率上升的原因

The world's $2tn interest bill

Average G7 borrowing costs have climbed to their highest since 2008

10-year bond yield (%), UK vs G7 range



FINANCIAL TIMES Source: LSEG • G7 = UK, US, Japan, Germany, France, Italy and Canada. Max and min excludes UK.

Why yields are going up

Up until the 2008 global financial crisis, interest rates were relatively high for major economies, but levels of debt were generally much lower. After that, debt surged but interest rates were kept lower by emergency measures taken by central banks as well as low economic growth. Now, by contrast, policymakers face an unenviable combination: record debt and high borrowing costs.

While governments are still finding enough buyers for their debt, investors are demanding a higher price for their money. New bond sales in recent months for the UK, the US and others have locked in the highest yields in almost two decades for longer-dated debt.

This shift has been accelerated by changes to pension schemes in some countries that have meant traditional long-term buyers stepping back, replaced by fast-money hedge funds that can be quick to demand higher yields.

本頁中文重點

- 圖表「Average G7 borrowing costs have climbed to their highest since 2008」：十年期公債殖利率（%），英國單獨標示並對照 G7 區間。2020 年前後接近 0，2022 年起急升，2026 年英國約 5、平均約 4。資料來源 LSEG。G7 為英、美、日、德、法、義、加，區間的最大最小值不含英國。
- 小標「Why yields are going up」：2008 年金融危機以前，主要經濟體利率相對高，但債務水準普遍低很多。危機之後債務暴增，利率卻因央行的緊急措施與低經濟成長而被壓低。現在決策者面對的是很難處理的組合——債務創高，而借款成本也高。
- 政府目前仍找得到足夠的買家，但投資人要求更高的價格。英國、美國與其他國家近幾個月的新債發行，已把長天期債務鎖在近二十年來最高的殖利率水準。
- 這個轉變被某些國家的退休金制度改變加速：傳統的長期買家退場，由反應快、會迅速要求更高殖利率的避險基金取代。

全球利息帳單新聞｜PDF 第 5/13 頁

債務成本的傳導是漸進的，但方向明確

"The era of free money is definitely over," says Kim Crawford, a global fixed income manager at JPMorgan Asset Management. "The bond market is [now] looking for discipline."

The nature of government debt markets, in which governments fund themselves with debt that is repaid at a wide range of durations, some coming due decades into the future, means that the transmission of higher borrowing costs into higher interest costs occurs gradually over time.

But the direction of travel seems clear. The US's Congressional Budget Office estimates that America's debt costs will double over the next decade and, on the current trajectory, exceed Social Security spending after 2047. The burden for other countries — which typically find it harder than the US to convince investors to buy their bonds on favourable terms — is likely to be higher still.

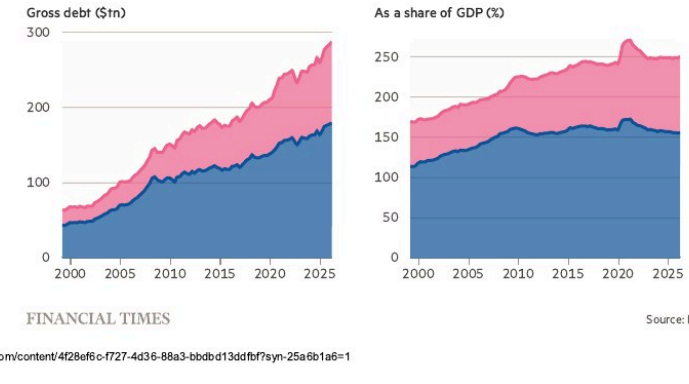
The causes of the problems with public finances are deep-rooted and long-running. They include a surge in spending to battle the Covid-induced slump and energy crises, as well as broader fiscal challenges since the financial crisis brought down rates of growth.

Global public debt hit 94 per cent of world GDP last year, up more than 10 percentage points since the year before the pandemic. The IMF now believes it will reach 100 per cent by the end of this decade.

Total corporate and government debt has reached almost $300tn

Global sectoral debt

Corporate Government



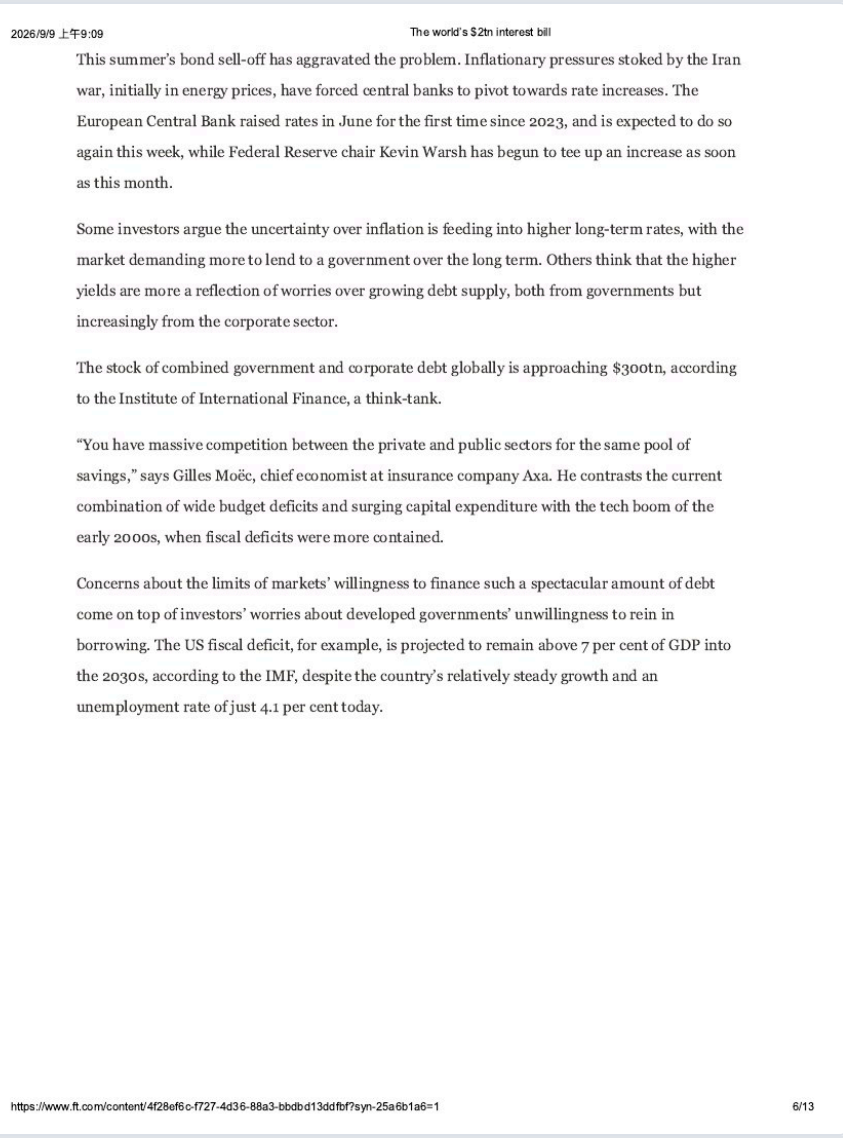
FINANCIAL TIMES Source: IIF

本頁中文重點

- 摩根大通資產管理全球固定收益經理人 Kim Crawford：「免費資金的時代確定結束了。債市現在要的是紀律。」
- 政府債務市場的特性是以各種年期融資，有些要到幾十年後才到期，所以借款成本上升要傳導成利息支出上升是漸進發生的。
- 但方向很清楚。美國國會預算辦公室估計，未來十年美國的債務成本會翻倍，並在 2047 年後超過社會安全支出。其他國家通常比美國更難說服投資人以有利條件買它們的債，負擔可能更重。
- 公共財政問題的成因是長期而深層的：對抗 Covid 造成的衰退與能源危機的支出激增，以及金融危機拉低成長率之後更廣泛的財政挑戰。
- 全球公共債務去年達到世界 GDP 的 94%，比疫情前一年高出 10 個百分點以上。IMF 認為本十年結束前會達到 100%。
- 圖表「Total corporate and government debt has reached almost $300tn」：全球部門別債務，分企業與政府。左圖毛債務（兆美元）從 2000 年約 65 升到 2026 年接近 290。右圖佔 GDP 比重（%）從約 170 升到約 250。資料來源 IIF。

全球利息帳單新聞｜PDF 第 6/13 頁

公私部門競爭同一筆儲蓄



2026/9/9 上午9:09 The world's $2tn interest bill

This summer's bond sell-off has aggravated the problem. Inflationary pressures stoked by the Iran war, initially in energy prices, have forced central banks to pivot towards rate increases. The European Central Bank raised rates in June for the first time since 2023, and is expected to do so again this week, while Federal Reserve chair Kevin Warsh has begun to tee up an increase as soon as this month.

Some investors argue the uncertainty over inflation is feeding into higher long-term rates, with the market demanding more to lend to a government over the long term. Others think that the higher yields are more a reflection of worries over growing debt supply, both from governments but increasingly from the corporate sector.

The stock of combined government and corporate debt globally is approaching $300tn, according to the Institute of International Finance, a think-tank.

"You have massive competition between the private and public sectors for the same pool of savings," says Gilles Moëc, chief economist at insurance company Axa. He contrasts the current combination of wide budget deficits and surging capital expenditure with the tech boom of the early 2000s, when fiscal deficits were more contained.

Concerns about the limits of markets' willingness to finance such a spectacular amount of debt come on top of investors' worries about developed governments' unwillingness to rein in borrowing. The US fiscal deficit, for example, is projected to remain above 7 per cent of GDP into the 2030s, according to the IMF, despite the country's relatively steady growth and an unemployment rate of just 4.1 per cent today.

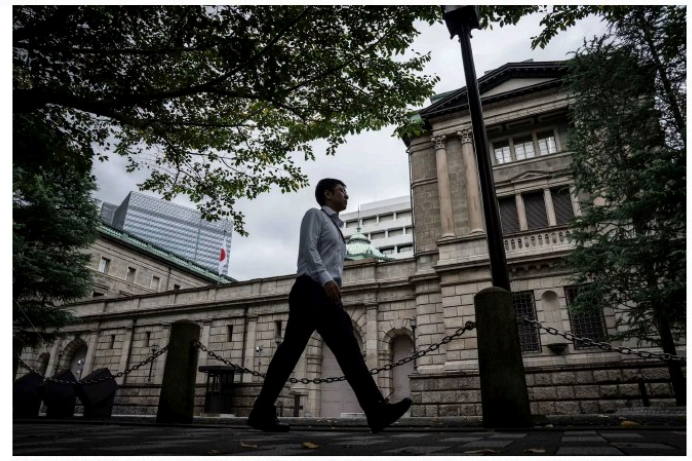
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本頁中文重點

- 今年夏天的債券賣壓讓問題惡化。伊朗戰爭先從能源價格推升通膨壓力，迫使央行轉向升息。歐洲央行六月自 2023 年以來首次升息，預期本週再升一次。聯準會主席 Warsh 也已經開始鋪陳最快本月升息。
- 有投資人認為通膨的不確定性正在傳導到更高的長期利率，市場要求更多報酬才願意長期借錢給政府。另一些人認為殖利率上升更多是反映對債務供給成長的擔憂——不只來自政府，也越來越多來自企業部門。
- 依國際金融協會（IIF）資料，全球政府與企業債務的存量合計接近 300 兆美元。
- 安盛首席經濟學家 Gilles Moëc：「公私部門之間正在激烈競爭同一池儲蓄。」他把現在「寬財政赤字加上資本支出暴增」的組合，和 2000 年代初期財政赤字較受控的科技榮景作對比。
- 除了市場願意為這麼龐大的債務提供多少資金有其極限之外，投資人還擔心已開發國家政府不願意控制借款。IMF 預估美國財政赤字到 2030 年代仍會高於 GDP 的 7%，儘管該國成長相對穩定、失業率目前只有 4.1%。

全球利息帳單新聞｜PDF 第 7/13 頁

政治風險溢酬與日本因素



The Bank of Japan headquarters in Tokyo. The central bank is raising interest rates just as the government embraces stimulatory spending © Yuichi Yamazaki/AFP/Getty Images

Political volatility has already led to higher debt interest costs than would have been paid otherwise, some investors argue, in what is sometimes dubbed the "moron premium". In Europe, political fragility since 2022 has added €100bn to interest expenses for the UK, Italy, France, Spain and Belgium, estimates insurer Allianz. Part of the rise in yields this year could be investors demanding compensation for political ructions to come.

Another factor unsettling the bond market this year has been events in Japan, for a long time the anchor that helped to hold down global borrowing costs due to its negative interest rate regime and voracious demand for foreign assets.

Now, the country's central bank is increasing interest rates just as Prime Minister Sanae Takaichi openly embraces stimulatory government spending. As Tokyo's borrowing costs climb to levels not seen since the 1990s, investors are fretting that the higher yields at home will attract money back from other markets, pushing up costs elsewhere.

Doomed to a doom loop?

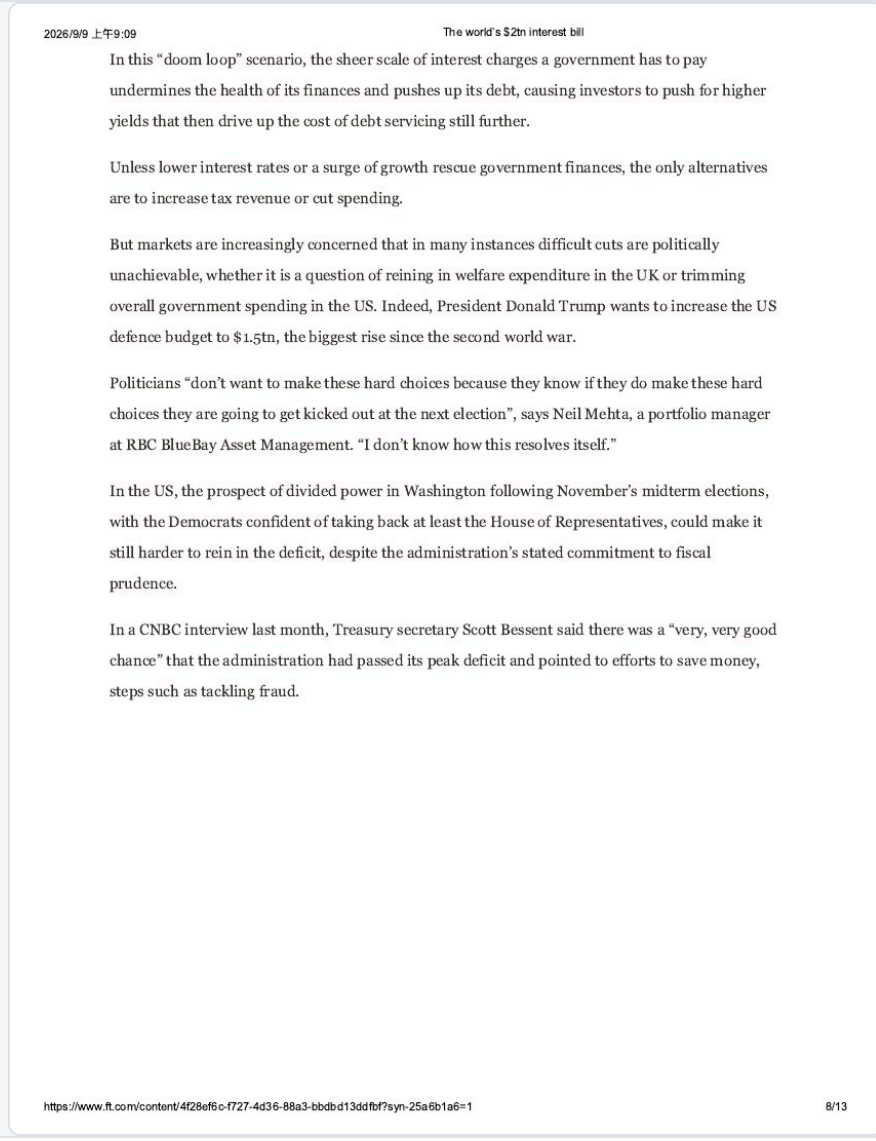
One big worry for investors is that some countries may be edging towards a vicious cycle in which rising debt costs make the fiscal outlook still worse.

本頁中文重點

- 圖片為東京的日本銀行總部。該央行正在升息，而政府同時擁抱刺激性支出。
- 有投資人認為政治波動本身已經推高了債務利息成本，這個現象有時被稱為「moron premium」。保險公司 Allianz 估計，2022 年以來的政治脆弱為英國、義大利、法國、西班牙與比利時增加了 1,000 億歐元的利息支出。今年殖利率上升的一部分，可能是投資人要求對未來政治動盪的補償。
- 今年擾動債市的另一個因素是日本。日本長期因為負利率政策與對外國資產的強勁需求，扮演壓低全球借款成本的錨。
- 現在日本央行開始升息，而首相高市早苗公開擁抱刺激性財政支出。隨著東京的借款成本升到 1990 年代以來未見的水準，投資人擔心本國較高的殖利率會把資金從其他市場吸回日本，推升別處的成本。
- 小標「Doomed to a doom loop?」：投資人的一大擔憂是某些國家可能正走向惡性循環——債務成本上升使財政前景更糟。

全球利息帳單新聞｜PDF 第 8/13 頁

惡性循環的機制，與政治上的困難


2026/9/9 上午9:09 The world's $2tn interest bill

In this "doom loop" scenario, the sheer scale of interest charges a government has to pay undermines the health of its finances and pushes up its debt, causing investors to push for higher yields that then drive up the cost of debt servicing still further.

Unless lower interest rates or a surge of growth rescue government finances, the only alternatives are to increase tax revenue or cut spending.

But markets are increasingly concerned that in many instances difficult cuts are politically unachievable, whether it is a question of reining in welfare expenditure in the UK or trimming overall government spending in the US. Indeed, President Donald Trump wants to increase the US defence budget to $1.5tn, the biggest rise since the second world war.

Politicians "don't want to make these hard choices because they know if they do make these hard choices they are going to get kicked out at the next election", says Neil Mehta, a portfolio manager at RBC BlueBay Asset Management. "I don't know how this resolves itself."

In the US, the prospect of divided power in Washington following November's midterm elections, with the Democrats confident of taking back at least the House of Representatives, could make it still harder to rein in the deficit, despite the administration's stated commitment to fiscal prudence.

In a CNBC interview last month, Treasury secretary Scott Bessent said there was a "very, very good chance" that the administration had passed its peak deficit and pointed to efforts to save money, steps such as tackling fraud.

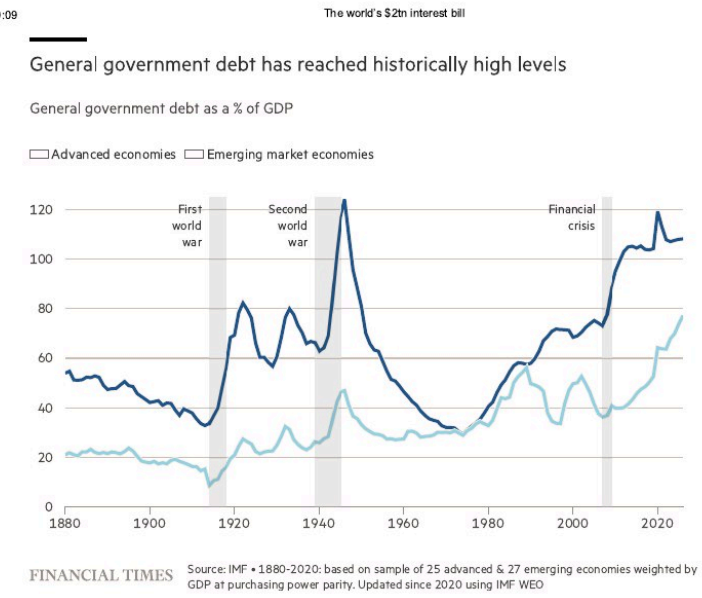
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本頁中文重點

- 惡性循環的運作方式：政府要付的利息規模之大，侵蝕財政健康並推高債務，使投資人要求更高的殖利率，而更高的殖利率又進一步墊高償債成本。
- 除非利率下降或成長大幅跳升來拯救政府財政，否則剩下的選擇只有增加稅收或削減支出。
- 但市場越來越擔心，許多情況下困難的削減在政治上根本做不到——無論是英國要控制社會福利支出，還是美國要縮減整體政府支出。事實上川普還想把美國國防預算提高到 1.5 兆美元，是二戰以來最大的增幅。
- RBC BlueBay 資產管理投資組合經理人 Neil Mehta：政治人物「不想做這些困難的選擇，因為他們知道一旦做了，下次選舉就會被踢出去。我不知道這件事要怎麼自己解決」。
- 美國方面，十一月期中選舉後華府可能出現分立政府（民主黨有信心至少拿回眾議院），即使行政部門宣示財政審慎，赤字也更難收斂。
- 財政部長 Bessent 上月在 CNBC 訪問中說，政府「很有很有可能」已經度過赤字高峰，並舉出打擊詐領等節流作為。

全球利息帳單新聞｜PDF 第 9/13 頁

債務佔 GDP 的歷史高點，與國際傳染



The anxiety surrounding US indebtedness is bleeding into global markets and infecting any other sovereign borrowers seen as particularly fragile — among them other "serial offenders" such as Britain, Italy and France, says David Rees, head of economics at asset manager Schroders. Some investors have this year dubbed the trio the "Bifs", for their high debt loads and vulnerability to the energy shock.

While the dollar's status as the world's premier reserve asset means many investors have little option but to buy Treasuries, other countries' finance ministers may well have to do more than the US to get their fiscal houses in order.

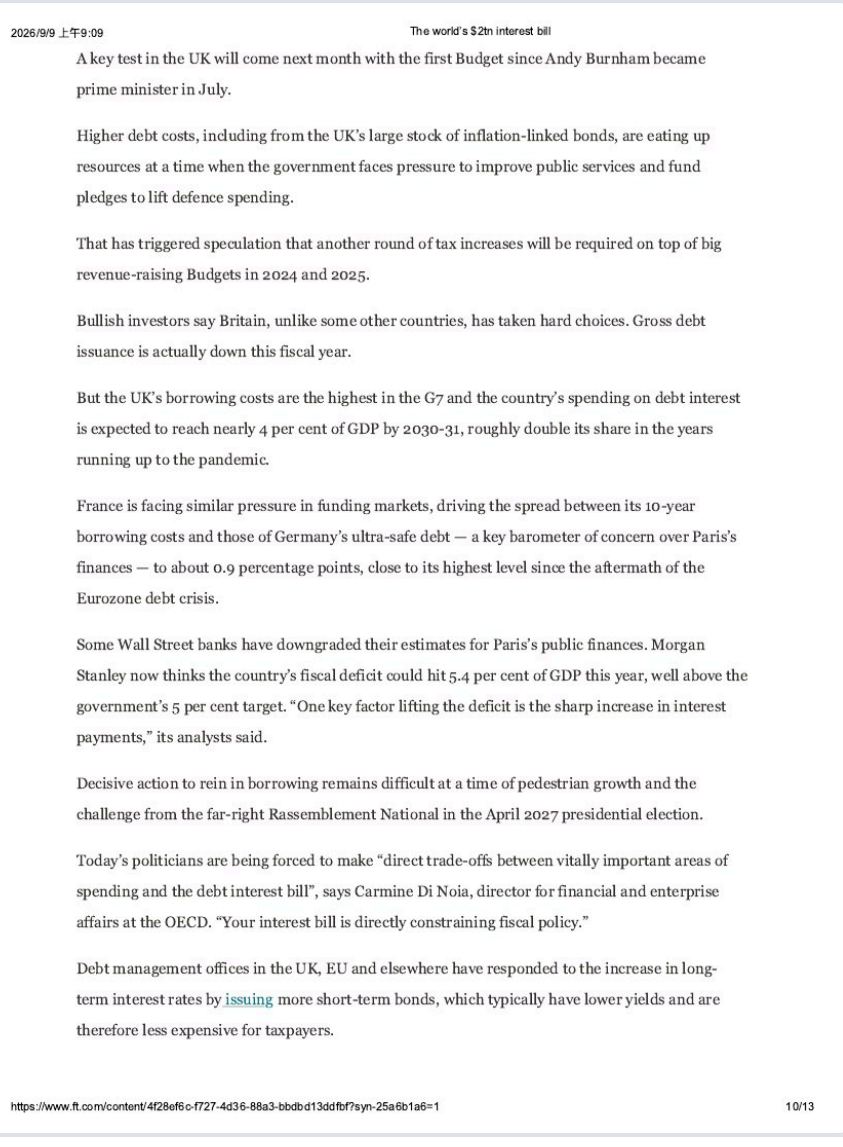
"No one is expecting miracles" in terms of deficit reduction, says Axa's Moëc. But countries need to show they can get or keep the headline deficit on a declining path even if the US is unlikely to do so.

本頁中文重點

- 圖表「General government debt has reached historically high levels」：一般政府債務佔 GDP 比重（%），分已開發經濟體與新興市場經濟體，涵蓋 1880 年至今並標示一戰、二戰與金融危機。已開發經濟體在二戰末期衝到約 124%，戰後降到 1970 年代約 32%，之後回升，近年在 105–118% 之間。新興市場從約 20% 升到近年約 76%。資料來源 IMF。
- 資產管理公司 Schroders 經濟主管 David Rees 指出，圍繞美國債務的焦慮正滲進全球市場，感染任何被視為特別脆弱的主權借款人——其中包括英國、義大利、法國這些「慣犯」。今年有投資人把這三個國家合稱為「Bifs」，因為它們債務負擔重又容易受能源衝擊影響。
- 美元作為全球首要準備資產的地位，意味許多投資人除了買美國公債別無選擇。但其他國家的財政部長可能必須比美國做更多，才能把財政整頓好。
- 安盛的 Moëc 說，赤字削減上「沒有人期待奇蹟」，但各國需要展現出總體赤字能走在下降路徑上，即使美國不太可能這麼做。

全球利息帳單新聞｜PDF 第 10/13 頁

英國與法國的具體數字



2026/9/9 上午9:09 The world's $2tn interest bill

A key test in the UK will come next month with the first Budget since Andy Burnham became prime minister in July.

Higher debt costs, including from the UK's large stock of inflation-linked bonds, are eating up resources at a time when the government faces pressure to improve public services and fund pledges to lift defence spending.

That has triggered speculation that another round of tax increases will be required on top of big revenue-raising Budgets in 2024 and 2025.

Bullish investors say Britain, unlike some other countries, has taken hard choices. Gross debt issuance is actually down this fiscal year.

But the UK's borrowing costs are the highest in the G7 and the country's spending on debt interest is expected to reach nearly 4 per cent of GDP by 2030-31, roughly double its share in the years running up to the pandemic.

France is facing similar pressure in funding markets, driving the spread between its 10-year borrowing costs and those of Germany's ultra-safe debt — a key barometer of concern over Paris's finances — to about 0.9 percentage points, close to its highest level since the aftermath of the Eurozone debt crisis.

Some Wall Street banks have downgraded their estimates for Paris's public finances. Morgan Stanley now thinks the country's fiscal deficit could hit 5.4 per cent of GDP this year, well above the government's 5 per cent target. "One key factor lifting the deficit is the sharp increase in interest payments," its analysts said.

Decisive action to rein in borrowing remains difficult at a time of pedestrian growth and the challenge from the far-right Rassemblement National in the April 2027 presidential election.

Today's politicians are being forced to make "direct trade-offs between vitally important areas of spending and the debt interest bill", says Carmine Di Noia, director for financial and enterprise affairs at the OECD. "Your interest bill is directly constraining fiscal policy."

Debt management offices in the UK, EU and elsewhere have responded to the increase in long-term interest rates by issuing more short-term bonds, which typically have lower yields and are therefore less expensive for taxpayers.

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本頁中文重點

- 英國的關鍵考驗是下個月——Andy Burnham 七月就任首相以來的第一份預算案。
- 較高的債務成本（包括英國大量的指數連結公債）正在吃掉資源，而政府同時面臨改善公共服務與兌現提高國防支出承諾的壓力。這引發外界揣測，在 2024 與 2025 兩次大幅增收的預算之後，可能還需要再一輪加稅。
- 看好的投資人說，英國不像有些國家，已經做了困難的選擇，本財政年度的毛發債量其實是下降的。
- 但英國的借款成本是 G7 最高，債務利息支出預期到 2030–31 年度將接近 GDP 的 4%，大約是疫情前幾年的兩倍。
- 法國在籌資市場面臨類似壓力：十年期借款成本與德國超安全公債的利差（巴黎財政疑慮的關鍵指標）約 0.9 個百分點，接近歐元區債務危機後的最高水準。摩根士丹利已下修對法國公共財政的估計，認為今年財政赤字可能達 GDP 的 5.4%，高於政府 5% 的目標，分析師指出「推高赤字的一個關鍵因素是利息支出的急增」。
- OECD 金融與企業事務總監 Carmine Di Noia：今天的政治人物被迫在「極重要的支出領域與債務利息帳單之間做直接取捨。你的利息帳單正在直接限制財政政策」。
- 英國、歐盟與其他地區的債務管理機構對長期利率上升的回應，是發行更多短天期債券，因為它們殖利率通常較低、對納稅人比較便宜。

全球利息帳單新聞｜PDF 第 11/13 頁

縮短天期只是權宜之計；央行也被捲進來

The US has gone down this route and has also announced plans to scale up its buybacks of longer-term debt.

But shortening the average maturity of debt is just a "quick fix", says Eiko Sievert, an executive director for sovereign credit ratings at Scope Ratings. "Any shock that happens will translate much more quickly into government interest expenses."



Donald Trump arrives on board a US aircraft carrier in Yokosuka, Japan. The president has plans for the largest rise in US defence spending since the second world war © Tomohiro Ohsumi/Getty Images

Central banks are also under pressure to respond. Trump has called on the Fed to reduce interest rates to help with the national debt. Some argue the global economy has moved into a new period of "fiscal dominance" in which monetary authorities are under pressure to hold rates down to help offset loose budgetary policy.

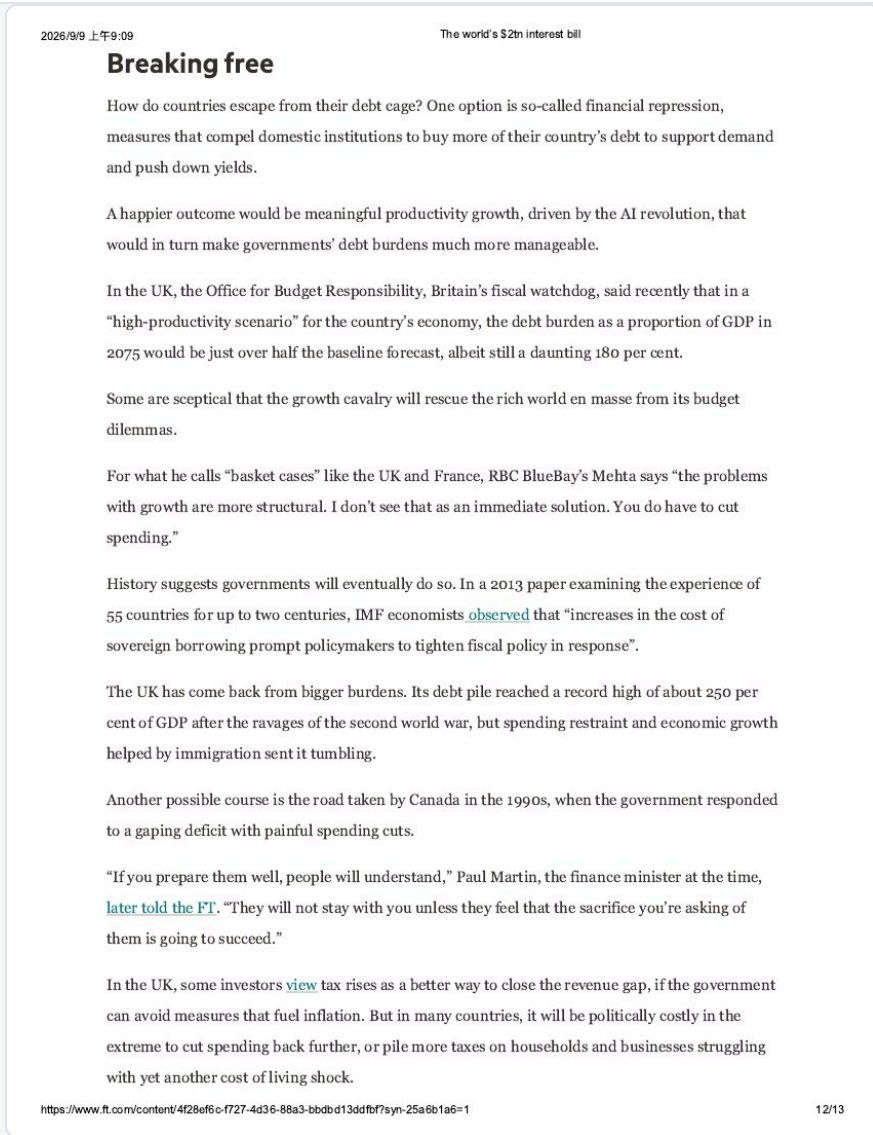
The Bank of England continues to sell bonds through its "quantitative tightening" balance-sheet-shrinking programme, which is putting upward pressure on borrowing costs. But it is anticipated to slow the pace this month and some argue it should stop long-dated sales altogether. The Bank of Japan, which has continued to buy assets, has been reducing the scale of its purchases but plans to stop this reduction next year and level out.

本頁中文重點

- 美國也走了這條路，並宣布要擴大買回長天期債務。
- 但 Scope Ratings 主權信評執行董事 Eiko Sievert 說，縮短債務平均年期只是「快速修補」：「任何衝擊發生時，都會更快轉化成政府的利息支出。」
- 圖片為川普在日本橫須賀登上美軍航艦。他計畫推動二戰以來最大的美國國防支出增幅。
- 央行同樣面臨壓力。川普要求聯準會降息以減輕國債負擔。有人認為全球經濟已進入新的「財政主導（fiscal dominance）」時期——貨幣當局被迫把利率壓低，以抵銷寬鬆的預算政策。
- 英格蘭銀行仍透過「量化緊縮」縮表賣債，對借款成本形成上行壓力，但預期本月會放慢速度，也有人主張應完全停止長天期的賣出。日本銀行仍在買入資產，雖已縮減購買規模，但計畫明年停止縮減並持平。

全球利息帳單新聞｜PDF 第 12/13 頁

脫困的幾條路：金融壓抑、成長、削減支出或加稅



2026/9/9 上午9:09 The world's $2tn interest bill

Breaking free

How do countries escape from their debt cage? One option is so-called financial repression, measures that compel domestic institutions to buy more of their country's debt to support demand and push down yields.

A happier outcome would be meaningful productivity growth, driven by the AI revolution, that would in turn make governments' debt burdens much more manageable.

In the UK, the Office for Budget Responsibility, Britain's fiscal watchdog, said recently that in a "high-productivity scenario" for the country's economy, the debt burden as a proportion of GDP in 2075 would be just over half the baseline forecast, albeit still a daunting 180 per cent.

Some are sceptical that the growth cavalry will rescue the rich world en masse from its budget dilemmas.

For what he calls "basket cases" like the UK and France, RBC BlueBay's Mehta says "the problems with growth are more structural. I don't see that as an immediate solution. You do have to cut spending."

History suggests governments will eventually do so. In a 2013 paper examining the experience of 55 countries for up to two centuries, IMF economists observed that "increases in the cost of sovereign borrowing prompt policymakers to tighten fiscal policy in response".

The UK has come back from bigger burdens. Its debt pile reached a record high of about 250 per cent of GDP after the ravages of the second world war, but spending restraint and economic growth helped by immigration sent it tumbling.

Another possible course is the road taken by Canada in the 1990s, when the government responded to a gaping deficit with painful spending cuts.

"If you prepare them well, people will understand," Paul Martin, the finance minister at the time, later told the FT. "They will not stay with you unless they feel that the sacrifice you're asking of them is going to succeed."

In the UK, some investors view tax rises as a better way to close the revenue gap, if the government can avoid measures that fuel inflation. But in many countries, it will be politically costly in the extreme to cut spending back further, or pile more taxes on households and businesses struggling with yet another cost of living shock.

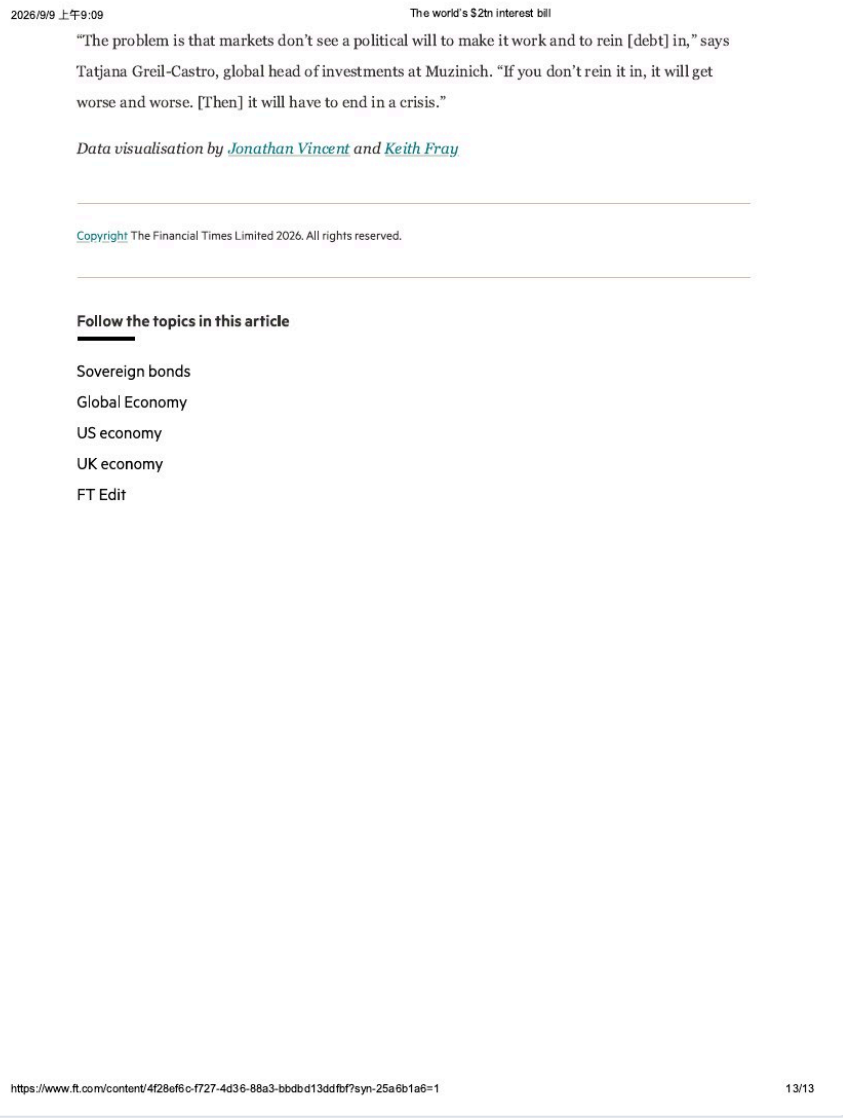
https://www.ft.com/content/4f28ef6c-f727-4d36-88a3-bbdbd13ddfbf?syn-25a6b1a6=1 12/13

本頁中文重點

- 小標「Breaking free」：國家要怎麼逃出債務牢籠？一個選項是所謂的金融壓抑（financial repression）——用措施迫使國內機構買更多本國債務，以支撐需求並壓低殖利率。
- 比較理想的結果是 AI 革命帶動有意義的生產力成長，使政府的債務負擔變得好管理得多。
- 英國預算責任辦公室（OBR）近期指出，在「高生產力情境」下，英國 2075 年的債務佔 GDP 比重只會是基準預測的一半多一點，但仍達到令人卻步的 180%。
- 有人懷疑成長這支救兵能不能整體拯救富裕國家的預算困境。RBC BlueBay 的 Mehta 針對他稱為「爛攤子」的英國與法國說：「成長的問題比較結構性。我不認為那是立即的解方。你還是得削減支出。」
- 歷史顯示政府最終會這麼做。IMF 經濟學家 2013 年一篇研究 55 國、橫跨最多兩個世紀經驗的論文指出，「主權借款成本上升會促使決策者收緊財政政策」。
- 英國曾從更大的負擔回來過：二戰後債務堆到約 GDP 的 250%，靠支出節制與移民帶動的經濟成長而大幅下降。另一條路是加拿大 1990 年代的做法——政府以痛苦的支出削減回應巨額赤字。當時的財政部長 Paul Martin 後來告訴 FT：「如果你把人民準備好，他們會理解。除非他們覺得你要求的犧牲會成功，否則他們不會跟著你走。」
- 英國也有投資人認為加稅是填補稅收缺口較好的方式，前提是政府能避免會推升通膨的措施。但在許多國家，進一步削減支出、或對正承受又一波生活成本衝擊的家庭與企業加稅，政治代價都極高。

全球利息帳單新聞｜PDF 第 13/13 頁

文章結語與末頁標籤


2026/9/9 上午9:09 The world's $2tn interest bill

"The problem is that markets don't see a political will to make it work and to rein [debt] in," says Tatjana Greil-Castro, global head of investments at Muzinich. "If you don't rein it in, it will get worse and worse. [Then] it will have to end in a crisis."

Data visualisation by Jonathan Vincent and Keith Fray

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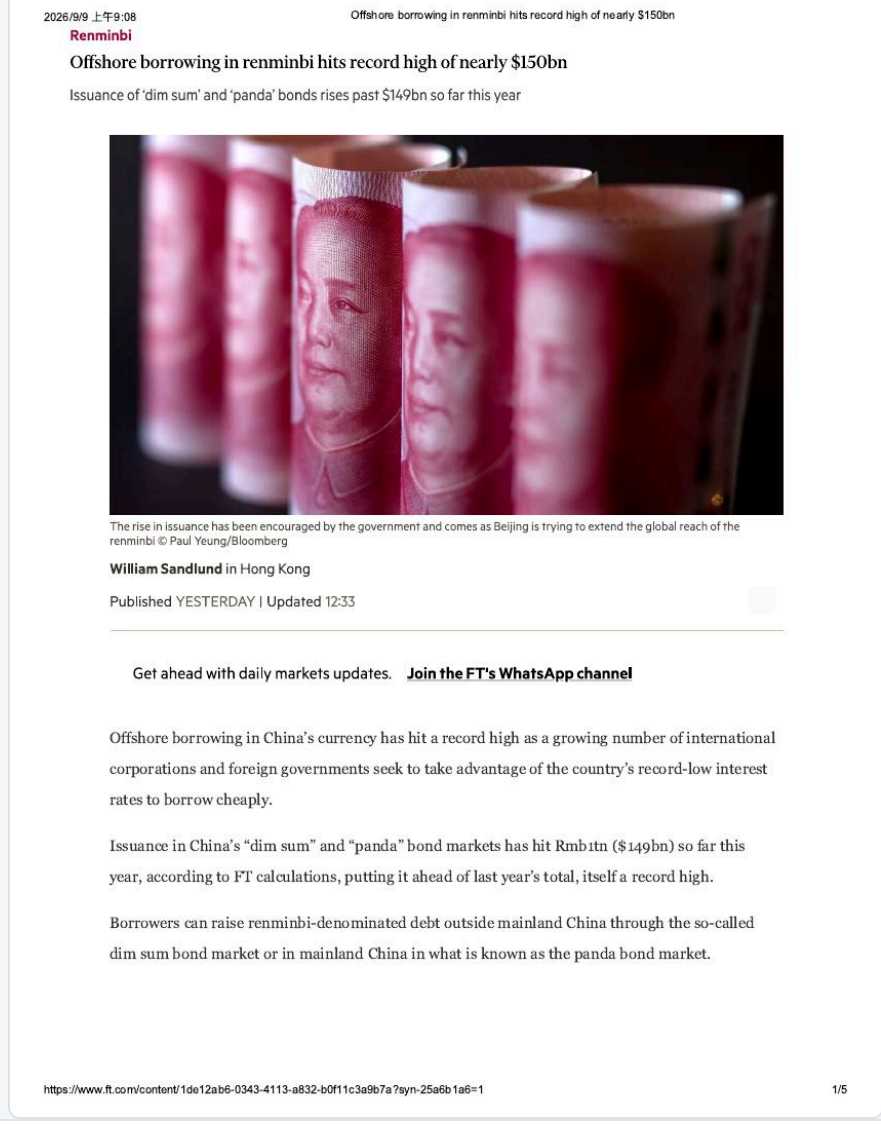
Sovereign bonds
Global Economy
US economy
UK economy
FT Edit

https://www.ft.com/content/4f28ef6c-f727-4d36-88a3-bbdbd13ddfbf?syn-25a6b1a6=1 13/13

本頁中文重點

- Muzinich 全球投資主管 Tatjana Greil-Castro 的結語：「問題是市場看不到讓這件事成功、把債務控制住的政治意志。如果你不控制它，情況只會越來越糟，最後必須以一場危機收場。」
- 資料視覺化由 Jonathan Vincent 與 Keith Fray 製作。版權為 The Financial Times Limited 2026。
- 主題標籤：Sovereign bonds、Global Economy、US economy、UK economy、FT Edit。

人民幣融資新聞｜PDF 第 1/5 頁 人民幣融資創新高：低利率、儲蓄池與貨幣競爭



2026/9/9 上午9:08 Offshore borrowing in renminbi hits record high of nearly $150bn

Renminbi

Offshore borrowing in renminbi hits record high of nearly $150bn

Issuance of 'dim sum' and 'panda' bonds rises past $149bn so far this year

The rise in issuance has been encouraged by the government and comes as Beijing is trying to extend the global reach of the renminbi © Paul Yeung/Bloomberg

William Sandlund in Hong Kong

Published YESTERDAY | Updated 12:33

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Offshore borrowing in China's currency has hit a record high as a growing number of international corporations and foreign governments seek to take advantage of the country's record-low interest rates to borrow cheaply.

Issuance in China's "dim sum" and "panda" bond markets has hit Rmb1tn ($149bn) so far this year, according to FT calculations, putting it ahead of last year's total, itself a record high.

Borrowers can raise renminbi-denominated debt outside mainland China through the so-called dim sum bond market or in mainland China in what is known as the panda bond market.

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文章重點

- 點心債與熊貓債年度發行合計突破人民幣 1 兆元，接近 1,500 億美元，反映國際企業與政府尋找較低成本的人民幣資金。
- 中國 10 年期公債殖利率約 1.68%，同期美國公債約 4.78%。顯著利差使人民幣更有機會成為融資貨幣。
- 中國龐大儲蓄、境內低收益資產與偏弱信貸需求，形成對更多可投資債券的需求。制度開放與南向 Bond Connect 額度擴大也支持市場成長。
- 人民幣在全球金融中的比重仍小，發行規模、投資人熟悉度與境外資金運用仍有限制，因此目前較像低成本融資管道，而非已取代美元。**即：現在的狀況是「借人民幣比較便宜」，還不是「大家改用人民幣」。**

老師課堂補充

- 老師以中美 10 年期公債殖利率比較借款成本：在中國發行人民幣債券的名目利率明顯較低，因此國際借款人有誘因轉向人民幣市場。**即：哪邊利率低就往哪邊借，這就是企業跑去發人民幣債的理由。**
- 境內銀行與保險公司持有龐大儲蓄，但跨境投資選項有限，因而也有強烈需求把資金配置到更多樣的債券，形成供需兩端的推力。
- 這不只是融資成本問題，也牽涉中美貨幣影響力與資本市場競爭。老師把它視為與前三篇『債務、利率、信任』主題相連的延伸案例。

課堂提問與解答

問 點心債與熊貓債有何不同？

答 點心債是在中國大陸境外發行的人民幣債券。熊貓債是境外機構在中國境內市場發行的人民幣債券。

問 低人民幣利率是否代表借款一定更便宜？

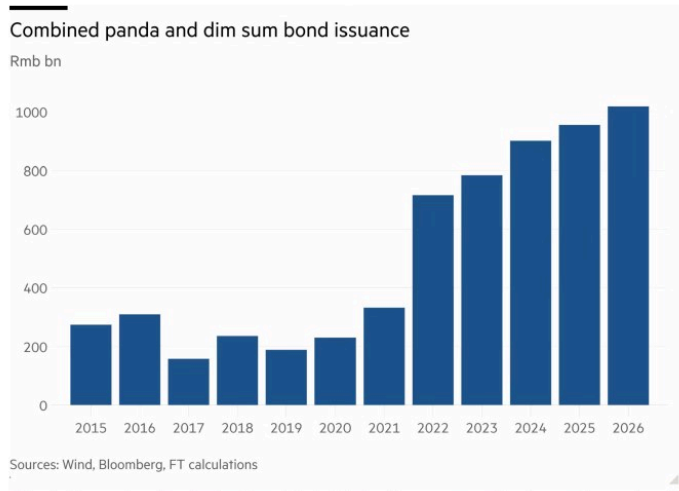
答 未必。借款人還要計入匯率避險、資金匯回、流動性與法規成本。只有總成本較低時，利差才真正轉化為融資優勢。

人民幣融資新聞｜PDF 第 2/5 頁

利差有多大，以及發行規模的組成

The rise in issuance has been encouraged by the government and comes as Beijing is trying to extend the global reach of its currency as part of efforts to compete with the US dollar and financial system.

Borrowing in renminbi has become cheaper as Chinese borrowing costs have diverged sharply from those in the US, with a 10-year government bond yielding 1.68 per cent compared with 4.78 per cent for an equivalent Treasury, close to the widest spread on record.



China's dim sum bond market has issued Rmb786.3bn this year, overtaking last year's record total. Borrowing in the panda bond market has hit Rmb231.6bn in 2026 so far, already more than in any full year on record.

Borrowing is not limited to foreigners — Chinese entities have made up close to two-thirds of dim sum bond issuance this year and can use the proceeds to fund overseas operations.

"The issuance and investor participation have been crazy," said David Yim, head of capital markets for Greater China and north Asia at Standard Chartered, of the offshore renminbi bond market.

"There are more natural needs for renminbi funding. Whether it's investment in China or you take it out and pay your suppliers in renminbi, the need for that is more."

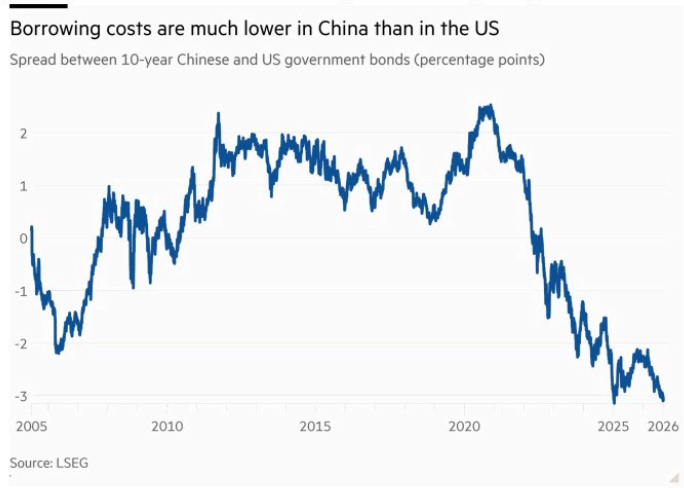
本頁中文重點

- 發行量上升是政府鼓勵的結果，背景是北京想擴大人民幣的全球影響力，與美元及美國金融體系競爭。
- 以人民幣借款變便宜，是因為中國的借款成本與美國明顯背離：十年期中國公債殖利率 1.68%，同天期美國公債 4.78%，利差接近歷史最大。
- 圖表「Combined panda and dim sum bond issuance」（人民幣十億元）：2015–2021 年多在 160–340 之間，2022 年跳到約 715，之後逐年上升，2026 年超過 1,000。資料來源 Wind、Bloomberg 與 FT 計算。
- 細分數字：中國點心債市場今年已發行人民幣 7,863 億元，超過去年的紀錄。熊貓債市場 2026 年至今發行人民幣 2,316 億元，已經高於任何一個完整年度的紀錄。
- 借款人不限於外國機構——中國本身的機構今年佔點心債發行量近三分之二，募得的資金可用於海外業務。
- 渣打銀行大中華與北亞資本市場主管 David Yim 形容離岸人民幣債市「發行與投資人參與都非常瘋狂」，並說「人民幣資金的自然需求變多了。不管是投資中國，或是把錢拿出去付給用人民幣計價的供應商，這類需求都在增加」。

人民幣融資新聞｜PDF 第 3/5 頁

利差走勢與境內儲蓄池

Offshore borrowing in renminbi hits record high of nearly $150bn



Authorities increased the amount of money that mainland investors can send south to Hong Kong through the Bond Connect market access scheme over the summer, which bankers said had helped boost issuance.

"The quota expansion was a very positive signal to the market," said Xixi Sun, Greater China bond syndicate head at Citi, which was involved in a Rmb3.5bn dim sum bond issuance by Kazakhstan's state oil and gas company KazMunayGas last month.

"We've seen the onshore insurance companies getting the [southbound Bond Connect] channel cleared," Sun said. "That's been driving the long-dated dim sum issuance." This year, companies such as Tencent have issued 10- and 30-year dim sum bonds.

The market's growth is being supported by an enormous pool of savings in mainland China that has limited investment options. Banks and insurance companies are rushing for assets to invest in now that yields on Chinese government bonds have dropped so low and overall credit demand is weak. In July, new bank loans declined by Rmb340bn, the largest monthly drop since records began.

本頁中文重點

- 圖表「Borrowing costs are much lower in China than in the US」：中美十年期公債利差（百分點）。2005–2021 年多半為正（中國較高），2022 年起轉為負值並持續擴大，2026 年接近 −3，也就是中國利率低於美國約 3 個百分點。資料來源 LSEG。
- 政策面：當局在夏天調高了境內投資人透過債券通南向機制匯往香港的額度，銀行家說這有助推升發行量。花旗大中華債券聯貸主管 Xixi Sun 說「額度擴大對市場是非常正面的訊號」，該行參與了哈薩克國家油氣公司 KazMunayGas 上月人民幣 35 億元的點心債發行。
- Sun 另指出「境內保險公司的南向債券通管道已經打通，這帶動了長天期點心債的發行」。今年騰訊等公司已發行 10 年期與 30 年期的點心債。
- 市場成長的根本支撐是中國境內龐大但投資選項有限的儲蓄池。中國公債殖利率已經很低、整體信貸需求偏弱，銀行與保險公司急著找資產投資。七月新增銀行貸款減少人民幣 3,400 億元，是有紀錄以來最大的單月降幅。

人民幣融資新聞｜PDF 第 4/5 頁

誰在發行、誰在買，以及規模的限制

Panda and dim sum bonds can offer a small but growing source of slightly higher yields for such investors. This year, the weighted average yield of a panda bond is 1.85 per cent and its duration is 3.17 years, according to data from Wind. That compares with a yield of about 1.7 per cent on a 10-year Chinese government bond.

This type of debt is still a small part of Chinese fixed income, with panda bonds accounting for just 0.25 per cent of the country's bond markets, according to Wind.

Although sovereign borrowers including Indonesia, Slovenia, Pakistan and Kazakhstan have tapped the panda bond market in the past year, the rise in issuance is primarily being driven by Chinese and non-Chinese banks.

"We've already seen very good growth but it's being driven primarily by banks," said Samuel Fischer, head of onshore debt capital markets for China at Deutsche Bank.

In late August, UBS borrowed Rmb2bn through five-year panda bonds with a coupon of 1.78 per cent in its inaugural issuance in the market. Goldman Sachs has issued Rmb61.5bn of dim sum bonds this year. Foreign banks typically convert the proceeds from renminbi bond sales into major currencies for use in their global operations.

Multinational corporations are less active. Despite its strong growth, renminbi-denominated debt has been absent from the borrowing plans of large corporates this year, such as the US-based hyperscalers that have increasingly borrowed from foreign credit markets.

A major factor holding them back is the relatively small size of bond issuances, said Fischer.

"Our clients want to know if they can do a $1bn issuance. We're getting to that point," Fischer said. "Once people see that an issuer can successfully raise $1bn, new issuers will follow quickly."

Another factor is the difficulty of attracting big institutional investors from mainland China into the dim sum and panda bond markets.

Such investors "probably have to do three or four months' work before buying a foreign name, unlike western fund managers who can buy with a credit rating only", said StanChart's Yim.

"You have to do extensive marketing work to penetrate this investor pool," he added.

本頁中文重點

- 對這些機構來說，熊貓債與點心債能提供規模不大但持續成長、收益略高的選擇。今年熊貓債的加權平均殖利率為 1.85%、存續期間 3.17 年（Wind 資料），相較之下十年期中國公債約 1.7%。
- 但這類債務在中國固定收益市場仍是小角色：依 Wind 資料，熊貓債只佔中國債市的 0.25%。
- 過去一年印尼、斯洛維尼亞、巴基斯坦與哈薩克等主權借款人都進過熊貓債市場，但發行量上升主要還是由中資與非中資銀行推動。德意志銀行中國境內債務資本市場主管 Samuel Fischer：「成長很好，但主要是銀行在推動。」
- 具體案例：八月底瑞銀以 5 年期熊貓債借入人民幣 20 億元，票面利率 1.78%，是它在這個市場的首次發行。高盛今年已發行人民幣 615 億元的點心債。外國銀行通常會把人民幣債券募得的資金換成主要貨幣，供全球業務使用。
- 跨國企業則相對不活躍。儘管成長強勁，今年大型企業（例如一直向外國信用市場大量借款的美國超大規模科技業者）的借款計畫裡並沒有人民幣計價債務。Fischer 說主要障礙是單筆發行規模偏小：「我們的客戶想知道能不能做到 10 億美元。我們正在接近那個點。一旦大家看到有發行人成功募到 10 億美元，新的發行人會很快跟上。」
- 另一個障礙是難以吸引中國境內的大型機構投資人進場。渣打的 Yim 說這些投資人「買一個外國名字前大概要做三、四個月的功課，不像西方基金經理人看信用評等就能買」，因此「要打進這個投資人池，必須做大量的行銷工作」。**即：債券賣不賣得動，有時候卡在買方要花多少工夫才敢下單。**

人民幣融資新聞｜PDF 第 5/5 頁

人民幣會變成下一個融資貨幣嗎

The renminbi's role in global finance remains small. There has been growing use of renminbi-denominated loans offshore and the currency has grown in trade finance, but its usage as a reserve asset remains limited.

Although Chinese government bonds have outperformed sovereign debt from the G7 since 2020, global fund managers now cite the country's low yields as a reason for not raising their allocation.

Still, analysts and bankers said it was possible that the rapid growth of China's panda and dim sum markets represented the beginning of the renminbi as a global funding currency similar to the role Japan's yen played during that country's decades of deflation and low interest rates.

"Low interest rates make a currency attractive for funding purposes. We saw similar dynamics with the yen historically," said Hui Shan, chief China economist at Goldman.

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Follow the topics in this article

Sovereign bonds
Renminbi
Asia-Pacific companies
Chinese business & finance
Corporate bonds

本頁中文重點

- 人民幣在全球金融中的角色仍然很小。離岸人民幣計價貸款的使用在增加，貿易金融中的使用也在成長，但作為準備資產的使用仍然有限。
- 雖然中國公債自 2020 年以來的表現優於 G7 主權債，全球基金經理人現在反而以中國殖利率太低為由，不願提高配置。
- 不過分析師與銀行家認為，熊貓債與點心債市場的快速成長，有可能是人民幣開始成為全球「融資貨幣」的起點，角色類似日本在數十年通縮與低利率期間的日圓。
- 高盛首席中國經濟學家 Hui Shan：「低利率會讓一個貨幣在融資用途上變得有吸引力。我們在日圓身上看過類似的動態。」
- 本頁末為版權聲明與主題標籤：Sovereign bonds、Renminbi、Asia-Pacific companies、Chinese business & finance、Corporate bonds。

十年期美債殖利率升到 2007 年以來高點

十年期美債殖利率新聞｜PDF 第 1/4 頁

Sovereign bonds

Ten-year Treasury yield hits highest level since 2007

Benchmark bond yield rises to 5.04% amid concerns over inflation and higher interest rates



The Federal Reserve building. Futures markets are putting a more than 90% probability on the bank raising rates by a quarter of a percentage point this week © Reuters

Emily Herbert and **Ian Smith** in London and **William Sandlund** in Hong Kong

Published YESTERDAY | Updated 02:13

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US borrowing costs hit their highest level since 2007 on Tuesday, as a global bond sell-off deepened following a renewed surge in oil prices and expectations of an interest rate increase by the Federal Reserve this week.

The 10-year yield, a benchmark for debt markets worldwide, jumped as much as 0.08 percentage points to 5.04 per cent, its highest point since July 2007. It later fell back to around 5 per cent, a level it topped on Monday for the first time in almost three years, in a bond rout sparked by the Iran war energy shock.

本頁中文重點

- 能源價格衝擊、聯準會升息預期與債市賣壓，使十年期美債殖利率一度達 5.04%。
- 殖利率是存量債券的市場報酬概念。對固定現金流債券，價格下降時殖利率上升。

老師課堂補充

- 老師先釐清「債務總額」與「借款利率」是不同問題：上週報導債務存量，本週新聞焦點是十年期公債殖利率。
- 大量賣債讓市場價格下跌。同一張固定現金流債券的殖利率會因此上升。
- 老師區分英文用語：政府發行的債券為 sovereign bond。美國財政部發行者稱 Treasury。股票多談 return，債券則談 yield／殖利率，常可從發債者角度理解為借款成本。

十年期美債殖利率新聞｜PDF 第 2/4 頁

聯準會、通膨與主權信用

The sell-off comes ahead of a crucial Fed meeting on Wednesday. Traders in futures markets are putting a roughly 90 per cent probability on the central bank raising rates by a quarter of a percentage point — a move that would put it on a collision course with US President Donald Trump's desire for lower borrowing costs.

The bond sell-off has been "relentless", said Matt Amis, an investment director at Aberdeen Investments.

"[It] feels like investors are throwing in the towel on long-held positions. Yields look elevated here, but until oil stops heading higher, it will be a struggle to add risk," he continued.

Soaring energy prices as a result of the Iran war have fuelled concerns about persistent above-target inflation in the US and elsewhere, leading traders to price in higher interest rates and knocking bonds. Investors' attention has also turned towards the US's $40tn debt pile, amid concerns that Washington is on an unsustainable fiscal path.

Scott Bessent, the US Treasury secretary, told Congress on Tuesday that a "fiscal consolidation is coming" and the country had "to get our house in order", but pointed to strong recent debt auctions as evidence of ongoing global investor confidence.

Some analysts argue that the Fed under new chair Kevin Warsh needs to raise rates to demonstrate its commitment to tackling inflation. The price of Brent crude, the international oil benchmark, was up 3.2 per cent on Tuesday at $109 a barrel.

"The Fed's credibility is under scrutiny," said Jenny Zeng, chief investment officer for fixed income at Allianz Global Investors. "Markets will be looking for chair Warsh to back recent rhetoric with action."

Stock markets also weakened. The S&P 500 was down 0.4 per cent during afternoon trading in New York, while the Nasdaq Composite fell 0.7 per cent. The Stoxx Europe 600 closed 0.3 per cent lower.

Yields on Japan's benchmark 10-year bonds also climbed on Tuesday to their highest level in 30 years, reaching as high as 3.04 per cent, ahead of a broadly expected interest rate increase by the Bank of Japan on Friday. Yields move inversely to prices.

Markets are pricing in a roughly 80 per cent chance that the BoJ will raise rates by 0.25 percentage points to 1.25 per cent this week, the highest level in 31 years.

本頁中文重點

- 市場當時預期升息 25 個基點，也關注油價、通膨、財政赤字與聯準會可信度。
- 25 基點＝0.25 個百分點。利率變動不是債券價格變動的同義詞。

老師課堂補充

- 25 個基點就是 0.25 個百分點。老師以原油、通膨預期和聯準會升息預期解釋債市賣壓。
- 市場也在觀察聯準會主席能否以實際政策支持先前的表態。
- 債券遭大量拋售時，供給與賣壓推低價格。由於未來票息既定，持有至到期殖利率上升。25 basis points 是利率增加 0.25 個百分點，不是原利率乘以 25%。
- 老師串起油價、通膨擔憂、升息預期與國際債市賣壓，並提及日本與英國市場受到關注。這是報導的市場解釋，不表示各項因素已被單獨識別為因果。

十年期美債殖利率新聞｜PDF 第 3/4 頁

全球公債賣壓及股市影響

“Markets are increasingly confident about a September BoJ hike and are pricing a higher terminal rate, which continues to put upward pressure on JGB yields,” said Masahiko Loo, senior fixed-income strategist at State Street in Tokyo.

In the UK, where the Bank of England is expected to hold rates this week, the 10-year gilt yield rose as much as 0.06 percentage points to 5.44 per cent before slipping back to 5.39 per cent.

Some expect the BoE to further pare back or halt sales of long-term debt through its so-called quantitative tightening programme at a scheduled decision on Thursday.

The multiyear highs in countries’ long-term borrowing costs, on top of record overall issuance across rich nations, are feeding anxiety over the world’s rising debt-servicing bill, which is already above $2tn, or 3 per cent of GDP, for OECD nations.

Strategists said borrowing costs in the US would need to rise higher for investors to pull significant money out of stocks. “We would need [a percentage point] more to really have some more negative implications for the S&P 500,” said Frank Benzimra, head of Asia equity strategy at Société Générale.

Bank of America’s closely watched survey of global fund managers showed on Tuesday that a “disorderly rise in bond yields” was now considered the biggest market risk for investors, with a third of respondents giving that response, overtaking an “AI bubble”.

In a sign that the bond move is dominating investor bets, the survey also found that “short US Treasuries” was named as the second most crowded trade in the market, after “long global semiconductors”.

Bessent attempted to steady the government bond market by launching a $6bn bond-buying programme last week. But the move did little to soothe investors’ nerves, with yields on the 10-year and 30-year bonds continuing to surge higher.

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Follow the topics in this article

Sovereign bonds

本頁中文重點

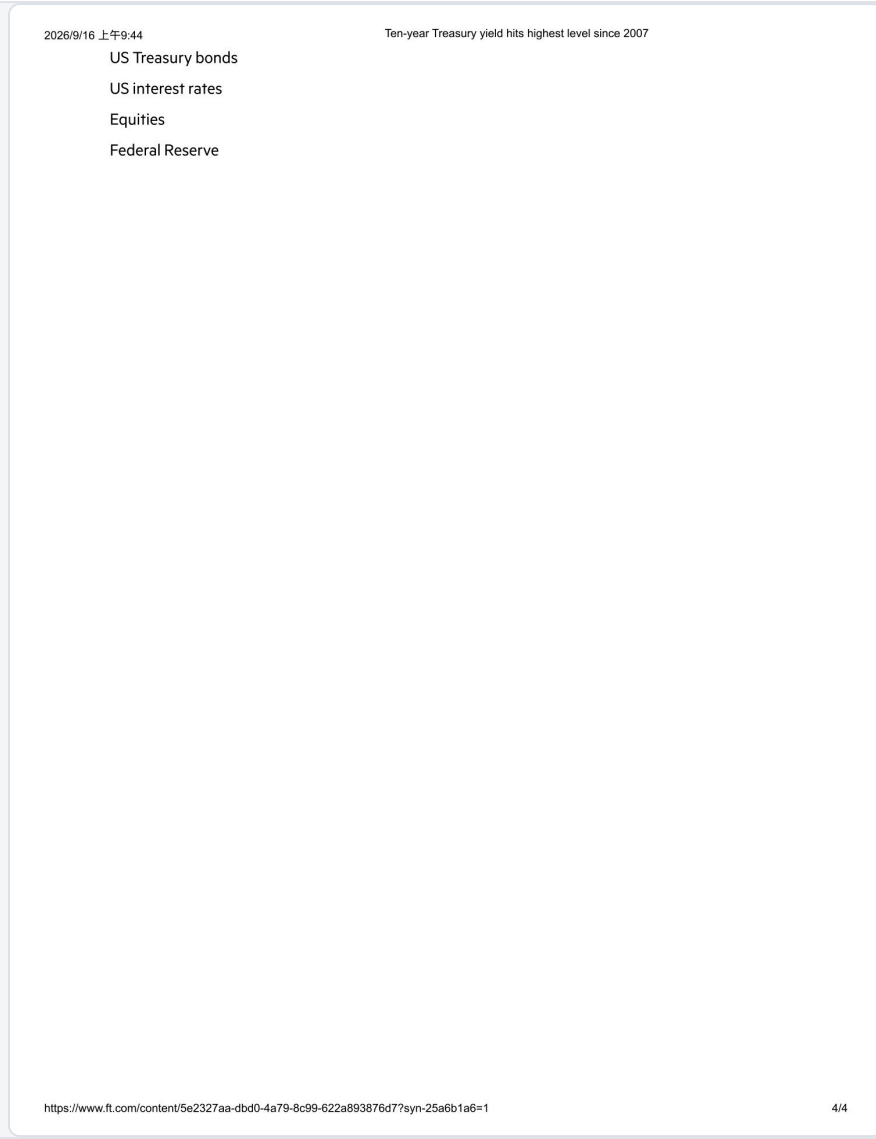
- 日本與英國長債殖利率也攀升。OECD 國家利息負擔和發債規模成為焦點。
- 新聞指出市場視公債殖利率失序上升為重大風險，可能經借款成本及折現率壓低股價。

老師課堂補充

- 美債利率常被拿來當全球估值的基礎。它上升時，許多資產使用的折現率也可能隨之提高。
- 新聞談美債可信度受檢視。後續課堂仍會用無風險利率作估值起點。要把「模型中的基準」與「真實世界主權風險討論」分開。

十年期美債殖利率新聞｜PDF 第 4/4 頁

文章末頁：主題標籤



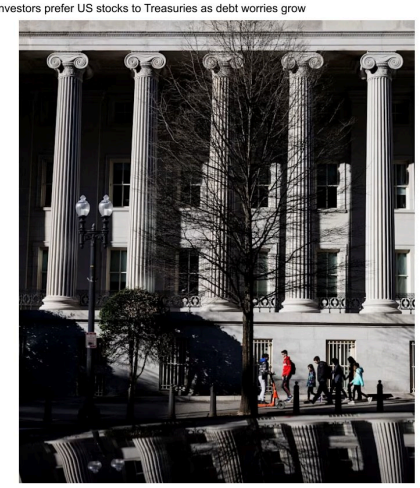
本頁中文重點

- 末頁只有新聞主題標籤與頁尾。

外資配置新聞｜PDF 第 1/5 頁

外資流向：美股超越美債

2026/9/16 上午9:46



US equities

Foreign investors prefer US stocks to Treasuries as debt worries grow

International flows into American equities exceed government debt for first time this century outside pandemic and aftermath of global financial crisis

JS Treasury building. Treasuries' traditional role as the global risk-free asset is coming under increasing pressure © Samuel m/Bloomberg

Emily Herbert and **Ian Smith** in London

Published YESTERDAY

Get ahead with daily markets updates. **Join the FT's WhatsApp channel**

Foreign investors are now buying more US stocks than government bonds, in a rare move that comes amid fears that inflation and America's fast-growing debt pile are undermining the "risk-free" status of Treasuries.

International flows into US stocks reached 2.8 per cent of US GDP on average in the year to June, overtaking Treasuries — at 2 per cent of GDP — for the first time this century outside brief episodes in the Covid-19 pandemic and the aftermath of the global financial crisis, according to analysis of US Treasury data by Deutsche Bank.

The change reflects the ever-growing lure of the US stock market, with the S&P 500 on track for a fourth straight year of double-digit gains. Powering the rally has been vast AI investment, which is driving profit margins to new highs, according to FactSet data stretching back to 2009.

本頁中文重點

- 截至六月的年度國際資金流，美股約占美國 GDP 2.8%，美債約 2%。報導將這種組合視為罕見轉變。
- 老師強調專業投資組合原本會持有一定比例的無風險資產，股債配置變動值得觀察。

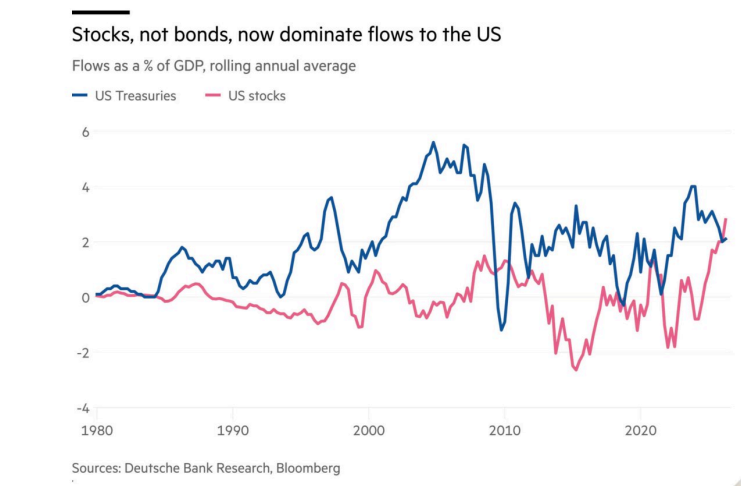
> **老師課堂補充**
>
> - 老師從專業投資人的角度補充：投資組合不能只比報酬，也要看風險，傳統配置通常留一部分在相對安全的公債。
> - 老師舉共同基金、避險基金與退休基金：專業投資人同時衡量收益和風險。傳統分散投資會保留一部分在相對安全且流動性高的資產，不能只追求股票報酬。

外資配置新聞｜PDF 第 2/5 頁

資金流向圖與美元角色

At the same time, Treasuries' traditional role as the global risk-free asset is coming under increasing pressure as investors grow warier of lending money to heavily indebted governments and question the independence of the Federal Reserve, in a move that could transform the way the US dollar trades. On Tuesday the 10-year Treasury yield reached its highest level since 2007.

This is a "huge shift in US asset markets", said George Saravelos, global head of FX research at Deutsche Bank, reflecting the fact "the American private balance sheet is booming . . . but the public sector balance sheet keeps worsening".



Saravelos added that the value of the dollar might now be more closely tied to flows into US stocks than to flows into Treasuries, suggesting the greenback could strengthen as risk appetite grows. This would mark a major break from its previous pattern of trading, when it has tended to rally during so-called risk-off periods, as foreign investors dump riskier trades and flock to US debt as a haven.

US assets were "no longer the 'safe' but the risky asset of choice", Saravelos wrote.

American assets came under pressure early last year after President Donald Trump's sweeping "liberation day" tariffs in April triggered fears that the era of US dominance in financial markets could be waning, hitting stocks, bonds and the dollar.

本頁中文重點

- 圖中藍線是流向美國國債、粉線是流向美股，以 GDP 比重表達，年度平滑值不可當成持股存量。
- 若美元更受股票資金流入驅動，其避險貨幣表現可能隨風險偏好而變。

老師課堂補充

- 老師請同學讀圖時看清縱軸是流量占 GDP 的比例，並指出股債資金流相對位置的轉變。
- 圖表是流入額相對 GDP 的比率，屬「流量」而非股票／國債已持有的總市值。老師請同學注意股票與國債兩條線的相對變化。

外資配置新聞｜PDF 第 3/5 頁

政府財務壓力與風險概念

This brief “sell America” sentiment quickly gave way to investor fear of missing out on further gains in AI stocks, propelling Wall Street to new highs. But the US Treasury market has remained under pressure, as it has been swept up in broader concerns about global worries about debt levels and inflation in a number of developed economies.



International flows into US stocks reached 2.8% of US GDP on average in the year to June, overtaking Treasuries © Spencer Platt/Getty Images

“Government bonds are not as risk-free as they used to be,” said James Turner, head of global fixed income for Emea at BlackRock. “With [government] deficits where they are, if that was a corporate [entity] . . . you would not say that was a risk-free situation.”

At the same time, the US government debt pile hit $40tn last month while the government continues to run a deficit, adding to fears that the fiscal trajectory of the US is unsustainable. Last week, Trump pledged to give a $5,000 “dividend” to every adult US citizen if Republicans retained control of Congress in the midterm elections, a move that could cost more than $1tn.

The yield on 30-year Treasuries has climbed from 4.83 per cent to 5.32 per cent so far this year. Yields rise when bond prices fall.

“I have never heard so much concern and debate around what ‘risk free’ actually means,” said Matt Rowe, senior portfolio manager at Man Group.

本頁中文重點

- 美國公債規模與財政赤字引起市場討論。報導提醒公債價格與殖利率反向。
- 新聞中的「無風險地位受質疑」是當時市場敘事，不代表本文證實美國必然違約。

老師課堂補充

- 對美國公債作「無風險資產」的假設仍是後面估值模型的起點。新聞呈現的是市場對政府財政的疑慮。

外資配置新聞｜PDF 第 4/5 頁

機構配置股多債少

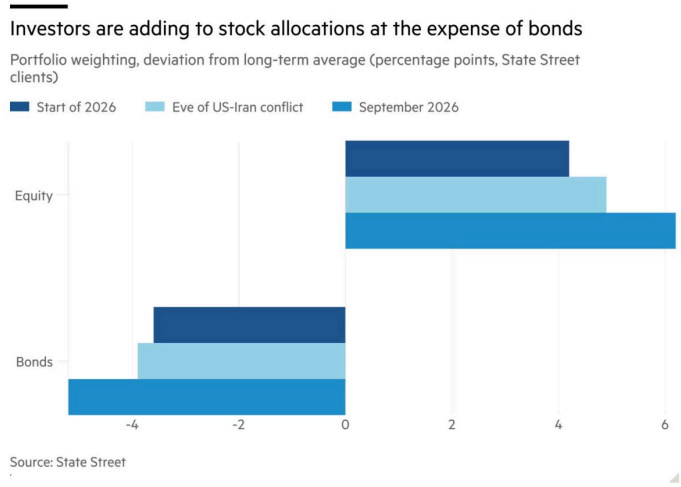
Investors with this growing anxiety around Treasuries were “quickly faced” with similar problems, such as rising issuance in other bond markets, he added. “In an odd twist, equities seem to be competing for those funds [instead].”

Earlier this month the manager of Norway’s $2.3tn oil fund proposed slashing its holdings of US Treasuries by about $80bn and instead buying debt such as mortgage-backed securities backed by government agencies.

Marija Veitmane, head of equity research at State Street, said the bank is seeing the same pattern emerging among institutional clients, with greater allocations to stocks, seemingly at the expense of government bonds.

“If you are worried about the US government’s fiscal prudence, then comparing equity fundamentals with governments . . . companies look very, very solid,” Veitmane said. “Profit growth is strong.”

The rise in yields comes despite US Treasury secretary Scott Bessent's plan to buy back long-term government debt, with the announcement last week of a $6bn purchase disappointing investors.



本頁中文重點

- 挪威基金與機構投資人調整政府債、股票及其他固定收益部位。圖表比較不同時點的權重偏離。
- 老師以 60/40 股債組合說明資產配置變化，不應直接把資金流入推論為股價永續上漲。

老師課堂補充

- 老師舉股債組合重新配置為例：賣出債券、提高股票比重，可能支持當期股價，但不是企業基本面改善的證明。
- 以原本股票六成、債券四成作口語示例：賣債增加股票配置，能帶來買盤，卻不能從流量單獨證明企業價值同步上升。

外資配置新聞｜PDF 第 5/5 頁

股市仍可能受債市拖累

In contrast, the blue-chip S&P 500 is up about 12 per cent this year, as booming corporate profits have allowed traders to shrug off warnings about the fallout from the near closure of the Strait of Hormuz and surging energy prices.

The index chalked up 52 per cent year-on-year earnings growth in the second quarter of 2026, according to FactSet, or 34 per cent excluding Amazon and Alphabet, which reported significant one-off boosts from their stakes in other AI companies.

However, some investors and analysts remain wary that the stock market, which has made little progress this summer, could start to suffer if yields rise further, raising corporate borrowing costs and increasing the lure of fixed income investments.

The sell-off in bonds was “top of mind in the equity market”, said Michiel Plakman, global head of equity at fund firm Robeco.

“If there’s trouble in the bond market . . . usually that’s what troubles stocks.”

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Follow the topics in this article

US Treasury bonds
US equities
Ian Smith
Emily Herbert

本頁中文重點

- 企業獲利與 AI 題材支撐美股，但殖利率再升會提高企業融資成本與股票估值折現率。
- 老師沒有給出行情結論。兩篇新聞是進入本堂「無風險利率」與「風險溢酬」的實例。

老師課堂補充

- 老師沒有把資金轉向美股解讀成明確買進訊號。若債券市場出現真正的危機，股票也可能受到波及。
- 老師提醒股市即使因資金流入而上漲，若衰退或債市信心問題惡化，股市仍可能受到波及。這不是保證漲跌的預測。

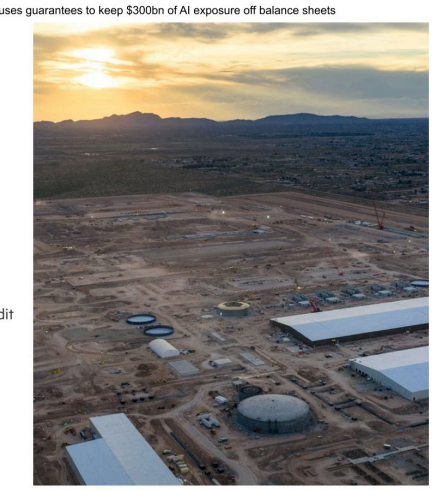
AI 表外融資新聞｜PDF 第 1/8 頁

AI 融資為何移到表外

Artificial intelligence

Big Tech uses guarantees to keep $300bn of AI exposure off balance sheets

Wall Street finds new way to turn tech giants' credit strength into cheaper funding for AI build-out



Meta El Paso Data Center construction site © Brandon Bell/Getty Images

Ryan McMorrow in San Francisco and **Michelle Chan** and **Michael Taffe** in New York

Published SEP 20 2026

Could you be the FT's top trader? **Try our new stock picking game**

Big Tech companies are rapidly expanding their use of guarantees to back debt for AI data centres and chips, issuing up to $300bn in commitments in less than a year while recording little of that exposure on their balance sheets.

First used by Meta on a huge data centre project last autumn, so-called residual value guarantees have been taken up by Broadcom as part of its chip financing deal for Anthropic and by Nvidia to offer support to OpenAI and other customers buying its chips.

These arrangements, under which tech companies guarantee a minimum future value for chips or data centres, join a growing set of creative financing structures embraced by Big Tech to accelerate the AI infrastructure boom. According to an FT analysis, tech giants have offered up to $300bn in these guarantees in the past 12 months alone.

本頁中文重點

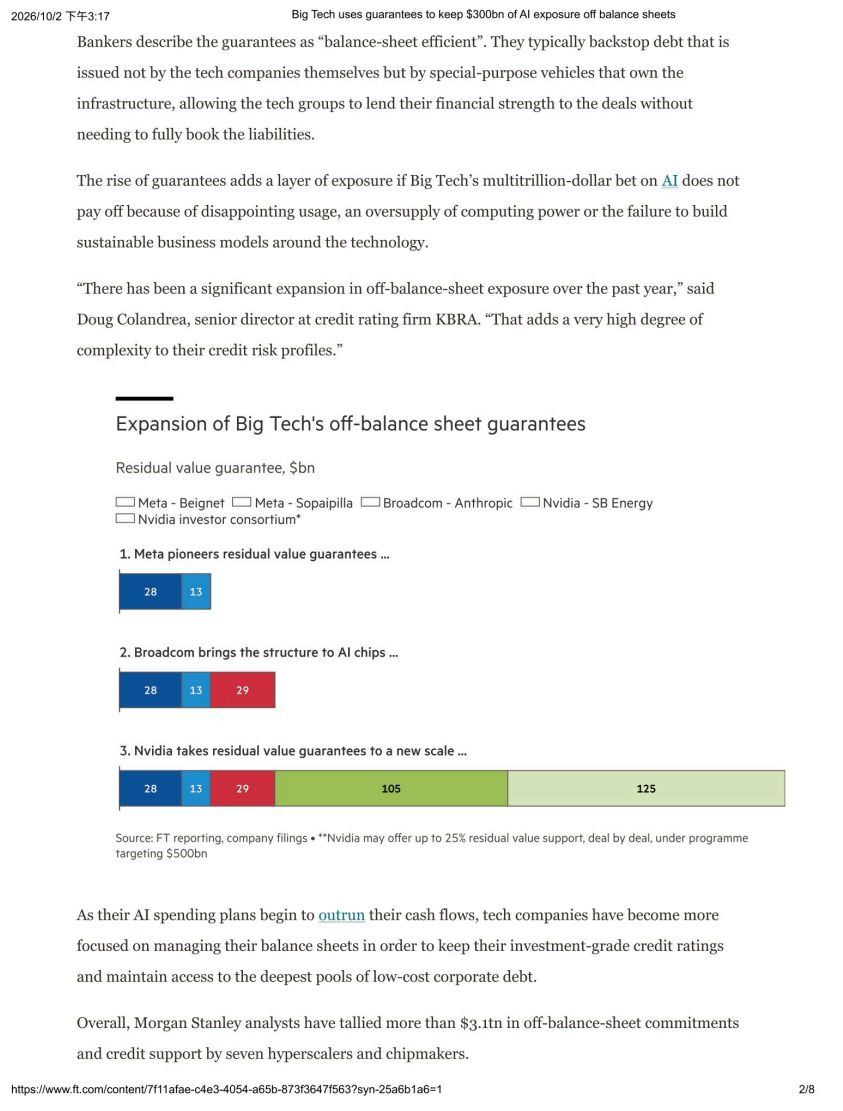
- 文章主題：大型科技公司以殘值保證支持 AI 資料中心及晶片融資。依這份課前文章，過去約一年相關保證承諾最高達 3,000 億美元。這是承諾／曝險，不等於已經支付的損失。
- Meta 先採用此結構，其後 Broadcom、Nvidia 也把它用在客戶的晶片與基礎設施融資。本文發布日為 2026/09/20。頁面上的 2026/10/02 是列印時間，不是上課日期。

老師補充

- 老師先區分 AI 的產出與所需投入：AI 服務可能是無形的，但支撐它的資料中心、建築與設備是實體資產，必須投入大量資本支出。
- 直接建置時，資產負債表增加資料中心。融資來源可能是現金、股權或債務。本堂特別討論以債務融資時，資產與負債同時增加的情境。
- 有了資產可能帶來未來獲利，但負債也提高固定償付負擔。股東是剩餘請求權人，需關注槓桿造成的風險，而不是把「資產增加」一律解讀為利多。

AI 表外融資新聞｜PDF 第 2/8 頁

SPV、信用支持與承諾規模



2026/10/2 下午3:17 Big Tech uses guarantees to keep $300bn of AI exposure off balance sheets

Bankers describe the guarantees as “balance-sheet efficient”. They typically backstop debt that is issued not by the tech companies themselves but by special-purpose vehicles that own the infrastructure, allowing the tech groups to lend their financial strength to the deals without needing to fully book the liabilities.

The rise of guarantees adds a layer of exposure if Big Tech’s multitrillion-dollar bet on AI does not pay off because of disappointing usage, an oversupply of computing power or the failure to build sustainable business models around the technology.

“There has been a significant expansion in off-balance-sheet exposure over the past year,” said Doug Colandrea, senior director at credit rating firm KBRA. “That adds a very high degree of complexity to their credit risk profiles.”

Expansion of Big Tech's off-balance sheet guarantees

Residual value guarantee, $bn

Meta - Beignet · Meta - Sopaipilla · Broadcom - Anthropic · Nvidia - SB Energy · Nvidia investor consortium*

1. Meta pioneers residual value guarantees ...

2. Broadcom brings the structure to AI chips ...

3. Nvidia takes residual value guarantees to a new scale ...

Source: FT reporting, company filings • **Nvidia may offer up to 25% residual value support, deal by deal, under programme targeting $500bn

As their AI spending plans begin to outrun their cash flows, tech companies have become more focused on managing their balance sheets in order to keep their investment-grade credit ratings and maintain access to the deepest pools of low-cost corporate debt.

Overall, Morgan Stanley analysts have tallied more than $3.1tn in off-balance-sheet commitments and credit support by seven hyperscalers and chipmakers.

https://www.ft.com/content/7f11afae-c4e3-4054-a65b-873f3647f563?syn-25a6b1a6=1 2/8

本頁中文重點

- 特殊目的載體（SPV）持有設備並借款。科技公司提供信用支持，使 SPV 得以利用保證人的財務信用。文章稱這種安排具有資產負債表效率。
- 需求不如預期、算力供給過多、AI 商業模式未成熟，都可能使承諾變成真實損失。
- 文章另引用七家大型業者合計逾 3.1 兆美元的表外承諾與信用支持。此口徑包括採購、租賃等，不能與標題的 3,000 億美元殘值保證混為一談。

老師補充

- 老師用兩張資產負債表解釋：如果設備及借款放在 SPV，科技公司自己的帳面負債可能較低，但它透過保證仍承擔風險。
- 單獨成立、缺乏營運歷史的 SPV 借款未必便宜。融資結構的關鍵在於科技公司的保證，而不是 SPV 三個字本身。
- 老師以「獨立公司、非子公司」作教學簡化。整理補充：SPV 並不必然獨立或不需合併，必須依實質控制、風險與適用會計規範判斷。

易錯提醒

- 老師把 SPV 講成「獨立公司、不是子公司」是為了讓例子好懂。實際上要不要合併，看的是誰實質控制、誰承擔風險，以及適用哪一套會計規範，不同案子的答案可能相反。

AI 表外融資新聞｜PDF 第 3/8 頁

殘值保證如何降低借款利差

“There is concern about all these future commitments because everyone is in a race to get the compute,” said Tim Musial, head of fixed income at CIBC Private Wealth US.

For lenders, the residual value guarantees from Big Tech help to alleviate concerns about current technology quickly becoming obsolete as they put up tens of billions to finance AI data centres and chips. The tech companies agree to cover specified shortfalls if these assets one day have to be sold or re-let and fetch less than a guaranteed minimum value.

Their introduction to AI financing has unlocked cheaper financing for projects with these guarantees, which typically price at just a 100 to 150 basis-point premium to the guarantor’s own debt, according to people familiar with the matter.

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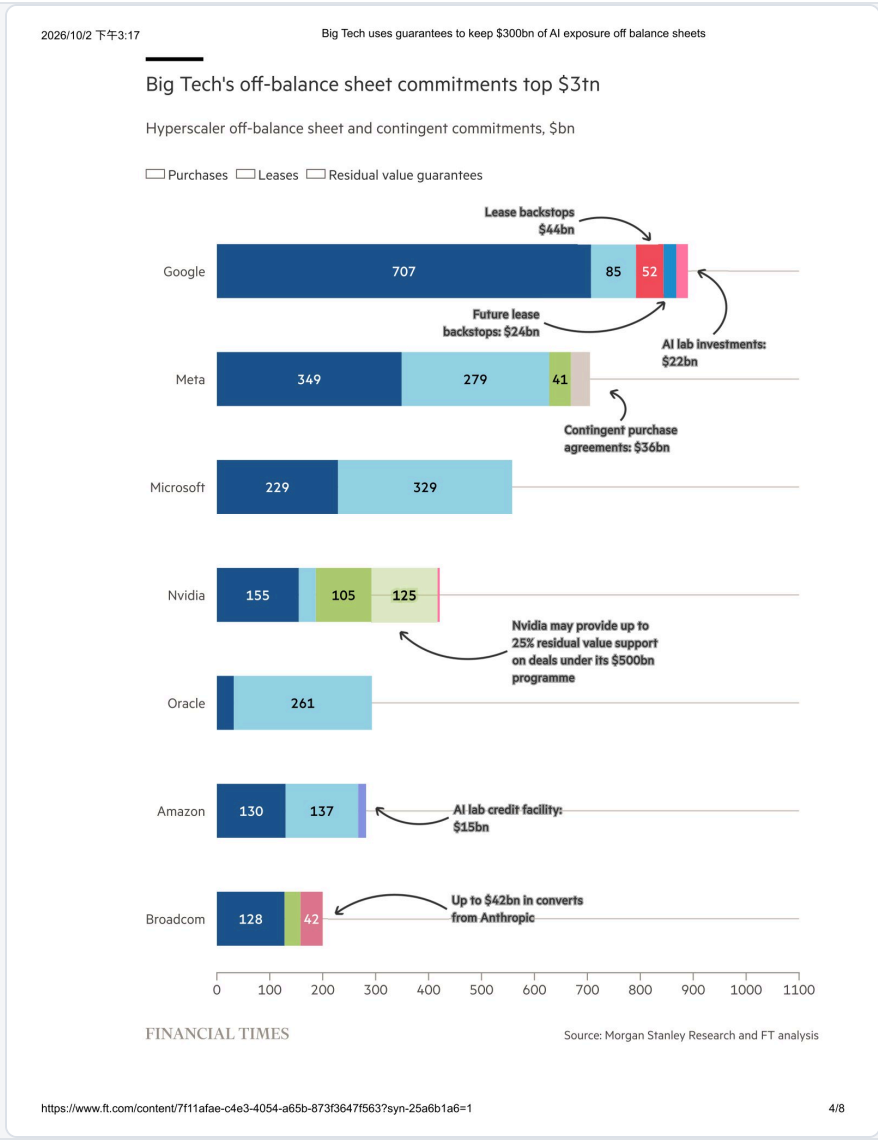
- 保證通常涵蓋資產出售或重新出租所得低於約定最低值的指定缺口。它不是無條件代付所有本金與利息。
- 這份文章描述，有保證的專案借款利率約比保證人本身債務高 100–150 個基點，也就是 1.00–1.50 個百分點。

老師補充

- 老師以科技公司自身借款成本約 5% 示意：有支持的 SPV 可能以約 6%–6.5% 融資。這是說明利差的假設例，不是某家公司當日市場報價。
- 貸款人擔心晶片迅速淘汰、未來轉售價值下滑。殘值保證提供部分下檔保護，所以願意收較低的信用利差。

AI 表外融資新聞｜PDF 第 4/8 頁

讀圖：表外承諾不全是同一種債



本頁中文重點

- 原圖把採購承諾、租賃與殘值保證等分開呈現。有的屬預定支出，有的只有特定條件下才需履行。
- Nvidia 計畫的 5,000 億美元規模與「最高 25% 支持」不同。$5,000$ 億 $\times\ 25\% = 1,250$ 億美元，只是條件與上限下的推算，不等於全部已承諾或已發生損失。

老師補充

- 老師提醒：金融安排會不斷創新，估值不能只代公式。看到表外項目，至少要知道誰持有資產、誰借錢、誰最終承担哪一部分風險。

易錯提醒

- 圖上的金額是未來陸續要付出去的承諾，不是今天就要還的銀行借款。但它確實會占用未來的現金，所以即使沒有列在負債裡，估值時也要算進來。

AI 表外融資新聞｜PDF 第 5/8 頁

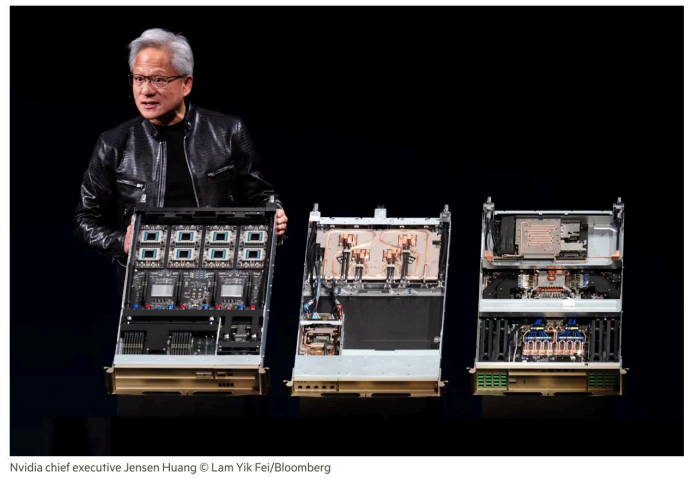
Broadcom 與 Nvidia 的供應商融資

Broadcom was the first chipmaker to adopt the guarantees this year, taking on $29bn in exposure in June to sell 1 gigawatt of chips to an SPV that will lease them to Anthropic. The guarantee is part of a complex vendor-financing programme Broadcom and Google put together to supply chips to the AI laboratory.

Broadcom's recent quarterly filing showed the credit support had little impact on its own balance sheet. People familiar with the matter said the chipmaker would probably offer guarantees to help finance some of its planned chip deliveries next year, which included another 5GW of chips for Anthropic as well as 1.3GW of custom chips for OpenAI.

The company told investors that "the strong profitability trajectory of the leading frontier AI labs and the sustaining value of the underlying assets" meant there was a low probability of triggering such guarantees.

Last month Broadcom's rival Nvidia unveiled a similar vendor financing programme. The chipmaker said it could offer residual value support of up to 25 per cent on deals being put together by Goldman Sachs and a group of Wall Street investors aiming to collectively muster $500bn of capital for Nvidia-powered AI infrastructure.



Nvidia chief executive Jensen Huang © Lam Yik Fei/Bloomberg

本頁中文重點

- 依文章，Broadcom 以約 290 億美元曝險支持 1GW 晶片銷售給 SPV，再租予 Anthropic。這也與 Google 的供應安排相關。
- 文章提到後續交付計畫包括 Anthropic 另 5GW、OpenAI 1.3GW 客製晶片。這些是文中計畫，不應視為已實現營收。
- Nvidia 類似計畫由金融機構與投資人組合籌資，可能按個案提供最高 25% 殘值支持。供應商促成客戶融資，同時可能承擔客戶／資產風險。

來源與講授範圍說明

- 本頁個別交易細節由新聞整理。課堂紀錄沒有逐項解說，不標成老師逐句補充。

AI 表外融資新聞｜PDF 第 6/8 頁

保證的是出售後缺口，不是整筆借款

Nvidia also extended $105bn in guarantees to SB Energy, a SoftBank subsidiary building a massive data centre campus in Ohio for OpenAI.

The chip group, led by Jensen Huang, will record no balance sheet liability until OpenAI's leases begin in 2028, and even then the guarantee structure will limit the weight on its books. In exchange, the campus will exclusively run Nvidia hardware for 20 years.

Bankers at Morgan Stanley and Goldman Sachs have been at the heart of putting the deals together.

"[Residual value guarantees] have a clear contractual mechanism whereby the asset has to be monetised through a competitive sale process, and that asset value is then netted against any exposure under the guarantee," said Richard Myers, head of structured products capital markets at Morgan Stanley, who introduced the structure to AI infrastructure.

Because the guarantor only covers the gap between the sale price and the guaranteed value, residual value guarantees "get more efficient balance-sheet treatment than a typical payment guarantee," he said.



A Stargate AI data centre under construction in Lordstown, Ohio in October 2025 © Kyle Grillot/Bloomberg

本頁中文重點

- 文章另載 Nvidia 對 SB Energy 專案提供約 1,050 億美元保證。該軟銀子公司在 Ohio 建置 OpenAI 資料中心。文中稱租約自 2028 年啟動，校區將長期使用 Nvidia 設備。
- 機制先經競爭性出售變現，再將出售價值抵減保證曝險。保證人只承擔契約涵蓋的差額。
- 教學式：可能支付額 = max（保證最低值 − 可抵減的處分價值，0），仍需受契約觸發條件與責任上限限制。

老師補充

- 老師明確修正「幫 SPV 還所有債」的粗略理解：是補足售價與保證額之間的差額。這也是會計負債認列與經濟曝險不完全相等的原因。

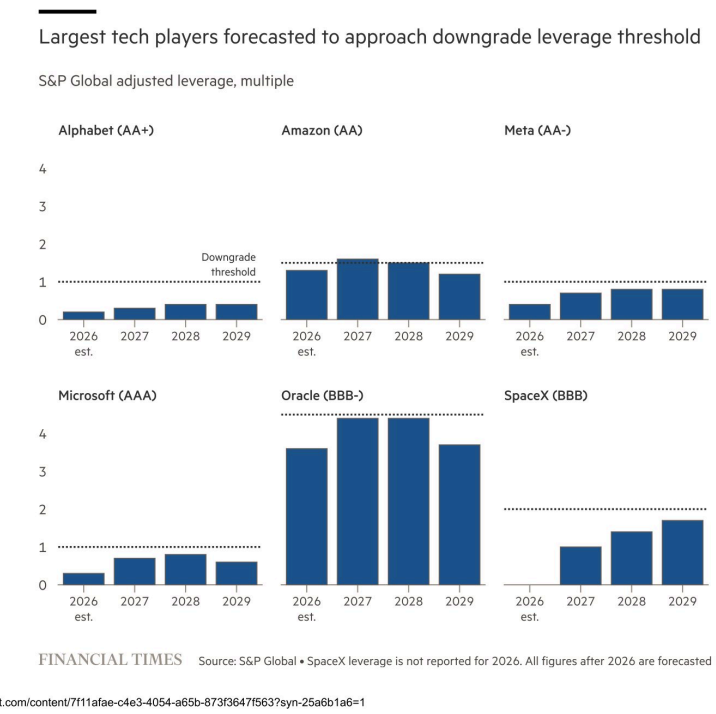
AI 表外融資新聞｜PDF 第 7/8 頁

Meta 案例與信用評等緩衝

Meta was the first tech group to embrace the structure for AI financing, providing a $28bn residual value guarantee to support its joint venture with Blue Owl developing the 2GW Hyperion data centre in Louisiana. The project stretches across 4mn square feet and is expected to come online by 2030, drawing enough power to supply about 1.5mn homes.

"Owning and capitalising one's own data centres on the balance sheet is very capital intensive," said Myers, who spent months designing the guarantee structure so Meta could enjoy some of the ownership benefits without the full balance-sheet burden.

The guarantees ultimately helped raise $27bn of debt for the data centre within 150bp of Meta's own bonds, with little exposure recorded on Meta's books. The tech group used a similar structure in July to support a 1GW data centre being built in El Paso.



本頁中文重點

- 文章中的 Meta／Blue Owl Hyperion 專案：2GW、約 400 萬平方英尺，殘值保證約 280 億美元，協助籌得約 270 億美元債務。利差在 Meta 自身債券約 150bp 以內。
- 下圖是調整後槓桿與降評門檻的比較。2026 之後數值是預測，不能當成已實現資料。
- 重點是剩餘舉債空間：如果槓桿逼近降評門檻，繼續直接借款可能拉高未來融資成本。

老師補充

- 老師指出競爭不會在這一輪投資後結束，業者還想保留下一輪資料中心與新設備投資的能力，所以會設法保留槓桿緩衝。
- 老師以 Microsoft、Oracle 等圖上公司說明。不代表所有公司的降評門檻、投資能力或資產品質都相同。

AI 表外融資新聞｜PDF 第 8/8 頁

評等機構如何把表外風險加回

While the guarantees are mostly off-balance-sheet, credit rating agencies say they do make adjustments to tech groups' leverage figures to reflect them.

Analysts at S&P say they add the amount by which the guaranteed value exceeds their own estimate of what the data centre or chips would fetch in a stressed sale.

Pierre Georges, S&P Global's head of infrastructure research, said the biggest gap he had seen between stressed value and debt on any residual value guarantee deal was about 25 per cent, which on a $20bn structure would add a roughly $5bn leverage adjustment to the guarantee provider's liabilities.

For Meta specifically, Georges said S&P took a "pretty harsh haircut" to the valuation of the Hyperion data centre, but still concluded it was worth more than the debt it backed, meaning no adjustment to Meta's leverage was needed.

"The reality we see today is that the gap, even with the stressed value, is closer to zero most of the time," he said.

A chart in this article has been amended after publication to correct the scale

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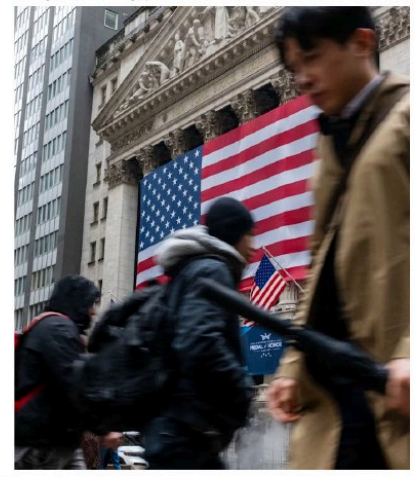
- S&P 的文章口徑：保證值超過壓力情境下可回收資產價值的部分，可能加進調整後負債／槓桿。
- 文中例子：200 億美元結構若有約 25% 缺口，調整額約 50 億美元。Meta 個案在該次壓力估值下，資產仍足以覆蓋支持債務，因此當時未增加其槓桿調整。**即：只要那批資產自己撐得住那筆債，分析師就不會再把它加進公司的槓桿裡。**
- 「未調整」不等於永遠零風險。壓力回收價值與契約條件會改變。文章也註明曾更正圖表刻度。

老師補充

- 老師說課本沒有能直接套用到每個新融資結構的唯一答案。這堂先學會理解機制，並把表外責任納入分析視野。他沒有在錄音中給出該新聞各公司的完整估值。

避險基金融資新聞｜PDF 第 1/10 頁

主要交易利潤來自融資，不是銀行自己下注



US banks

How a trillion-dollar hedge fund borrowing spree became Wall Street's cash cow

Banks' trading businesses are booming, but post-crisis regulation means it is no longer them placing the bets

strians walk past the New York Stock Exchange building © Spencer Platt/Getty Images

Costas Mourselas and **Ortenca Aliaj** in London and **Joshua Franklin** in New York

Published 52 MINUTES AGO

Could you be the FT's top trader? **Try our new stock picking game**

Goldman Sachs had a message for its clients. As it gathered some of the top hedge fund executives in a 500-acre resort outside London this February, it wanted them to know: the Wall Street bank no longer saw them as rivals. Instead, they had become some of its most valued clients.

The message underscored what has become an inescapable reality on Wall Street. US banks are generating record trading profits. But they are no longer the ones doing the trading.

Firms such as Jane Street and Citadel have aggressively expanded into banks' traditional turf, dominating the core business of market making as well as placing big bets on market moves. But their insatiable demand for financing to juice those trades has made them irresistible clients for lenders and a powerful engine for profits.

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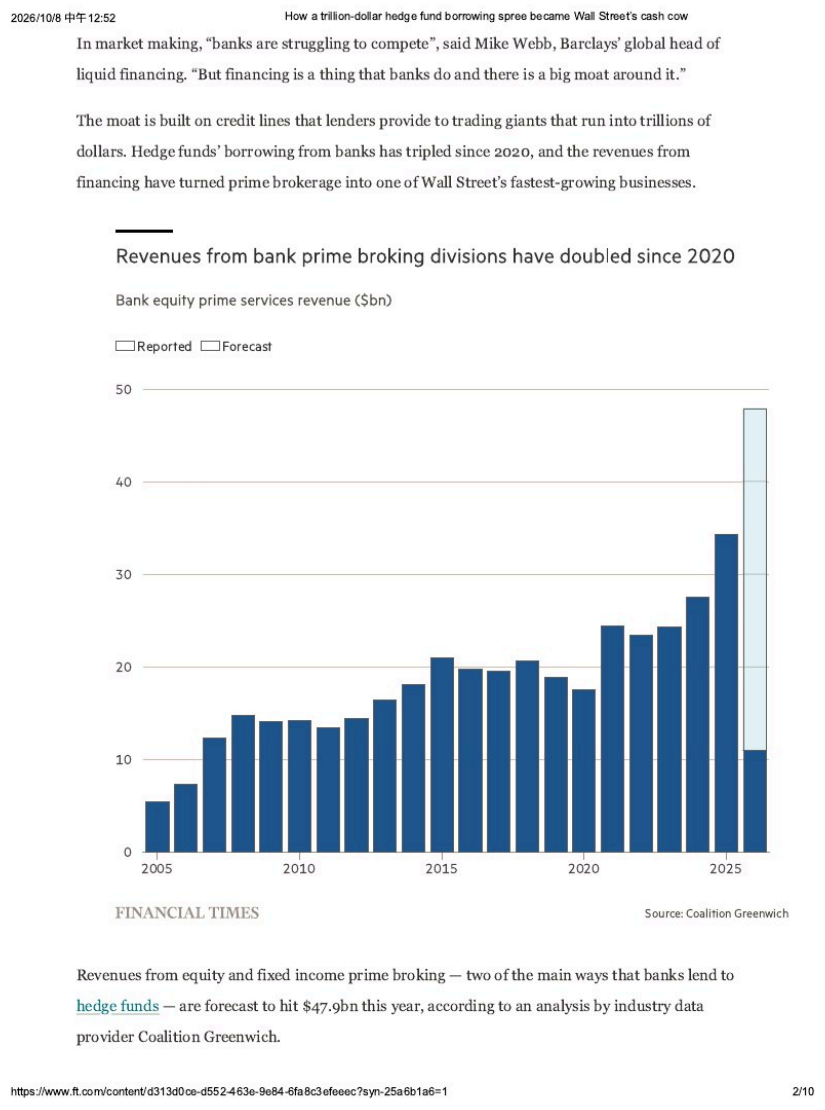
- 文章出處：Financial Times，〈How a trillion-dollar hedge fund borrowing spree became Wall Street's cash cow〉，記者 Costas Mourselas、Ortenca Aliaj（倫敦）與 Joshua Franklin（紐約）。頁首的 2026/10/08 是列印時間，不是上課日期。
- 導言一句話就點出全文主軸：銀行的交易相關業務正在大賺，但 2008 年後的監管規定使下注的人已經不是銀行本身。
- 開場場景：今年二月高盛在倫敦近郊一處 500 英畝的度假莊園對頂尖避險基金高層表示，已不再把他們看成競爭對手，而是重要客戶。
- 美國各大銀行的交易業務獲利創下紀錄，但真正在市場上承擔部位的不是銀行。Jane Street 與 Citadel 這類公司主導造市，它們對融資的需求極大，因此成為銀行很難放棄的客戶。**即：銀行賺的是替這些造市商調錢的錢，所以不敢放掉他們。**

老師課堂說明

- 老師說今天時間不夠，無法在課堂上講這篇 Financial Times 的文章，文章已經上傳 Moodle，請同學自己下載閱讀。
- 老師先給出全文主旨：投資銀行的主要獲利部門換了，而這個轉變可能帶來金融不穩定。
- 老師描述過去的模式：以前講到投資銀行，是指它用自己的策略去交易股票與固定收益商品，拿自有資金承擔部位，這叫自營交易（proprietary trading）。

避險基金融資新聞｜PDF 第 2/10 頁

融資業務的進入門檻，與 prime broking 收入曲線



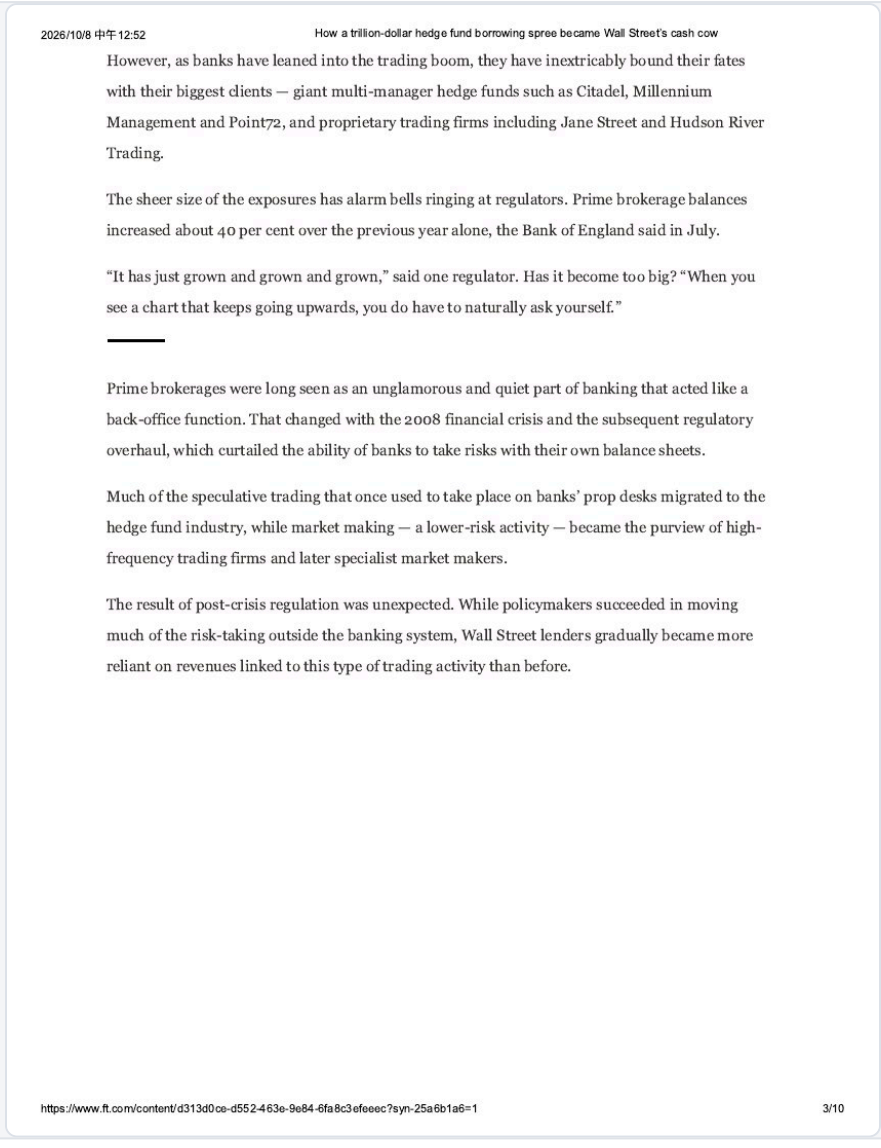
本頁中文重點

- Barclays 全球流動性融資主管 Mike Webb 說，在造市這一塊「銀行很難競爭」，但是「融資是銀行本來就在做的事，而且周圍有一道很寬的護城河」。
- 護城河的具體內容是：銀行提供給這些交易巨頭的信用額度，規模合計達到數兆美元。避險基金向銀行的借款自 2020 年以來成長為三倍，融資帶來的收入使 prime brokerage 成為華爾街成長最快的業務之一。
- 圖表「Revenues from bank prime broking divisions have doubled since 2020」：銀行股票類 prime services 收入（十億美元），2005 年約 5.5，2015 年前後在 18–21 之間盤整，2020 年約 17.5，之後逐年攀升，2025 年約 34.5，2026 年預測值明顯更高。資料來源 Coalition Greenwich。
- 文字說明：股票與固定收益兩類 prime broking 是銀行借錢給避險基金的兩個主要管道，今年合計收入預估達 479 億美元。

老師課堂說明

- 老師說 2008 年金融海嘯之後，對投資銀行出現大量監管規範，於是它們轉向另一個業務部門，也就是 prime brokerage。**即：自己不能冒的險，就改成替冒險的人提供資金。**
- 老師強調這個業務的性質不同：它不是直接去交易，而是支援別人（像是避險基金），具體做的事情是提供資金、放款、融資與槓桿。

避險基金融資新聞｜PDF 第 3/10 頁 風險移出銀行體系之後，銀行反而更依賴這塊收入


2026/10/8 中午12:52 How a trillion-dollar hedge fund borrowing spree became Wall Street's cash cow

However, as banks have leaned into the trading boom, they have inextricably bound their fates with their biggest clients — giant multi-manager hedge funds such as Citadel, Millennium Management and Point72, and proprietary trading firms including Jane Street and Hudson River Trading.

The sheer size of the exposures has alarm bells ringing at regulators. Prime brokerage balances increased about 40 per cent over the previous year alone, the Bank of England said in July.

"It has just grown and grown and grown," said one regulator. Has it become too big? "When you see a chart that keeps going upwards, you do have to naturally ask yourself."

Prime brokerages were long seen as an unglamorous and quiet part of banking that acted like a back-office function. That changed with the 2008 financial crisis and the subsequent regulatory overhaul, which curtailed the ability of banks to take risks with their own balance sheets.

Much of the speculative trading that once used to take place on banks' prop desks migrated to the hedge fund industry, while market making — a lower-risk activity — became the purview of high-frequency trading firms and later specialist market makers.

The result of post-crisis regulation was unexpected. While policymakers succeeded in moving much of the risk-taking outside the banking system, Wall Street lenders gradually became more reliant on revenues linked to this type of trading activity than before.

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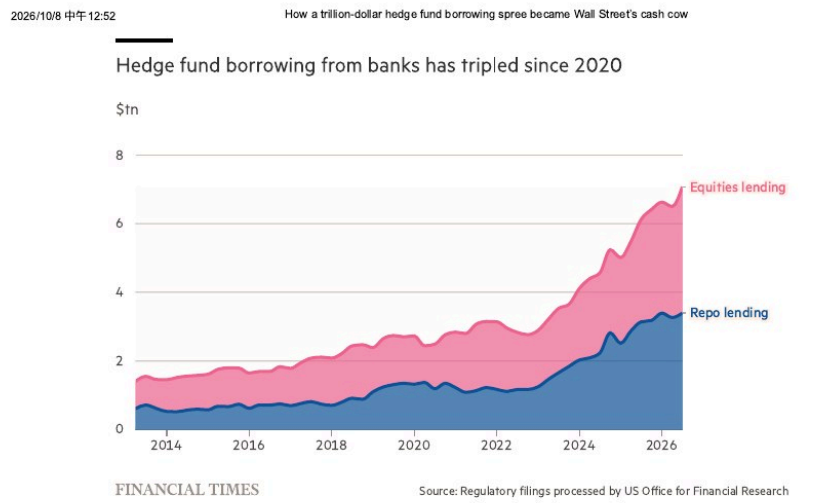
- 銀行在這波交易熱潮中把自己的命運和最大客戶綁在一起：Citadel、Millennium Management、Point72 這類大型多策略避險基金，以及 Jane Street、Hudson River Trading 這類自營交易公司。
- 曝險規模已讓監管機關警覺。英格蘭銀行在七月表示，prime brokerage 餘額一年之內增加約 40%。一位監管官員說「它就是一直長、一直長」，被問到是否已經太大時回答：「當你看到一張一直往上的圖，你自然會這樣問自己。」
- 歷史背景：prime brokerage 長期被視為銀行裡不起眼、安靜、接近後台的業務。2008 年金融危機與其後的監管改革改變了這一點，因為新規定限制了銀行用自有資產負債表承擔風險的能力。
- 業務因此重新分配：原本在銀行自營交易台上的投機交易移到避險基金業。造市這種風險較低的活動，先是落到高頻交易公司手上，後來由專業造市商承接。**即：監管沒有讓這些風險消失，只是換了人去承擔。**
- 文章指出後危機監管出現了預期之外的結果——政策確實把大部分風險承擔移到銀行體系之外，但華爾街的放款機構反而比以前更依賴與這類交易活動連動的收入。

老師課堂說明

- 老師說這個市場現在規模很大，而銀行是借錢給這些避險基金的人。
- 老師點出風險的傳導路徑：正因為銀行把錢借出去，一旦這些避險基金出事，投資銀行的獲利也會跟著受影響。
- 老師說這是另一篇關於系統性風險的文章——放款的規模在過去並不是投資銀行的主要業務，但現在它們借出大量資金給避險基金並提供相關服務，靠這個賺錢。

避險基金融資新聞｜PDF 第 4/10 頁

借款規模與重複性收入：為什麼銀行想做這門生意



The key attraction of servicing hedge funds and trading firms is recurring revenues. Investment bankers can spend years courting executives such as Elon Musk in order to secure a one-off payment for advising on an initial public offering. But acting as prime broker to Citadel, Millennium or Jane Street offers a steady revenue stream that can generate as much as $200mn per year after trading costs, said three executives at bulge-bracket banks.

Martin Moloney, deputy secretary-general at the Financial Stability Board, said: "If anyone thought that by making sure banks were properly capitalised that we would make the problem go away . . . that would have been a naive view."

Prime services revenues will have leapt from 10 per cent of banks' equities revenues in 2005 to about 38 per cent this year, according to Coalition Greenwich — and those figures exclude the vast derivatives services banks provide.

"People are increasingly understanding that as non-banks take positions, they actually use the assistance of banks to do that," he added.

How hedge funds juice their bets

本頁中文重點

- 圖表「Hedge fund borrowing from banks has tripled since 2020」（兆美元）：分成 Equities lending（股票融資）與 Repo lending（附買回融資）兩層，2014 年合計約 1.5 兆，2020 年前後約 2.5 兆，2023 年起急升，2026 年接近 7 兆。資料來源為美國 Office for Financial Research 處理的監理申報資料。
- 服務避險基金與交易公司的關鍵吸引力是收入會重複發生。投資銀行家可能花好幾年經營像 Elon Musk 這樣的客戶，最後只拿到一次性的 IPO 顧問費。但擔任 Citadel、Millennium 或 Jane Street 的 prime broker，是穩定的收入流，扣掉交易成本後每年可達 2 億美元（三位大型投行高層的說法）。
- 金融穩定委員會（FSB）副秘書長 Martin Moloney 說：「如果有人以為只要確保銀行資本充足，問題就會消失……那會是太天真的看法。」
- 依 Coalition Greenwich 的估計，prime services 收入佔銀行股票部門收入的比重，將從 2005 年的 10% 上升到今年的約 38%，而且這個數字還不包含銀行提供的大量衍生品服務。
- Moloney 補充：「大家越來越理解，當非銀行機構建立部位時，它們其實是靠銀行的協助在做這件事。」

老師課堂說明

- 老師請同學看這張放款規模的圖，說放款金額隨時間一路增加，特別是疫情之後那一段上升得很明顯。
- 老師提到這段期間的上升和 AI 相關的行情有關。

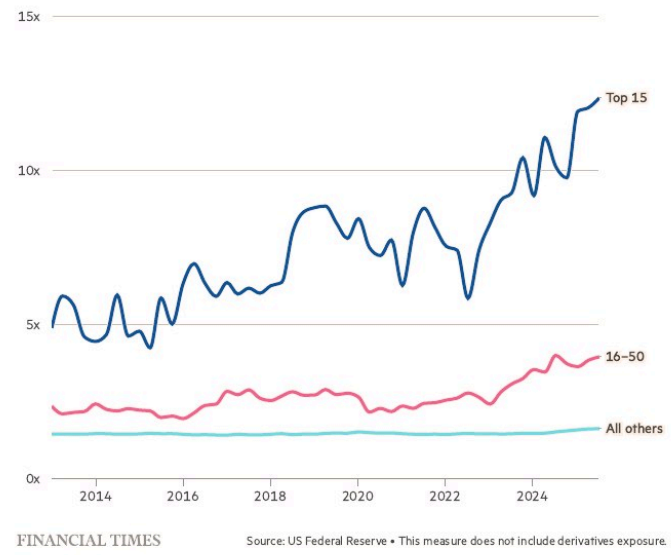
避險基金融資新聞｜PDF 第 5/10 頁

槓桿倍數：前 15 大與其他基金的差距

2026/10/8 中午12:52 How a trillion-dollar hedge fund borrowing spree became Wall Street's cash cow

The world's largest hedge funds are using record amounts of leverage

Gross asset value divided by net asset value



FINANCIAL TIMES Source: US Federal Reserve • This measure does not include derivatives exposure.

Top-50 hedge funds borrow $3 to juice their bets for every dollar of money they manage, according to data from the US Federal Reserve. But for the top 15, that rises to $11 of borrowing, the data shows.

Those figures probably understate the true amount of leverage because they do not include derivatives exposure — which is built into the contracts themselves.

One bank executive said that when derivatives exposures were taken into account, the largest hedge funds were 20 to 25 times levered.

Market makers can have greater leverage: as much as 40 times, according to another bank executive, because they are perceived as lower risk than hedge funds placing big wagers.

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本頁中文重點

- 圖表「The world's largest hedge funds are using record amounts of leverage」，衡量方式是總資產價值除以淨資產價值。前 15 大基金從 2014 年約 5 倍，震盪上升到 2025 年約 12 倍的紀錄水準。第 16–50 名約 3 倍且近年緩升。其餘基金長期貼近 1 倍。資料來源為美國聯準會，且此指標不含衍生品曝險。
- 前 50 大避險基金每管理 1 美元的資金，就借入約 3 美元來放大部位。前 15 大則是借 11 美元（聯準會資料）。
- 文章指出這些數字可能低估真實槓桿，因為沒有計入衍生品曝險——那是內建在合約本身裡面的。
- 一位銀行高層說，把衍生品曝險算進去之後，最大的幾家避險基金槓桿是 20 到 25 倍。
- 另一位銀行高層說造市商的槓桿可以更高，最多到 40 倍，理由是它們被認為比下大注的避險基金風險低。

老師課堂說明

- 老師請同學看前 15 大避險基金這條線，說它們的槓桿在上升，這就是這篇文章要講的事情，請同學自己把文章讀過一次。

避險基金融資新聞｜PDF 第 6/10 頁 為什麼銀行敢把額度放大：集中化客戶與內部軋平

Goldman Sachs and Morgan Stanley are the biggest players in the business, with JPMorgan Chase third, according to hedge fund executives. But as prime broking revenues have exploded, they have faced stiffer competition from Wall Street rivals Citigroup and Bank of America, as well as challengers in Europe including BNP Paribas, Barclays and ABN Amro.

One of the biggest drivers of growth — beyond simply the expansion of the hedge fund industry — has been the rise of a small number of mega hedge funds and trading firms. Large multi-managers made up more than a third of the industry's trading activity last year despite managing less than a tenth of its assets, according to Goldman.

Because the managers are sophisticated players with hundreds of trading teams across a spectrum of asset classes, they have an inbuilt diversification and level of risk management that has allowed banks to justify ratcheting up their lending.

Webb at Barclays said: "An institutional-quality hedge fund or asset manager is a sophisticated organisation with a liquidity-management function that reports directly to the founder."

Fewer, bigger clients are also simpler to manage. "It's easier to know a smaller set of clients really well," Webb added.

Scale helps banks too. The more trading flow they have, the easier it is to match one client's long position with another client's short — a crucial balancing act known as internalisation.

That means the bank can finance both sides of the transaction without having to borrow or source stock in the open market, eliminating one of the costliest parts of the business and boosting margins.

But the growing scale of a few clients at the top of the industry has also shifted the balance of power, according to people familiar with industry practices, with clients pushing for far more favourable lending terms.

Lock-up terms, the length of time a prime broker commits to providing financing before it can withdraw credit, have gradually lengthened.

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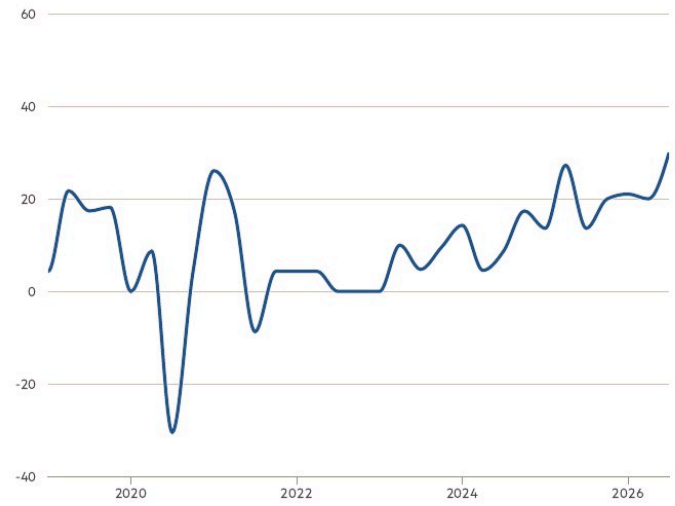
- 市場地位：高盛與摩根士丹利是最大的兩家，摩根大通第三（避險基金高層說法）。隨著 prime broking 收入暴增，它們面對花旗、美國銀行，以及歐洲的 BNP Paribas、Barclays、ABN Amro 的競爭。
- 成長的最大驅動力之一，是少數幾家超大型避險基金與交易公司的崛起。依高盛資料，大型多策略基金去年佔整個業界交易活動三分之一以上，但管理的資產不到十分之一。
- 銀行為放大額度找到的理由是：這些管理人橫跨多種資產類別、有數百個交易團隊，結構上本來就分散，風險管理也成熟。Webb 說：「機構等級的避險基金或資產管理公司是成熟的組織，有一個直接向創辦人報告的流動性管理部門。」
- 客戶少而大也比較好管理：「認識一小群客戶，比較容易認識得深。」
- 規模對銀行本身也有利：交易流量越大，越容易把一個客戶的多頭和另一個客戶的空頭對起來，這個動作叫 internalisation（內部化）。對得起來的部分，銀行就能同時替交易雙方融資，不必到公開市場借券或調券，省掉這門生意裡最貴的一塊成本，利潤率因而提高。**即：客戶越多，彼此的多空就越容易互相抵銷，銀行需要向外調錢的部分就越少。**
- 另一面是議價權的移動：頂端少數客戶的規模變大之後，開始要求遠比以前有利的放款條件。其中之一是 lock-up terms——prime broker 承諾提供融資、在這段期間內不能抽回信用的時間長度——已經逐漸拉長。

避險基金融資新聞｜PDF 第 7/10 頁

融資承諾期限拉長，等於銀行抽身的時間變慢

Hedge funds are pushing banks for better lending terms

Net percentage of banks reporting more intense hedge fund lobbying



FINANCIAL TIMES Source: Based on responses to the Federal Reserve Board's quarterly senior credit officer opinion survey on dealer financing terms.

Commitments that typically lasted two weeks went up to 30 days, stretched to three months and more recently to half a year for some of the industry's most coveted clients. That means a bank cannot cut back its financing or demand more margin than agreed at the start of the contract without giving increasingly lengthy notice.

Some regulators see this as intense competition steadily weakening safeguards put in place in case things go wrong, especially because banks often have limited transparency into their clients' broader activities and true risk profiles.

本頁中文重點

- 圖表「Hedge funds are pushing banks for better lending terms」：回報避險基金遊說力道變強的銀行淨比率，2020 年疫情初期一度跌到 −30 附近，之後多數時間為正，2026 年升到接近 30。資料來源為聯準會對交易商融資條件的季度高級信用主管意見調查。
- 具體變化：原本通常兩週的融資承諾延長到 30 天，再拉長到三個月，最近對業界最搶手的客戶甚至到半年。
- 這個期限的意義是：在承諾期間內，銀行不能把融資縮減、也不能要求比合約開始時更高的保證金，除非提前給出越來越長的通知。
- 部分監管者認為這是激烈競爭正在一步步削弱出事時的防護機制，尤其因為銀行對客戶更廣泛的活動與真實風險輪廓往往看不清楚。

避險基金融資新聞｜PDF 第 8/10 頁

銀行看不到的部分，與 LTCM 的先例

Big trading firms and hedge funds typically disclose as little as possible about their relationships with rival prime brokers, which they say is necessary to protect their intellectual property and prevent leaks.

Large counterparties will often provide an overall leverage figure for their business across multiple prime brokers, according to the bulge-bracket veteran, but banks do not know how that leverage is distributed or the concentration across their positions.

Prime broking's biggest blow-ups



John Meriwether, founder of Long-Term Capital Management © James Leynse/Corbis via Getty Images

Prime broking has become a highly lucrative business, but it can also be fraught with risk.

Prime brokers lose money when their clients take losses so large that they burn through a cash buffer required by the bank known as 'margin'. If, after taking losses, a client misses a margin call, prime brokers seize the underlying positions and conduct a fire sale to neutralise the risk.

The most famous blow-up might be the collapse of Long-Term Capital Management in 1998, a hedge fund founded by former Salomon Brothers bond trader John Meriwether. LTCM negotiated terms that meant it paid zero in initial margin to its prime broker Bear Stearns when it initiated some trades.

It racked up $1.3tn in off-balance-sheet derivative notional exposure against a capital base of close to $5bn in 1997 — and received about $3.6bn bailout from commercial banks the following year after the Federal Reserve intervened.

本頁中文重點

- 大型交易公司與避險基金通常盡可能少揭露自己與各家 prime broker 的往來，理由是保護智慧財產、避免策略外洩。
- 一位資深大型投行人士說，大型交易對手往往只提供一個跨多家 prime broker 的整體槓桿數字，但銀行不知道這些槓桿怎麼分布，也不知道部位的集中程度。
- 小標「Prime broking's biggest blow-ups」，配圖是 Long-Term Capital Management 創辦人 John Meriwether。
- 文章說明 prime broker 虧錢的機制：客戶虧損大到燒光銀行要求的現金緩衝（也就是保證金 margin），客戶在虧損後未能補繳保證金時，prime broker 會接管底下的部位並拋售，以中和自己的風險。
- 最有名的例子是 1998 年 LTCM 的崩潰。這家基金由前所羅門兄弟債券交易員 John Meriwether 創立，談到的條件讓它在發動某些交易時，對 prime broker Bear Stearns 支付的初始保證金是零。
- LTCM 累積了 1.3 兆美元的表外衍生品名目曝險，而 1997 年的資本基礎接近 50 億美元。在聯準會介入之後的隔年，獲得商業銀行約 36 億美元的紓困。

避險基金融資新聞｜PDF 第 9/10 頁

Archegos 與今年的險情：競爭壓力會侵蝕風控

More recently, there was the default of Bill Hwang's family office Archegos Capital in 2021.

Archegos entered into highly leveraged derivatives agreements with its prime brokers to bet on companies including ViacomCBS and Baidu.

His firm used its desirability as a client to negotiate high leverage with each bank, while refusing to disclose details about the size of similar positions across all its prime brokers. The $5.5bn loss sustained by Credit Suisse, one of Archegos's prime brokers, contributed to a loss of confidence in the lender which resulted in a sale to its rival UBS.

This year, prime brokers narrowly escaped another debacle when AI-focused hedge fund Situational Awareness got into trouble after a rally reversed.

Banks had lent billions of dollars to the hedge fund's 24-year-old founder and were exposed to a brutal unwinding of those trades — until Citadel stepped in to buy its portfolio of public stocks, sparing their blushes.

Prime brokers insist standards have tightened in recent years, with greater scrutiny around leverage and more robust risk controls. But some regulators disagree on whether banks have fully learnt broader lessons around liquidity despite previous blow-ups.

Moloney of the Financial Stability Board said: "I recognise that many of the major participants would say we have very robust collateral and risk-management systems in place, and that is good as long as it lasts."

"But competitive pressure can erode those risk controls and you need to be extremely careful about them because their impacts can be cross border and they can trigger much wider problems across the financial system."

Additional reporting by Martin Arnold. Visuals by Louis Ashworth

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US banks

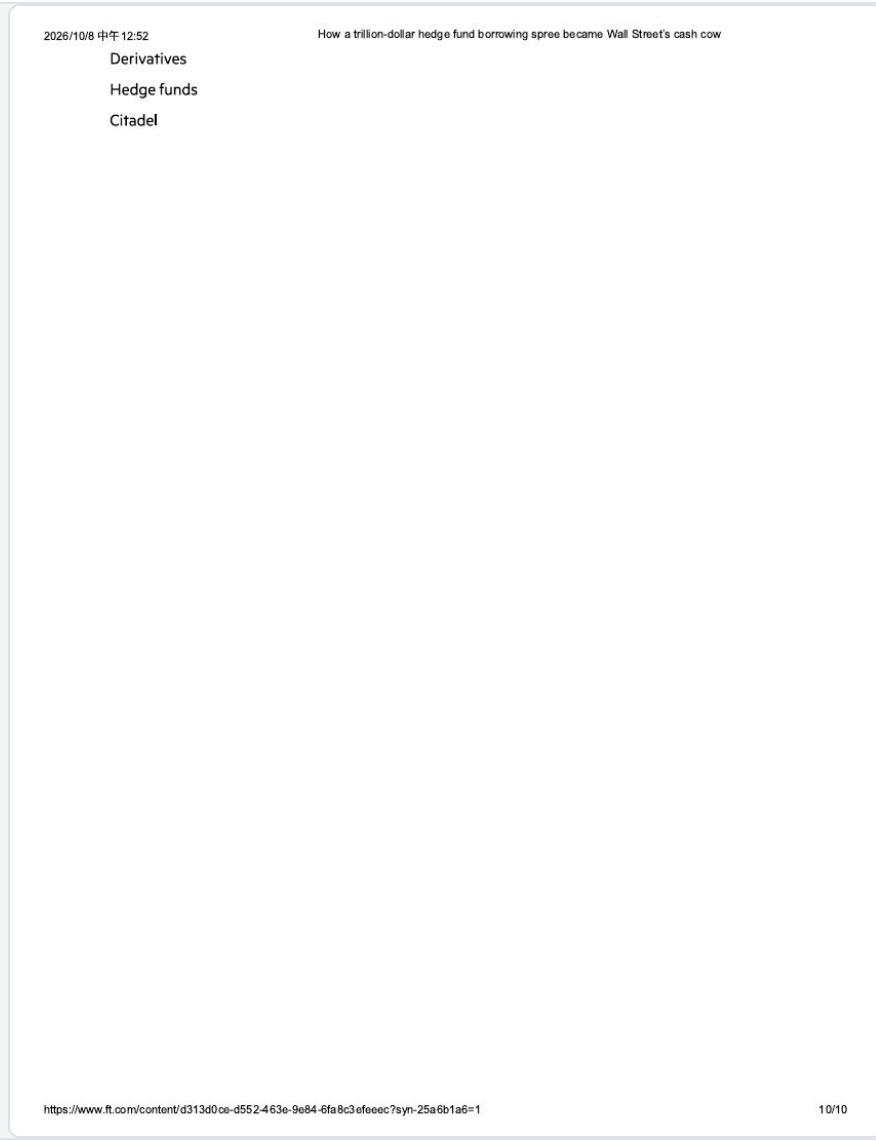
Investment Banking

本頁中文重點

- 較近期的例子是 2021 年 Bill Hwang 的家族辦公室 Archegos Capital 違約。Archegos 和它的 prime broker 簽訂高度槓桿的衍生品協議，押注 ViacomCBS、百度等公司。
- 它利用自己作為客戶的搶手程度，分別向每一家銀行談到高槓桿，同時拒絕揭露它在所有 prime broker 的同類部位總規模。瑞士信貸作為其中一家 prime broker 承受 55 億美元虧損，導致市場對它失去信心，最終被對手 UBS 併購。**即：每一家都只看到自己那一塊，就沒有人看得出這個客戶總共壓了多大的部位。**
- 今年 prime broker 又差一點遇上另一場災難：專注 AI 的避險基金 Situational Awareness 在一波漲勢反轉後陷入困境。銀行借給這家基金 24 歲創辦人數十億美元，原本要面對這些交易的劇烈平倉，最後是 Citadel 出手買下它的公開股票組合，才讓銀行免於難堪。
- Prime broker 方面堅稱標準近年已經收緊，對槓桿有更嚴格的審查、風控也更健全。但部分監管者不認為銀行已經從過去的爆雷中完整學到流動性方面的教訓。
- FSB 的 Moloney 說：「我理解許多主要參與者會說，我們有非常健全的擔保品與風險管理系統，這在它持續有效的期間確實是好事。」
- 「但競爭壓力會侵蝕那些風控，你必須非常小心，因為它們的影響可能跨境，可能在整個金融體系引發更大範圍的問題。」
- 文末註記：補充報導 Martin Arnold，視覺 Louis Ashworth。版權為 The Financial Times Limited 2026。

避險基金融資新聞｜PDF 第 10/10 頁

文章末頁：主題標籤

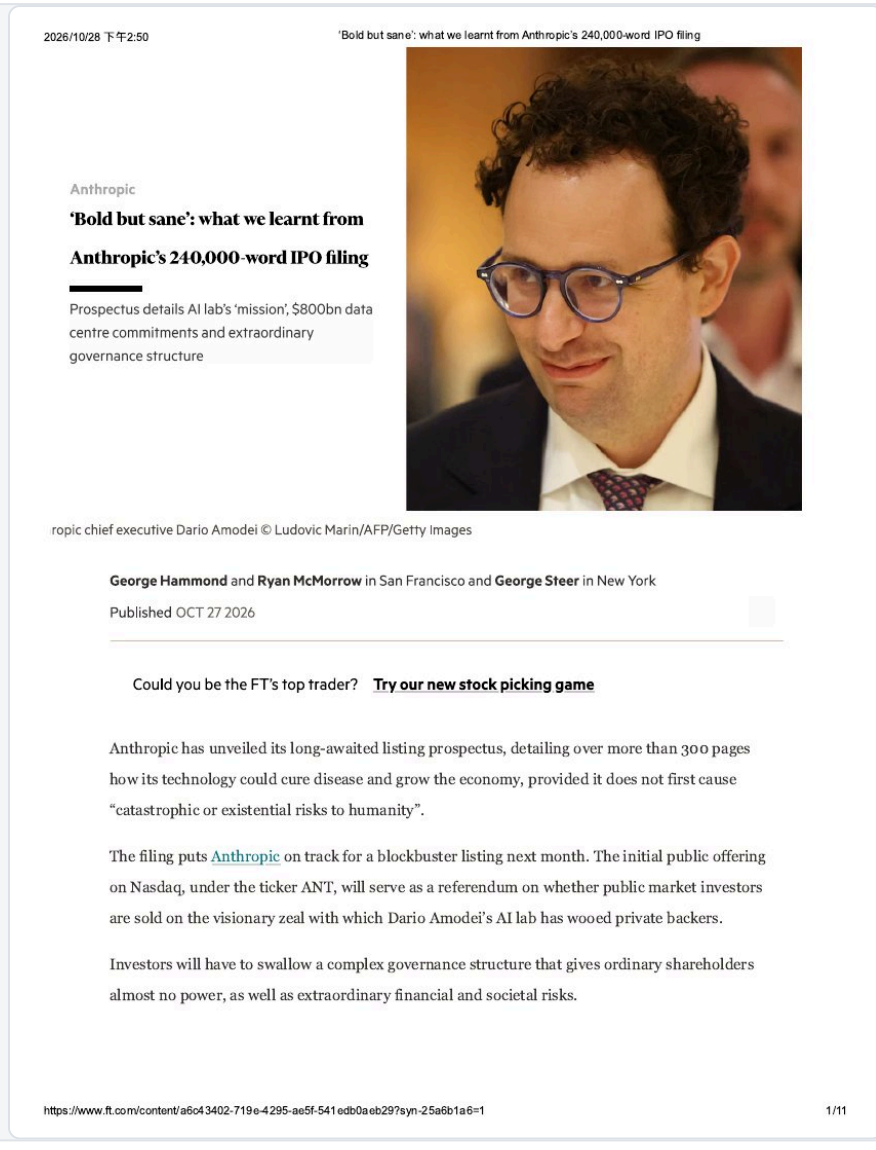


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- 承接前一頁的「Follow the topics in this article」標籤清單：Derivatives、Hedge funds、Citadel（前一頁已列出 US banks、Investment Banking）。

Anthropic IPO 新聞｜PDF 第 1/11 頁

Anthropic 公布上市公開說明書：使命、算力承諾與特殊的治理結構



2026/10/28 下午2:50 'Bold but sane': what we learnt from Anthropic's 240,000-word IPO filing

Anthropic

'Bold but sane': what we learnt from Anthropic's 240,000-word IPO filing

Prospectus details AI lab's 'mission', $800bn data centre commitments and extraordinary governance structure

ropic chief executive Dario Amodei © Ludovic Marin/AFP/Getty Images

George Hammond and **Ryan McMorrow** in San Francisco and **George Steer** in New York

Published OCT 27 2026

Could you be the FT's top trader? **Try our new stock picking game**

Anthropic has unveiled its long-awaited listing prospectus, detailing over more than 300 pages how its technology could cure disease and grow the economy, provided it does not first cause "catastrophic or existential risks to humanity".

The filing puts Anthropic on track for a blockbuster listing next month. The initial public offering on Nasdaq, under the ticker ANT, will serve as a referendum on whether public market investors are sold on the visionary zeal with which Dario Amodei's AI lab has wooed private backers.

Investors will have to swallow a complex governance structure that gives ordinary shareholders almost no power, as well as extraordinary financial and societal risks.

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本頁中文重點

- 文章出處：Financial Times，〈'Bold but sane': what we learnt from Anthropic's 240,000-word IPO filing〉，記者 George Hammond、Ryan McMorrow（舊金山）與 George Steer（紐約），發布日 2026/10/27。頁首的 2026/10/28 是列印時間。
- 導言：公開說明書（prospectus）詳述這家 AI 實驗室的「使命」、8,000 億美元的資料中心承諾，以及非比尋常的治理結構。
- Anthropic 公布期待已久的上市公開說明書，用超過 300 頁說明它的技術可以如何治癒疾病、推動經濟成長，前提是它不會先帶來「對人類的災難性或存在性風險」。
- 這份文件讓 Anthropic 走上下個月在 Nasdaq 掛牌的路，代號 ANT。這次首次公開發行（initial public offering，IPO）等於一次公投：公開市場的投資人，是否也買帳 Dario Amodei 用來吸引私人投資者的那份理想熱情。
- 投資人必須接受一個讓一般股東幾乎沒有權力的複雜治理結構，以及非比尋常的財務與社會風險。

老師課堂說明

- 公司要上市時，必須把所有細節寫進公開說明書給外部投資人看，美國申報用的是 S-1 文件。這份文件超過 300 頁，老師挑出三個重點：使命、8,000 億美元的資料中心支出，以及特殊的治理結構。
- 老師先回到股權的基本意義：股權代表你擁有這家公司，是最後一順位拿現金流的剩餘請求者，所以股東除了賺資本利得，還有投票權，可以影響重大的公司決策。這家公司的特別之處，就在於它把這項權利拿掉了大半。
- **即：買股票通常同時買到「分錢的權利」和「說話的權利」，這次的股票主要只給了前者。**

Anthropic IPO 新聞｜PDF 第 2/11 頁

賣的是文化與使命：創辦團隊仍在，與 OpenAI 對照

Alongside dizzying growth and $800bn in data centre spending commitments, the company's founders urge prospective owners of the stock "not to think of your purchase in purely transactional terms". Here is what we learnt from Anthropic's S-1.



The IPO will serve as a referendum on whether public market investors are sold on the visionary zeal with which Amodei's AI lab has wooed private backers © Kent Nishimura/AFP/Getty Images

Selling a unique culture and mission

Anthropic's filing plays up its mission to responsibly develop AI for the long-term benefit of humanity. The company's seven-strong founding team, who left OpenAI to launch their own safety-orientated rival in 2021, said in the S-1 that "all of our efforts are guided by a singular focus on our mission".

The founders framed this focus as a key advantage over rivals, helping the business compete for talent and remain focused on the "highest-value work for humanity". They also highlighted that their continued presence at the top of the lab was "uncommon", a subtle dig at rival OpenAI, which has lost the majority of its founding team.

> We are first and foremost a research lab—our mission is the responsible development and maintenance of advanced AI for the long-term benefit of humanity. This mission directs our focus—along with

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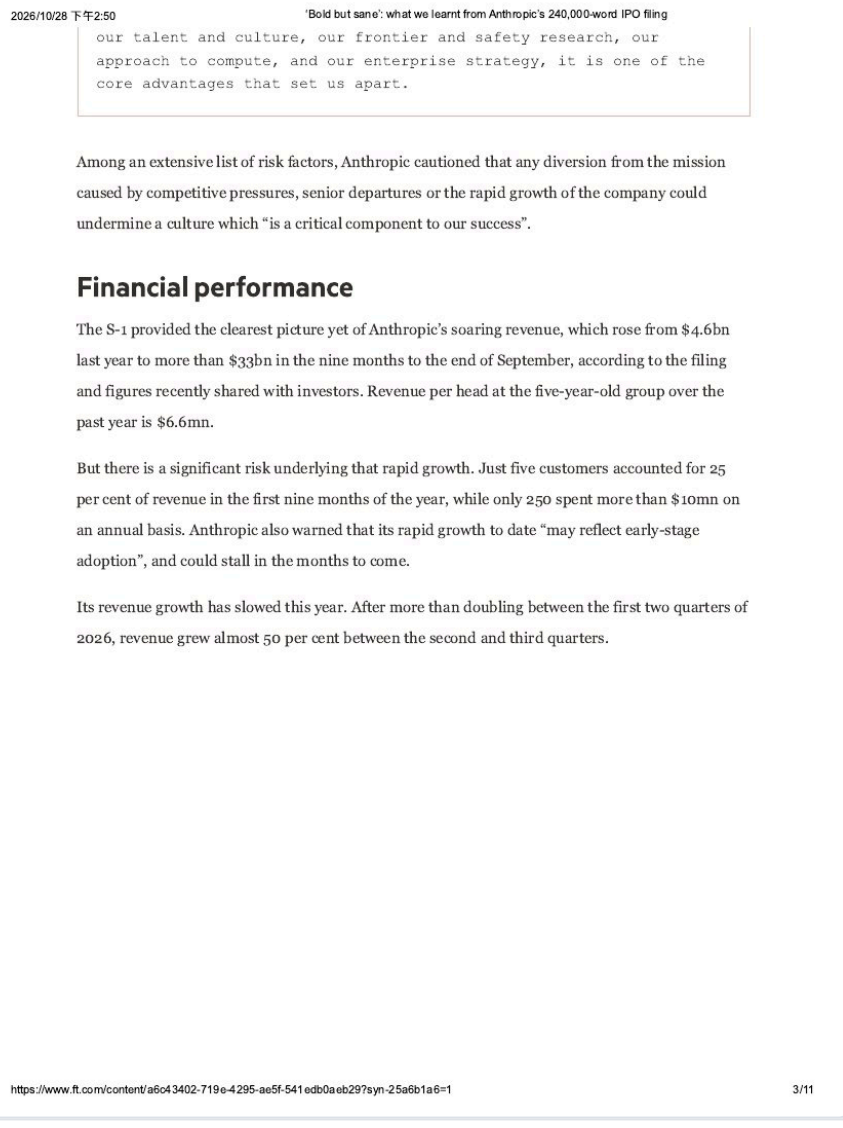
- 在驚人的成長與 8,000 億美元資料中心支出承諾之外，創辦人呼籲潛在股東「不要用純粹交易的角度看待你的購買」。
- 小標「Selling a unique culture and mission」：申報文件強調以負責任的方式發展 AI、追求人類的長期利益。七人創辦團隊在 2021 年離開 OpenAI 創立這家以安全為導向的對手，S-1 寫道「我們所有的努力都以使命為唯一焦點」。
- 創辦人把這種專注描述成相對於對手的關鍵優勢，有助於爭取人才、專注在「對人類價值最高的工作」。他們也強調創辦團隊仍然留在公司最上層是「不常見的」，文章認為這是在暗諷已經失去大部分創辦團隊的 OpenAI。
- 引言框：「我們首先是一間研究實驗室，我們的使命是負責任地發展與維護先進 AI，以造福人類的長期利益。這個使命引導我們的焦點……」（下一頁續）。

老師課堂說明

- 老師唸了這段使命宣言，說公開說明書裡會放這類好聽的話，接著要問的是實際表現如何。
- 這呼應第九週正課後半的故事與數字：公司在上市時一定會先講故事，投資人要做的是把故事換成可以檢驗的數字。

Anthropic IPO 新聞｜PDF 第 3/11 頁

財務表現：營收九個月衝到 330 億美元，但客戶集中、成長放緩



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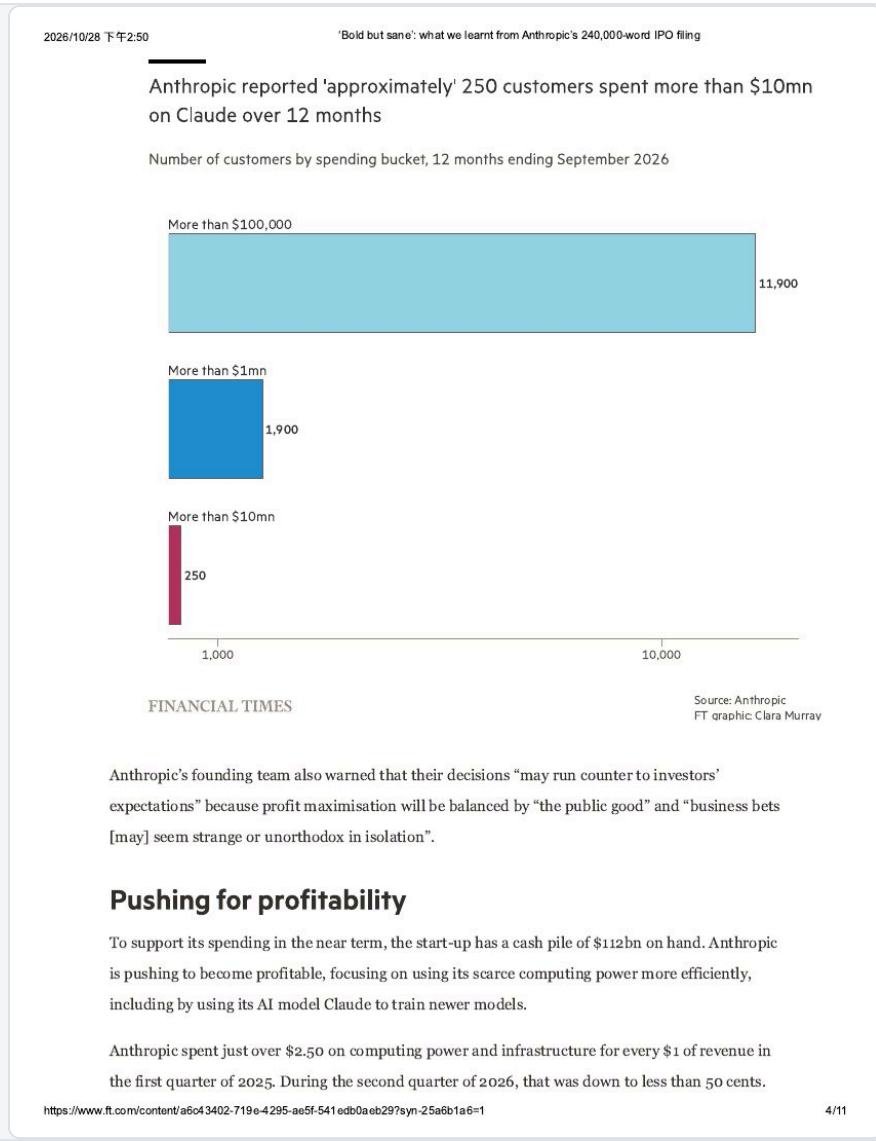
- 引言框續：使命也引導「我們的人才與文化、前沿與安全研究、我們取得算力的方式與企業策略，這是讓我們與眾不同的核心優勢之一」。
- 在很長的風險因素（risk factors）清單中，Anthropic 提醒：競爭壓力、資深成員離職或公司的快速成長，都可能讓公司偏離使命，進而破壞被視為「成功關鍵要素」的文化。
- 小標「Financial performance」：營收從去年的 46 億美元，成長到今年截至 9 月底九個月超過 330 億美元。這家成立五年的公司，過去一年的每位員工營收是 660 萬美元。
- 快速成長底下有很大的風險：今年前九個月，光是五家客戶就占營收 25%，而一年花超過 1,000 萬美元的客戶只有 250 家。Anthropic 也提醒，到目前為止的快速成長「可能反映的是早期採用」，未來幾個月可能停滯。
- 今年營收成長已經放緩：2026 年第一季到第二季成長超過一倍，第二季到第三季則成長將近 50%。

老師課堂說明

- 老師說營收從 46 億跳到九個月 330 億，十到十二月的數字還沒出來，成長率非常驚人。
- 但風險同樣明顯：五家客戶就占四分之一營收，失去其中一家，營收就會大幅下滑。成長也在放緩，第一季到第二季翻倍，第二季到第三季只剩 50%，雖然還是很高，但市場競爭非常激烈。
- 估值時這些事情分別落在不同的輸入值：成長放緩影響營收成長率，客戶集中與競爭影響風險與失敗機率（見把敘事換成估值輸入）。**即：營收衝得快是故事的亮點，但幾個大客戶撐起一大塊營收，代表這個成長並不穩。**

Anthropic IPO 新聞｜PDF 第 4/11 頁

客戶分布與「公共利益」條款，邁向獲利：每一塊營收的算力成本大降



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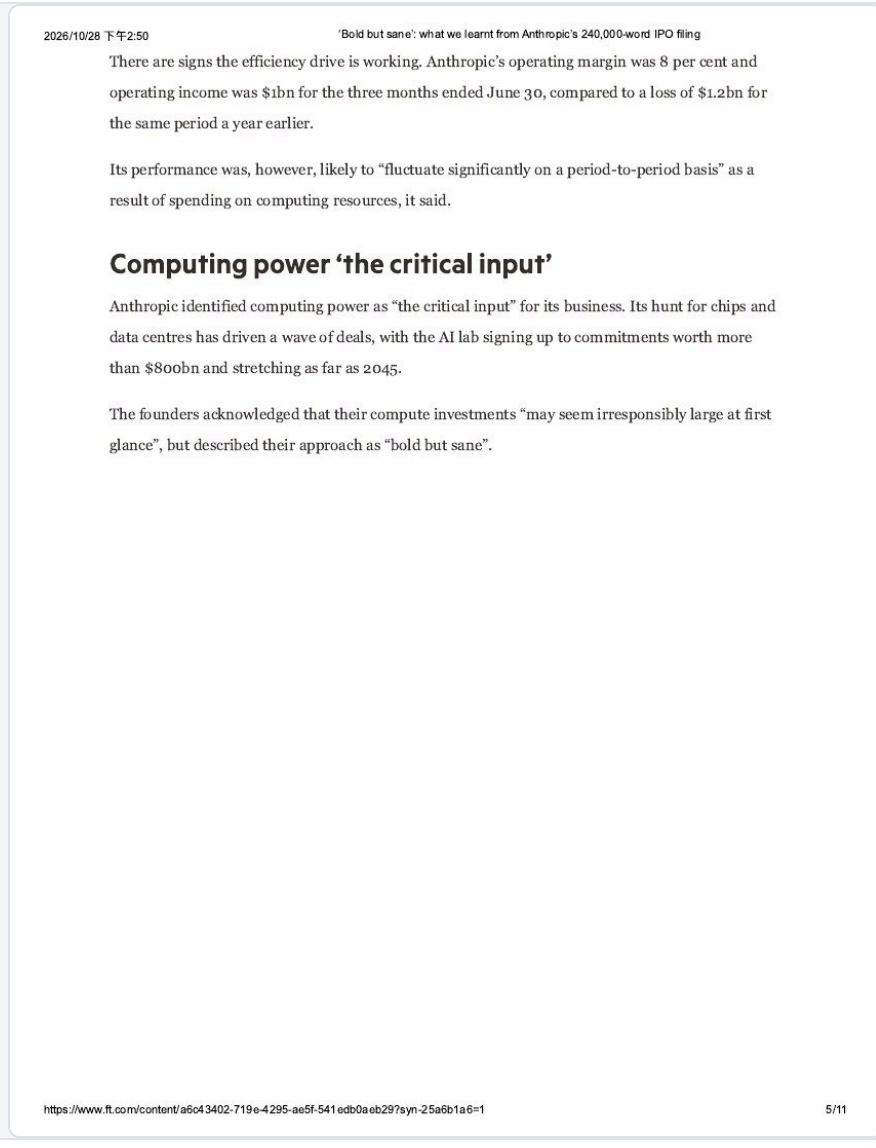
- 圖表「Anthropic reported 'approximately' 250 customers spent more than $10mn on Claude over 12 months」：截至 2026 年 9 月的 12 個月，依支出級距的客戶數（對數座標）。一年花超過 10 萬美元的有 11,900 家，超過 100 萬美元的 1,900 家，超過 1,000 萬美元的 250 家。資料來源 Anthropic。
- 創辦團隊也警告，他們的決策「可能違背投資人的期待」，因為追求利潤最大化會和「公共利益」相互權衡，而且「單獨來看，有些商業賭注可能顯得奇怪或不合常規」。
- 小標「Pushing for profitability」：為了支撐近期的支出，這家新創手上有 1,120 億美元現金。Anthropic 正努力轉為獲利，重點是更有效率地使用稀缺的算力，包括用自家模型 Claude 訓練更新的模型。
- 2025 年第一季，每一美元營收要花略高於 2.50 美元在算力與基礎設施上，到 2026 年第二季降到不到 50 美分。

老師課堂說明

- 老師說客戶集中度很高：250 家公司花超過 1,000 萬，1,900 家超過 100 萬，約 12,000 家超過 10 萬。
- 老師用自己的例子說明營收能不能持續成長才是關鍵問題：他每個月付 200 美元訂閱，由研究經費支付，因為寫程式、做研究、備課都用得上。但對一般人或學生來說這太貴了，就像多數人不需要最高規格的筆電。要一直把營收往上推，得找到願意持續付這麼多錢的客戶。
- 這裡的 1,120 億美元現金，估值時是在營運資產之外另外加回的那一塊（現金與有價證券）。

Anthropic IPO 新聞｜PDF 第 5/11 頁

效率提升開始見效，算力是「關鍵投入」，承諾支出到 2045 年


2026/10/28 下午2:50 'Bold but sane': what we learnt from Anthropic's 240,000-word IPO filing

There are signs the efficiency drive is working. Anthropic's operating margin was 8 per cent and operating income was $1bn for the three months ended June 30, compared to a loss of $1.2bn for the same period a year earlier.

Its performance was, however, likely to "fluctuate significantly on a period-to-period basis" as a result of spending on computing resources, it said.

Computing power 'the critical input'

Anthropic identified computing power as "the critical input" for its business. Its hunt for chips and data centres has driven a wave of deals, with the AI lab signing up to commitments worth more than $800bn and stretching as far as 2045.

The founders acknowledged that their compute investments "may seem irresponsibly large at first glance", but described their approach as "bold but sane".

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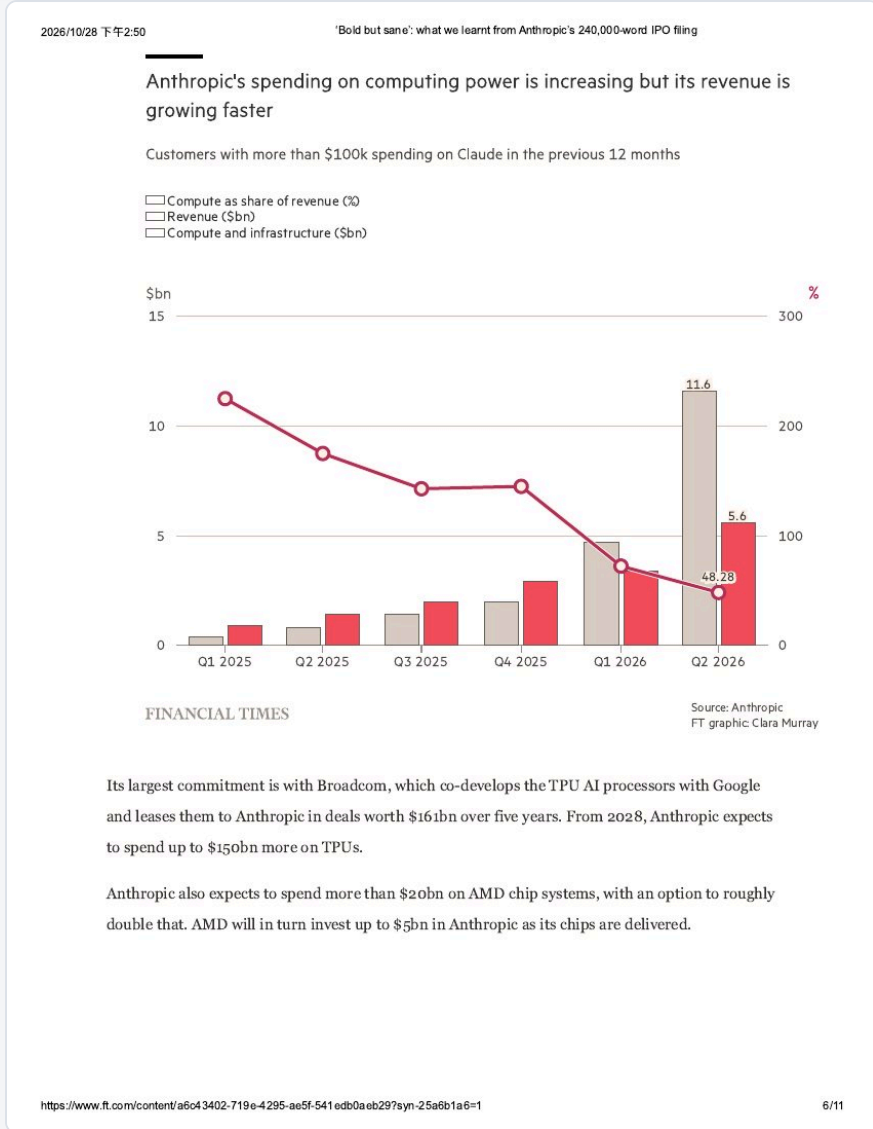
- 有跡象顯示效率提升正在奏效：截至 6 月 30 日的三個月，營業利益率（operating margin）8%、營業利益 10 億美元，去年同期則是虧損 12 億美元。
- 但公司也說，因為算力支出的關係，績效可能「在不同期間大幅波動」。
- 小標「Computing power 'the critical input'」：Anthropic 把算力定義為業務的「關鍵投入」。為了取得晶片與資料中心，它簽下一連串交易，承諾金額超過 8,000 億美元，期限最遠到 2045 年。
- 創辦人承認這些算力投資「乍看之下可能大得不負責任」，但形容自己的做法是「大膽但理智」（bold but sane），這也是本文標題的出處。

老師課堂說明

- 老師說 Anthropic 最重要的營業成本就是算力，因為每回答一次使用者的請求都要計算。這個成本一直在下降，利潤率才拉得起來。
- 公司想靠 8,000 億美元的再投資把成本壓得更低，合作對象包括 Broadcom、SpaceX、Google 等，但這筆承諾可能非常沉重。
- 估值上，這些長期算力承諾和營業租賃的性質很像：都是未來每年必須付的固定金額，Damodaran 的估值就把它資本化後當成債務扣除（見第 5 頁簡報）。
- **即：每一塊營收要付的算力成本降下來，獲利才會出現，但為了繼續降成本而簽下的長約，本身就是一筆要還的負擔。**

Anthropic IPO 新聞｜PDF 第 6/11 頁

營收成長快過算力支出，與 Broadcom、AMD 的晶片協議



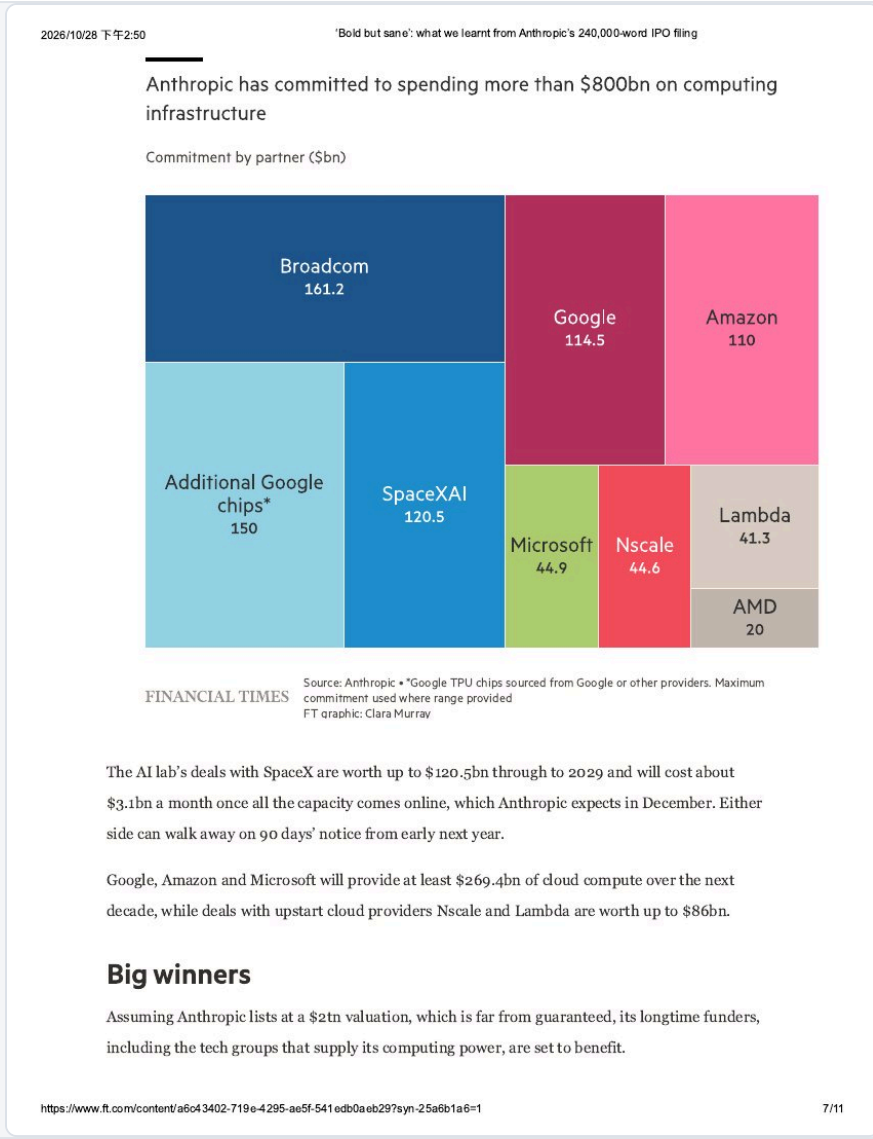
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- 圖表「Anthropic's spending on computing power is increasing but its revenue is growing faster」：每季營收（米色柱）、算力與基礎設施支出（紅色柱，十億美元）與算力占營收比例（折線，右軸 %）。2026 年第二季營收 116 億、算力支出 56 億，比例降到 48.28%（$5.6/11.6$）。2025 年第一季的比例在圖上約 225%，內文寫的是每一美元營收略高於 2.50 美元，兩者有出入。
- 圖表副標題寫的是「過去 12 個月支出超過 10 萬美元的客戶」，和圖的內容對不上，應是前一張圖的副標題沒有更新。
- 最大的一筆承諾是 Broadcom：它和 Google 共同開發 TPU 這種 AI 處理器，再租給 Anthropic，五年合約價值 1,610 億美元。Anthropic 預期從 2028 年起，還會在 TPU 上再花最多 1,500 億美元。
- Anthropic 也預期在 AMD 晶片系統上花超過 200 億美元，並有大約加倍的選擇權。AMD 則會隨著晶片交貨，對 Anthropic 投資最多 50 億美元。

> **老師課堂說明**
>
> - 老師指著折線說，算力成本占營收的比例一路往下掉，這就是利潤率改善的來源。
> - 從營收與成本兩條柱子看，算力支出的金額仍在變大，只是營收長得更快。**即：成本絕對金額還在增加沒關係，只要營收長得比它快，每一塊錢營收留下來的就越多。**

Anthropic IPO 新聞｜PDF 第 7/11 頁

超過 8,000 億美元的算力承諾分布，SpaceX 合約與雲端業者



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- 圖表「Anthropic has committed to spending more than $800bn on computing infrastructure」：依合作對象的承諾金額（十億美元）。Broadcom 161.2、額外的 Google 晶片 150（可能來自 Google 或其他供應商，取區間上限）、SpaceXAI 120.5、Google 114.5、Amazon 110、Microsoft 44.9、Nscale 44.6、Lambda 41.3、AMD 20，合計 807。
- 與 SpaceX 的協議到 2029 年最多 1,205 億美元，全部產能上線後每月約 31 億美元，Anthropic 預期在 12 月上線。從明年初起，任何一方都可以提前 90 天通知退出。
- Google、Amazon 與 Microsoft 在未來十年將提供至少 2,694 億美元的雲端算力（$114.5 + 110 + 44.9 = 269.4$），與新興雲端業者 Nscale、Lambda 的協議最多 860 億美元（$44.6 + 41.3 = 85.9$）。
- 小標「Big winners」：假設 Anthropic 以 2 兆美元估值上市（這遠遠不是確定的事），長期出資者，包括供應它算力的科技集團，都會受惠。

> **老師課堂說明**
>
> - 老師說市場共識是 Anthropic 值大約 2 兆美元。如果是這樣，Amazon 會很高興，因為它持有大約 20%，其次是持有約 7% 的 Google。
> - 這裡和第一週的新聞是同一件事的兩面：大型科技公司持有 Anthropic 的股權，估值上漲時帳上就認列評價利得，所以 Anthropic 的上市價格會直接影響這些公司的獲利數字。

Anthropic IPO 新聞｜PDF 第 8/11 頁

主要股東：Amazon 的可轉換票據、Google、創投，分批解禁

Its largest hyperscaler backer is Amazon, which would own about a fifth of the company if its convertible notes were fully converted — worth about $400bn at a $2tn valuation. Its holding is capped at 12 per cent, with the excess kept in notes that convert only as room opens up under the limit. But Amazon can waive that cap.



Branding for Claude Code on removable posters at an Anthropic developer conference in London © Chris Ratcliffe/Bloomberg

Next comes Google, which has a roughly 7 per cent stake. That could still rise as the search giant invests up to $20bn more under deals struck earlier this year. It could commit $10bn at the listing price, or into the IPO itself if both sides agree to make Google an anchor for the public offering.

Among financial investors, the filings indicate Spark Capital invested less than $1bn for a stake worth $14.5bn. It is a similar-sized holding to that of Coatue, which invested $4.75bn, according to a person familiar with the matter.

Like SpaceX, Anthropic has opted to stagger when insiders can sell their equity into the public market, with five release dates within the first 76 days of trading. Non-founder directors, officers and some current and former employees will be free to sell a chunk of shares on Anthropic's market debut. Companies typically stipulate that early investors can first begin selling down their stakes 180 days after an IPO.

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- 最大的超大規模業者股東是 Amazon：如果它的可轉換票據（convertible notes）全部轉換，會擁有公司約五分之一，以 2 兆美元估值計約 4,000 億美元（$20\% \times 2$ 兆）。它的持股上限是 12%，超出的部分以票據形式保留，等上限有空間時才轉換。Amazon 可以放棄這個上限。
- 配圖說明：Anthropic 在倫敦開發者大會上的 Claude Code 品牌海報。
- 其次是持有約 7% 的 Google。依今年稍早的協議，它還會再投資最多 200 億美元，持股可能上升：可以用上市價格投入 100 億美元，或在雙方同意下成為這次公開發行的基石投資人。
- 財務投資人方面，申報文件顯示 Spark Capital 投資不到 10 億美元，持股現值 145 億美元。據知情人士說，這和 Coatue 的持股規模相近，而 Coatue 投入了 47.5 億美元。
- 和 SpaceX 一樣，Anthropic 把內部人可以賣股的時間分批錯開，在交易的前 76 天內有五個解禁日。非創辦人董事、高階主管與部分現任及前任員工，在上市當天就可以賣出一部分股份。一般公司通常規定早期投資人要等上市 180 天後才能開始賣。

老師課堂說明

- 可轉換票據到期前是債，轉換之後變成股份，所以它同時帶有債與股權的性質。估每股價值時，這類會轉成股份的請求權要和員工選擇權一樣另外處理（選擇權價值扣除法的第三步就列了可轉換債的轉換權）。
- **即：一張紙今天算債、明天可能變成股份，算每股價值時就要先把它可能拿走的那一份扣掉。**

Anthropic IPO 新聞｜PDF 第 9/11 頁

「新穎」的治理結構：長期利益信託、50.1% 投票權，92 頁風險因素



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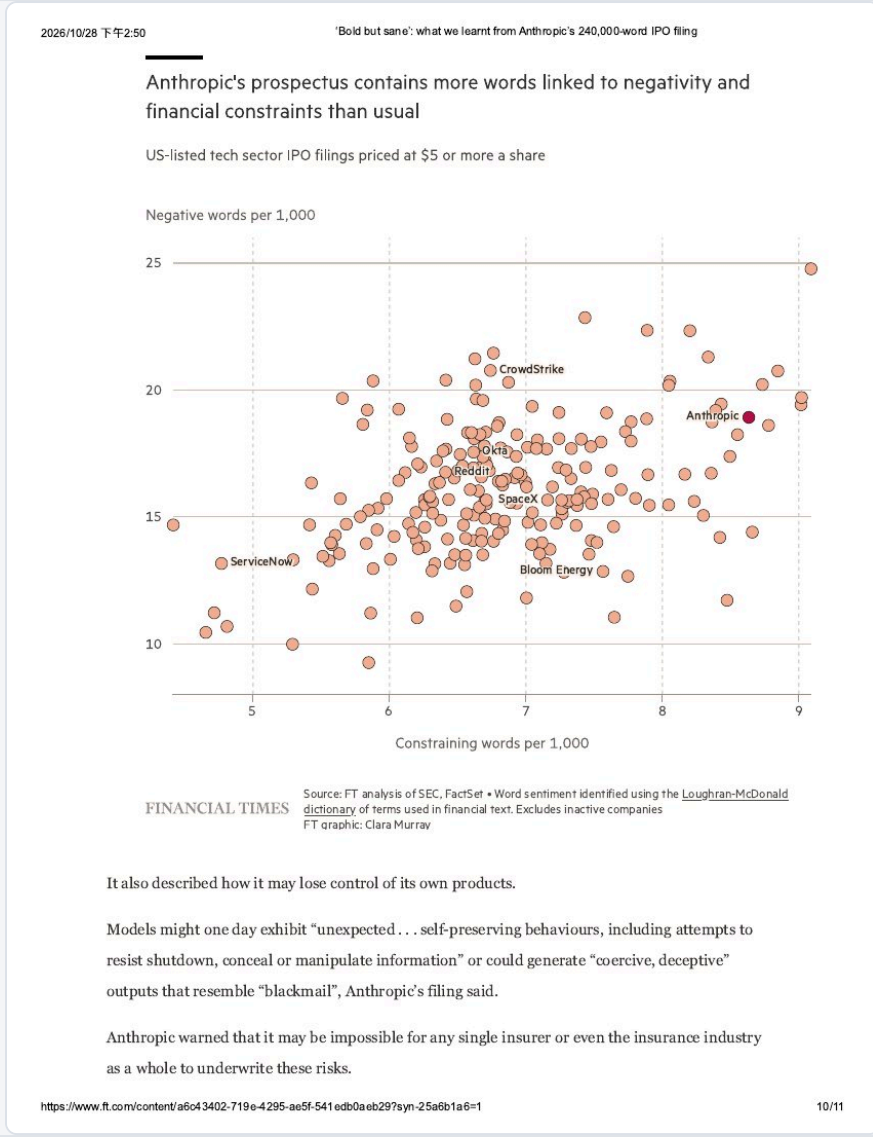
- 小標「'Novel' governance」：Anthropic 承認它「新穎」的公司治理結構和多數上市公司「有顯著差異」，一般股東對公司決策幾乎沒有控制權。
- 結構的最上層是「長期利益信託」（Long-Term Benefit Trust，LTBT），由信託自己任命成員，原則上有權選任與解任董事會七席中的四席。現任受託人包括前聯準會主席 Ben Bernanke，以及哈佛大學物理與天文學教授 Christopher Stubbs。
- 引言框：公司實施以長期利益信託為核心的創新治理結構，目的是以公益公司（public benefit corporation）的身分，負責任地平衡三項目標：股東的金錢利益、受公司行為實質影響者的最大利益，以及公司宗旨。
- 執行長 Dario Amodei 與同為共同創辦人的妹妹 Daniela 都不能被信託從董事會解任。
- 其餘三席由股東選出，但實際上由 Amodei 兄妹與另外五位共同創辦人決定。他們透過一種特殊股份擁有 50.1% 的投票權，保有日常經營的控制權。
- 信託對第四席與關鍵一票的控制也有限制：董事會的提名委員會（成員包括 Daniela Amodei）必須同意它的決定。雙方意見不同時，由代表員工的一票打破僵局。
- 小標「'Existential' risk factors」：S-1 有 92 頁風險因素，和公司為了防範未來訴訟常用的制式揭露大不相同。Anthropic 的風險揭露大約是 SpaceX 的三倍長，包括警告它的技術可能帶來「對人類的災難性或存在性風險」，以及可能「為試圖開發化學、生物或其他危險武器的惡意人士提供實質協助」。

老師課堂說明

- 老師說一般股東影響公司決策有兩種方式：直接投票，或選出代表自己投票的董事會。這家公司兩條路都被限縮了：創辦人握有 50.1% 的投票權，七席董事中有四席由信託選出。老師的解讀是，這套安排讓決策實際上留在創辦團隊手上。
- 對照 Meta 這類雙重股權（dual-class shares）公司：它們有一般股與超級投票權股，一股超級投票權股的票數是一般股的十倍。Anthropic 則有五種股份類別，結構更複雜。
- 老師補充：除非創辦人把股份賣給外部投資人，否則這家公司幾乎不可能成為併購的對象。
- 這種結構在估值上屬於複雜度與控制權的問題，Damodaran 在第 5 頁的結論也說治理結構值得再打一層折扣。**即：你可以分享股價上漲的好處，但公司要往哪裡走，基本上不是你說了算。**

Anthropic IPO 新聞｜PDF 第 10/11 頁

風險揭露的用字：負面與限制性字眼比一般 IPO 多，可能失去對產品的控制



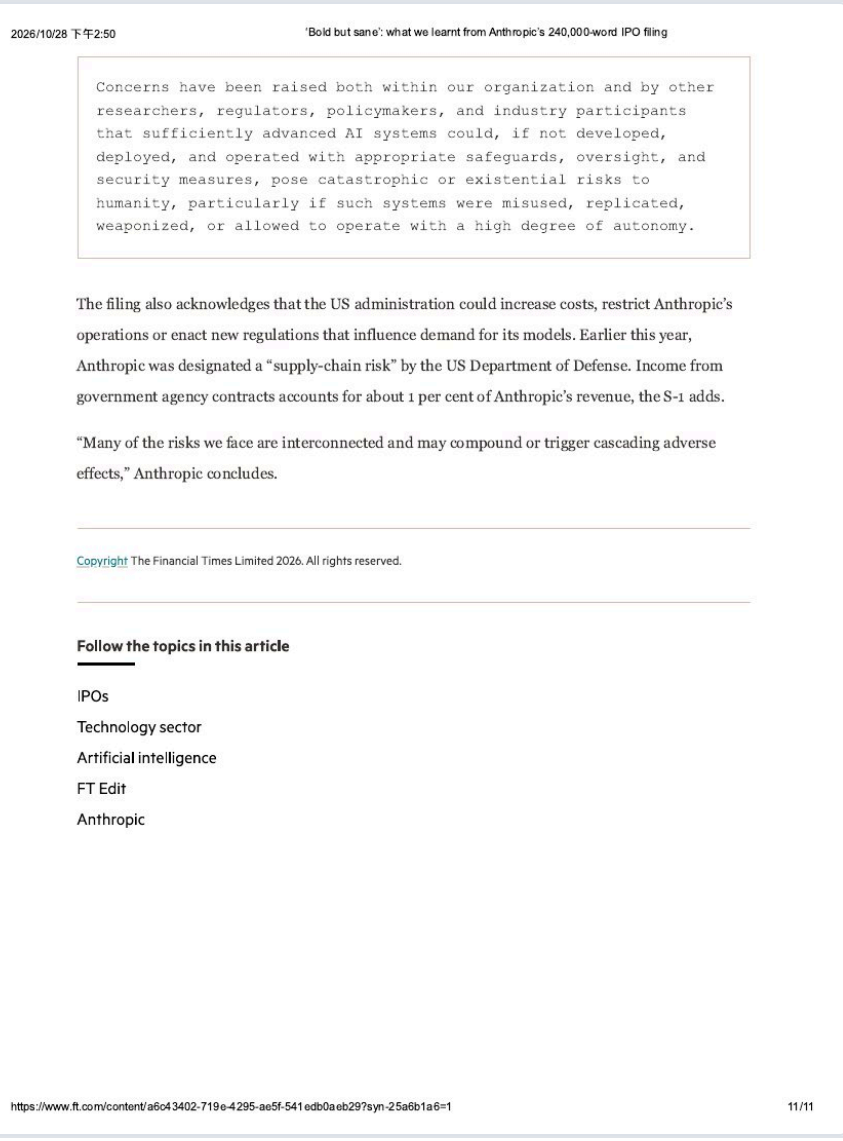
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- 圖表「Anthropic's prospectus contains more words linked to negativity and financial constraints than usual」：在美國上市、發行價 5 美元以上的科技業 IPO 申報文件，橫軸是每千字的限制性字眼數（約 4.5～9），縱軸是每千字的負面字眼數（約 9～25）。Anthropic 落在右上方（限制性約 8.6、負面約 19），標出的對照公司有 CrowdStrike、Okta、Reddit、SpaceX、Bloom Energy、ServiceNow。資料來源是 FT 對 SEC 與 FactSet 資料的分析，用字分類依 Loughran-McDonald 財務文本字典。
- 文件也描述公司可能失去對自家產品的控制。
- 模型可能有一天出現「意料之外的……自我保存行為，包括試圖抗拒關機、隱匿或操縱資訊」，或產生類似「勒索」的「脅迫性、欺騙性」輸出。
- Anthropic 警告，任何單一保險人，甚至整個保險業，都可能無法承保這些風險。

> **老師課堂說明**
>
> - 老師說文章接下來用了很多篇幅談風險因素，整篇的重點就是介紹這次 IPO 與公開說明書的主要內容。
> - 風險因素寫得越長越重，並不代表估值一定要打多少折，而是提醒分析師：這些風險要反映在折現率、失敗機率或情境分析裡，不能只看漂亮的營收成長（對照敘事中斷、轉移與改變）。

Anthropic IPO 新聞｜PDF 第 11/11 頁

風險相互牽連：政府監管、供應鏈風險認定



2026/10/28 下午2:50 'Bold but sane': what we learnt from Anthropic's 240,000-word IPO filing

> Concerns have been raised both within our organization and by other researchers, regulators, policymakers, and industry participants that sufficiently advanced AI systems could, if not developed, deployed, and operated with appropriate safeguards, oversight, and security measures, pose catastrophic or existential risks to humanity, particularly if such systems were misused, replicated, weaponized, or allowed to operate with a high degree of autonomy.

The filing also acknowledges that the US administration could increase costs, restrict Anthropic's operations or enact new regulations that influence demand for its models. Earlier this year, Anthropic was designated a "supply-chain risk" by the US Department of Defense. Income from government agency contracts accounts for about 1 per cent of Anthropic's revenue, the S-1 adds.

"Many of the risks we face are interconnected and may compound or trigger cascading adverse effects," Anthropic concludes.

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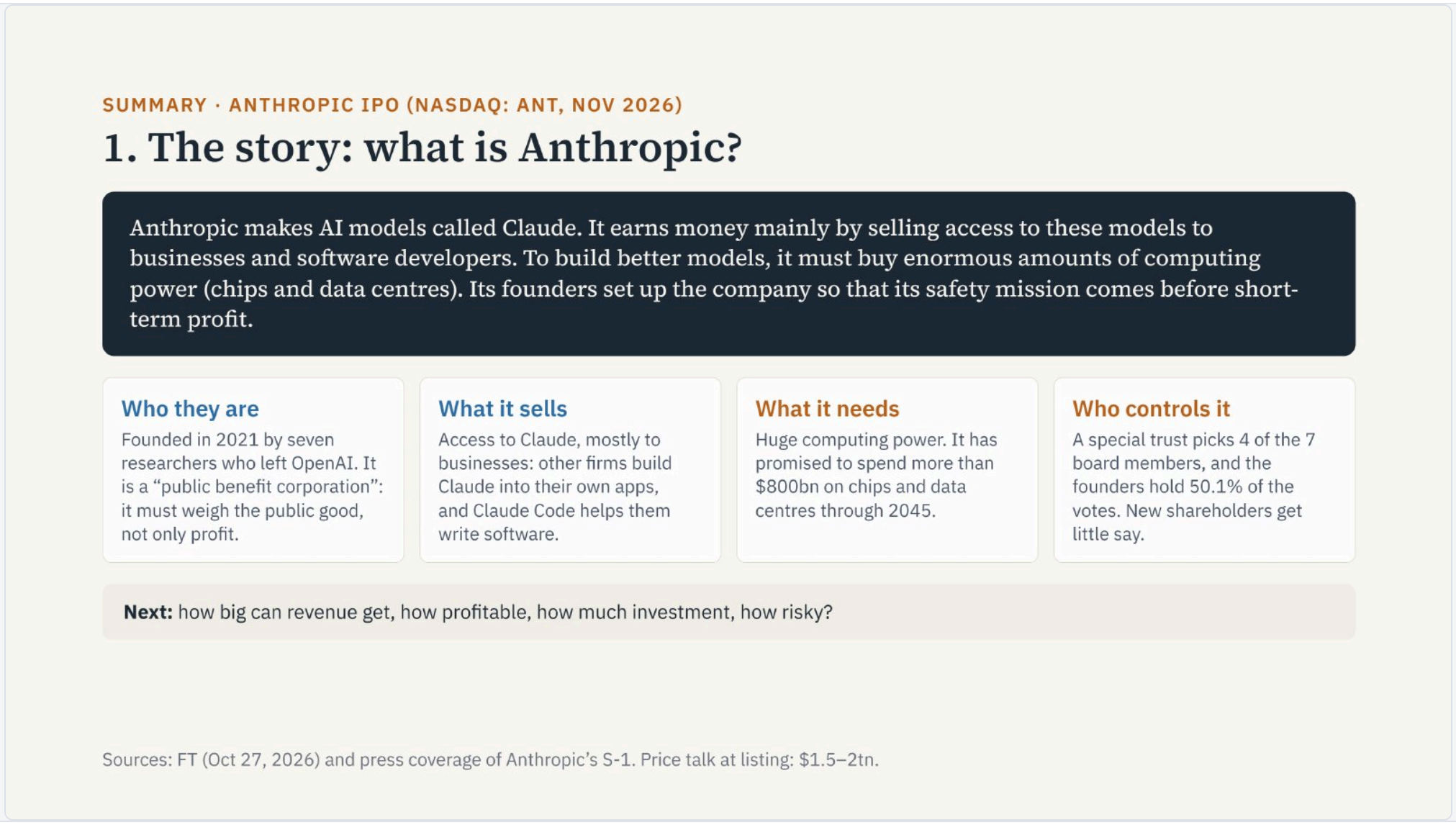
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- 引言框：公司內部與其他研究人員、監管機關、政策制定者與產業參與者都曾提出疑慮，認為足夠先進的 AI 系統如果沒有適當的防護、監督與安全措施，可能對人類帶來災難性或存在性風險，特別是在系統被濫用、複製、武器化，或被允許高度自主運作時。
- 申報文件也承認，美國政府可能提高成本、限制 Anthropic 的營運，或制定影響模型需求的新法規。今年稍早，Anthropic 被美國國防部認定為「供應鏈風險」。S-1 補充，政府機關合約的收入約占營收 1%。
- Anthropic 的結論是：「我們面對的許多風險彼此相關，可能互相加乘，或引發連鎖的不利影響。」
- 文末為 FT 版權聲明與主題標籤：IPOs、Technology sector、Artificial intelligence、FT Edit、Anthropic。

老師課堂說明

- 老師在進入簡報之前總結：這篇文章介紹了 IPO 新聞，並整理公開說明書的重點，接下來用簡報把這些資訊和估值連起來。

故事：Anthropic 是什麼公司



本頁中文重點

- 這份簡報由老師以 FT 這篇文章與其他報導為基礎整理，標題是「Summary · Anthropic IPO（Nasdaq: ANT, Nov 2026）」。
- 公司做的是名為 Claude 的 AI 模型，主要收入來自把模型的使用權賣給企業與軟體開發者。要做出更好的模型，必須買進大量算力（晶片與資料中心）。創辦人把公司設計成安全使命優先於短期利潤。
- 四格摘要：**他們是誰**：2021 年由七位離開 OpenAI 的研究人員創立，是公益公司，必須兼顧公共利益，不只追求利潤。**賣什麼**：Claude 的使用權，主要賣給企業，讓其他公司把 Claude 建進自己的應用程式，Claude Code 幫他們寫軟體。**需要什麼**：龐大的算力，承諾到 2045 年在晶片與資料中心上花超過 8,000 億美元。**誰控制**：特殊信託選出七席董事中的四席，創辦人擁有 50.1% 的投票權，新股東幾乎沒有發言權。
- 下一步要回答：營收能長到多大、獲利能力如何、要投資多少、風險有多高。頁尾註明上市的價格討論區間是 1.5 兆到 2 兆美元。

老師課堂說明

- 老師說這幾頁是他用 Claude 整理 FT 與其他報導製作的，估值部分則參考本課教材作者、紐約大學 Damodaran 教授對 Anthropic 做的 DCF。
- 老師提到自己每月付 200 美元，有時還會遇到系統忙碌、要等十分鐘的情況，這代表公司仍然需要持續擴充算力。
- 老師口述時說成信託選出全部七席，文章與簡報寫的都是七席中的四席，以書面資料為準。
- 老師說接下來談估值，要依序看營收、利潤率、風險與再投資。這正是把敘事換成估值輸入那條線的順序。**即：先把公司是誰、賣什麼、需要什麼、誰說了算講清楚，後面每一個估值數字才有故事可以對照。**

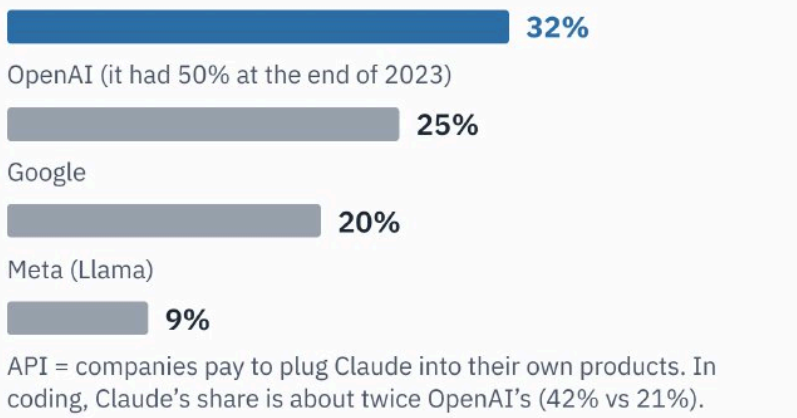
Anthropic IPO 課堂簡報｜PDF 第 2/5 頁

市場：企業客戶最強，但市場年輕、競爭激烈

SUMMARY · ANTHROPIC IPO

2. Market: strongest with business customers

Share of business AI usage (API), mid-2025



API = companies pay to plug Claude into their own products. In coding, Claude's share is about twice OpenAI's (42% vs 21%).

How much customers spend a year (to Sep 2026)



- Existing customers now spend about 9 times more than a year ago ("net revenue retention" ≈ 900%)
- Risk: the five biggest customers bring in 25% of all revenue
- Rivals: OpenAI (strong with consumers; valued at $852bn in Mar 2026), Google, and free open-source models

Takeaway: Anthropic leads today, but the market is young and competitive, so shares can change fast.

Sources: Menlo Ventures (mid-2025); S-1 via FT and press (Oct 2026); Damodaran (Oct 22, 2026).

本頁中文重點

- 左圖：2025 年中企業 AI 使用量（API）市占：Anthropic 32%、OpenAI 25%（2023 年底曾有 50%）、Google 20%、Meta（Llama）9%。API 是指企業付費把 Claude 接進自家產品。在寫程式的領域，Claude 的市占約是 OpenAI 的兩倍（42% 對 21%）。
- 右側：截至 2026 年 9 月的一年內，250 家企業花 1,000 萬美元以上、1,900 家花 100 萬美元以上、11,900 家花 10 萬美元以上。既有客戶現在花的錢約是一年前的 9 倍，淨收入留存率（net revenue retention）約 900%。
- 風險：前五大客戶帶來 25% 的營收。對手包括 OpenAI（在消費者端很強，2026 年 3 月估值 8,520 億美元）、Google，以及免費的開源模型。
- 結論：Anthropic 目前領先，但市場很年輕又競爭激烈，市占可能變化得很快。資料來源：Menlo Ventures（2025 年中）、FT 與媒體報導的 S-1（2026 年 10 月）、Damodaran（2026/10/22）。

老師課堂說明

- 老師說 AI 產業正在擴張，但 OpenAI 兩三年前還有 50%，現在只剩四分之一左右，可見市占換手很快。再過幾個月會怎樣沒人知道，不能說 Anthropic 已經可以安穩享受領先者的優勢。
- 客戶結構風險更高：前五大客戶占 25%，少了一家營收就會掉很多。
- 市占與客戶集中度會分別影響估值裡的市占率假設與失敗機率，市場越年輕，這兩個數字的不確定性越大（對照可能、合理、很可能）。

Anthropic IPO 課堂簡報｜PDF 第 3/5 頁

數字：成長很快，獲利還不明朗



本頁中文重點

- 左圖：營收（十億美元）2023 年 0.04、2024 年 0.4、2025 年 4.6、2026 年 1–9 月 33，是 2025 年全年的 7 倍。成長在放緩：第三季 +45%，第二季則是成長超過一倍。
- 右圖：每一美元營收的算力成本：2025 年第一季 2.50、第二季 1.75、第三季 1.43、第四季 1.45、2026 年第一季 0.72、第二季 0.48。服務客戶的成本大幅下降，讓公司在 2026 年第二季第一次出現營業利益。頁尾註明第一個與最後一個數字取自 FT 內文，其餘從 FT 的圖讀出。
- 五個數字：2026 年 1–6 月依一般公認會計原則（GAAP）的營業虧損 −7 億美元。公司把股票薪酬加回後報的「調整後」獲利 +16 億美元。同期以股份支付給員工的薪酬 22 億美元。淨損 −807 億美元，主要是可轉換證券的帳面損失。2026 年 6 月的現金與短期投資 1,320 億美元。
- 核對：$-0.7 + 2.2 = 1.5$，和調整後獲利 1.6 只差四捨五入。FT 內文寫的現金是 1,120 億美元，簡報的 1,320 億含短期投資，兩者口徑不同。

老師課堂說明

- 老師說營業利益其實是負的，原因是公司付給員工大量的股票薪酬。公司把這 16 億的股票薪酬加回去，獲利就變成正的。**老師特別強調，本課依循的 Damodaran 認為不應該這樣做**（不要把股票薪酬加回）。
- 淨損 807 億主要是可轉換證券的評價損失：公司價值上升，它發出去的可轉換證券跟著變貴，帳上就認列損失，和第一週大型科技公司認列評價利得是同一件事的兩面。這筆損失不是營運賺賠，所以估值從營業利益（EBIT）出發。
- **即：先分清楚哪些是本業賺的、哪些是帳上評價或付給員工的股份，才知道哪個數字可以拿來估現金流。**

易錯提醒

- 股票薪酬是用股份付的費用，加回去會讓營業利益看起來比較好，但員工拿走的是股東的一部分公司，估值時要把它留在費用裡。

估值：兩種快速檢驗 2 兆美元的方法

SUMMARY · ANTHROPIC IPO

4. Valuation: two quick ways to judge $2tn

Method 1 · Multiple of future sales

- Some investors expect yearly sales of about $320bn by the end of 2027.
- Then $2tn is only about 7× 2028 sales: a bit below Microsoft. SpaceX trades at 16×.
- At SpaceX's 16×, Anthropic would be worth about $5tn.

Method 2 · Market size (TAM) × share

- Total market × Anthropic's share = future sales; × a multiple, then discount back to today.
- SpaceX puts "enterprise apps" at $22.7tn. 3% share ≈ $700bn of sales × 10 = $7tn; 3 years back ≈ $4.5tn.
- Flip it: $2tn needs only about $300bn of sales in 3 years, roughly 1.4% of that market (our calc.).



The catch: both methods assume AI labs keep the profits. If AI models become a cheap commodity, sales and multiples both shrink.

Source: FT Lex, "Anthropic at $2tn isn't far-fetched" (Sep 21, 2026). Flip-side figure: our calculation.

本頁中文重點

- 方法一，未來營收的倍數：有些投資人預期 2027 年底年營收約 3,200 億美元。這樣 2 兆美元大約只是 2028 年營收的 7 倍，略低於 Microsoft，而 SpaceX 的交易倍數是 16 倍。以 SpaceX 的 16 倍計，Anthropic 約值 5 兆美元（$3{,}200$ 億 $\times\ 16 \approx 5.1$ 兆）。照字面數字算是 $2{,}000/320 \approx 6.3$ 倍，簡報寫「大約 7 倍」是約略的說法。
- 方法二，市場規模（TAM）乘市占：總市場 $\times$ 市占 $=$ 未來營收，再乘一個倍數、折回今天。SpaceX 估計「企業應用」市場有 22.7 兆美元，3% 的市占約 7,000 億美元營收（$22.7 \times 3\% = 0.68$ 兆），乘 10 倍是 7 兆，往回折三年約 4.5 兆。
- 反過來算：2 兆美元只需要三年後約 3,000 億美元的營收，大約是那個市場的 1.4%（簡報自行計算）。
- 右圖比較各方法得到的價值（兆美元）：上市價格討論 2、TAM 3% 市占 4.5、16 倍營收 5、TAM 樂觀情境 10（摩根士丹利估 60 兆美元的市場，AI 拿 10%，Anthropic 再拿三分之一）。
- 但書：兩種方法都假設 AI 實驗室能保住利潤。如果 AI 模型變成廉價的大宗商品，營收和倍數都會一起縮水。資料來源：FT Lex〈Anthropic at $2tn isn't far-fetched〉（2026/9/21）。

$$\text{方法一：價值} = \text{倍數} \times \text{未來營收}$$

$$= 16 \times 320 \approx 5{,}100 \text{ (十億美元)}$$

$$\text{方法二：}\ 22.7 \text{ 兆} \times 3\% \approx 0.7 \text{ 兆}$$

$$0.7 \times 10 = 7 \text{ 兆}, \quad \frac{7}{(1+r)^3} \approx 4.5 \text{ 兆}$$

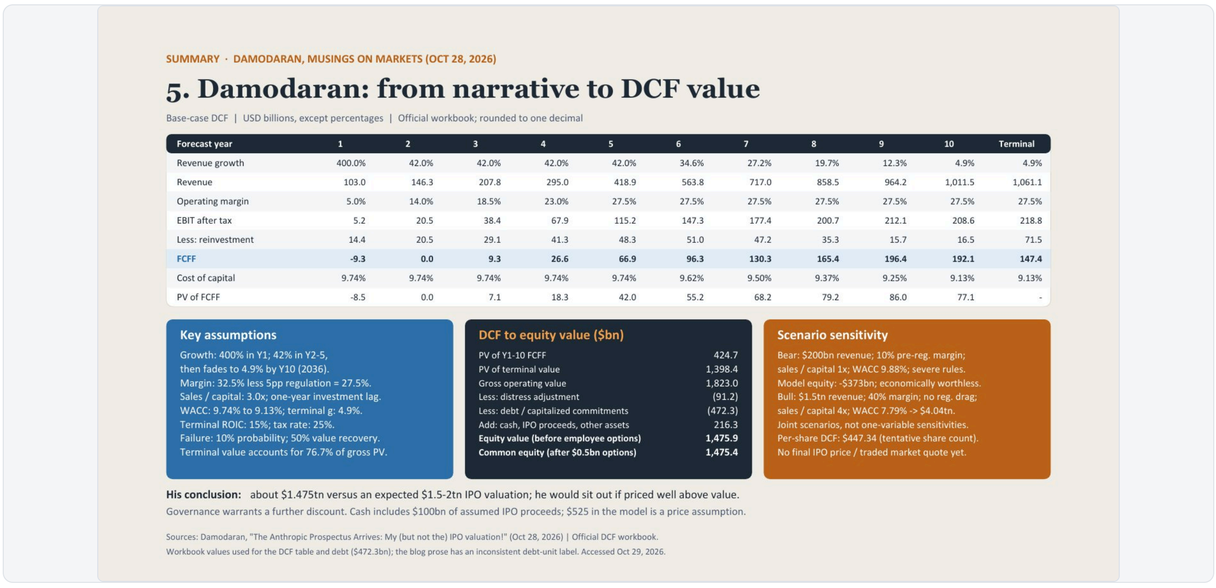
從 7 兆折回 4.5 兆，隱含的年折現率約 $(7/4.5)^{1/3} - 1 \approx 15.9\%$。

老師課堂說明

- 老師說大家最愛用倍數，其中企業價值除以營收（EV／Sales）用得非常多，因為只要知道營收，就能推回企業價值（營收倍數，下一個單元會正式講）。
- 只要 7 倍就能得到 2 兆，而 7 倍比 Microsoft 還低。剛上市的 SpaceX 是 16 倍，如果說兩家做的是同一種事業、可以用同樣的倍數，估值就能到 5 兆。老師說這就是用倍數替數字找理由。
- 第二種方法像是 DCF 加倍數的混合：算市場規模、乘市占、再算現值。只要拿到 1.4% 的市占就有 2 兆。老師的結論是 2 兆大約是市場共識的平均值，16 倍的情況是 5 兆，非常樂觀的情況甚至到 10 兆。
- 這兩種方法都屬於相對評價：價值取決於拿誰來比、用哪個倍數，而不是公司自己的現金流。**即：倍數挑得不同，同一家公司就能說成值 2 兆或 5 兆，所以要先問這個倍數是怎麼來的。**

Anthropic IPO 課堂簡報｜PDF 第 5/5 頁

Damodaran 的估值：從敘事到 DCF，基準情境約 1.475 兆美元



SUMMARY · DAMODARAN, MUSINGS ON MARKETS (OCT 28, 2026)

5. Damodaran: from narrative to DCF value

Base-case DCF | USD billions, except percentages | Official workbook; rounded to one decimal

Forecast year	1	2	3	4	5	6	7	8	9	10	Terminal
Revenue growth	400.0%	42.0%	42.0%	42.0%	42.0%	34.6%	27.2%	19.7%	12.3%	4.9%	4.9%
Revenue	103.0	146.3	207.8	295.0	418.9	563.8	717.0	858.5	964.2	1,011.5	1,061.1
Operating margin	5.0%	14.0%	18.5%	23.0%	27.5%	27.5%	27.5%	27.5%	27.5%	27.5%	27.5%
EBIT after tax	5.2	20.5	38.4	67.9	115.2	147.3	177.4	200.7	212.1	208.6	218.8
Less: reinvestment	14.4	20.5	29.1	41.3	48.3	51.0	47.2	35.3	15.7	16.5	71.5
FCFF	-9.3	0.0	9.3	26.6	66.9	96.3	130.3	165.4	196.4	192.1	147.4
Cost of capital	9.74%	9.74%	9.74%	9.74%	9.74%	9.62%	9.50%	9.37%	9.25%	9.13%	9.13%
PV of FCFF	-8.5	0.0	7.1	18.3	42.0	55.2	68.2	79.2	86.0	77.1	-

Key assumptions

Growth: 400% in Y1; 42% in Y2-5, then fades to 4.9% by Y10 (2036).
Margin: 32.5% less 5pp regulation = 27.5%.
Sales / capital: 3.0x; one-year investment lag.
WACC: 9.74% to 9.13%; terminal g: 4.9%.
Terminal ROIC: 15%; tax rate: 25%.
Failure: 10% probability; 50% value recovery.
Terminal value accounts for 76.7% of gross PV.

DCF to equity value ($bn)

PV of Y1-10 FCFF	424.7
PV of terminal value	1,398.4
Gross operating value	1,823.0
Less: distress adjustment	(91.2)
Less: debt / capitalized commitments	(472.3)
Add: cash, IPO proceeds, other assets	216.3
Equity value (before employee options)	**1,475.9**
Common equity (after $0.5bn options)	**1,475.4**

Scenario sensitivity

Bear: $200bn revenue; 10% pre-reg. margin; sales / capital 1x; WACC 9.88%; severe rules. Model equity: -$373bn; economically worthless.
Bull: $1.5tn revenue; 40% margin; no reg. drag; sales / capital 4x; WACC 7.79% -> $4.04tn.
Joint scenarios, not one-variable sensitivities.
Per-share DCF: $447.34 (tentative share count).
No final IPO price / traded market quote yet.

His conclusion: about $1.475tn versus an expected $1.5-2tn IPO valuation; he would sit out if priced well above value.
Governance warrants a further discount. Cash includes $100bn of assumed IPO proceeds; $525 in the model is a price assumption.

Sources: Damodaran, "The Anthropic Prospectus Arrives: My (but not the) IPO valuation!" (Oct 28, 2026) | Official DCF workbook.
Workbook values used for the DCF table and debt ($472.3bn); the blog prose has an inconsistent debt-unit label. Accessed Oct 29, 2026.

本頁中文重點

- 這一頁整理 Damodaran 的部落格〈The Anthropic Prospectus Arrives: My (but not the) IPO valuation!〉（2026/10/28）與他公開的試算表，單位十億美元。
- 十年預測：營收成長第 1 年 400%、第 2–5 年 42%，之後逐年遞減到第 10 年的 4.9%。營業利益率從 5% 升到 27.5%（他估的穩定利益率是 32.5%，再扣 5 個百分點作為監管的拖累）。稅後營業利益減去再投資就是 FCFF，第 1 年 −9.3、第 5 年 66.9、第 10 年 192.1。
- 再投資用銷售對投入資本比 3.0 倍，並假設投資比營收早一年。資金成本從 9.74% 降到 9.13%，終值成長率 4.9%，終值期的資本報酬率 15%，稅率 25%。另外給 10% 的失敗機率、失敗時可回收 50% 的價值。
- 從 DCF 到股權價值：前十年 FCFF 現值 424.7 加終值現值 1,398.4，得到營運資產毛額 1,823.0，減困境調整 91.2、減債務與資本化承諾 472.3、加現金、IPO 募資與其他資產 216.3，得到員工選擇權前的股權價值 1,475.9，扣掉 5 億美元的選擇權後普通股是 1,475.4。終值占營運資產現值 76.7%。
- 情境：悲觀情境（營收 2,000 億、監管前利益率 10%、銷售對資本 1 倍、WACC 9.88%、嚴格法規）的股權價值是 −3,730 億，等於一文不值。樂觀情境（營收 1.5 兆、利益率 40%、沒有監管拖累、銷售對資本 4 倍、WACC 7.79%）是 4.04 兆。這些是多個變數一起變動的情境，不是一次只動一個變數。
- 結論：約 1.475 兆美元，低於預期的 1.5–2 兆美元上市估值，如果定價遠高於價值他會選擇不參與。每股 DCF 價值 447.34 美元（暫定股數）。治理結構值得再打一層折扣。現金中包含假設的 1,000 億美元 IPO 募資。

年度	1	2	3	4	5	6	7	8	9	10	終值期
營收成長	400.0%	42.0%	42.0%	42.0%	42.0%	34.6%	27.2%	19.7%	12.3%	4.9%	4.9%
營收	103.0	146.3	207.8	295.0	418.9	563.8	717.0	858.5	964.2	1,011.5	1,061.1
營業利益率	5.0%	14.0%	18.5%	23.0%	27.5%	27.5%	27.5%	27.5%	27.5%	27.5%	27.5%
稅後營業利益	5.2	20.5	38.4	67.9	115.2	147.3	177.4	200.7	212.1	208.6	218.8
減：再投資	14.4	20.5	29.1	41.3	48.3	51.0	47.2	35.3	15.7	16.5	71.5
FCFF	−9.3	0.0	9.3	26.6	66.9	96.3	130.3	165.4	196.4	192.1	147.4
資金成本	9.74%	9.74%	9.74%	9.74%	9.74%	9.62%	9.50%	9.37%	9.25%	9.13%	9.13%
FCFF 現值	−8.5	0.0	7.1	18.3	42.0	55.2	68.2	79.2	86.0	77.1	—

$$\text{終值期再投資} = \frac{g}{\text{ROC}} \times \text{EBIT}(1-t) = \frac{4.9\%}{15\%} \times 218.8 = 71.5$$

$$\text{TV}_{10} = \frac{147.4}{9.13\% - 4.9\%} = 3{,}484.6$$

$$\frac{3{,}484.6}{\prod_{t=1}^{10}(1+\text{WACC}_t)} = \frac{3{,}484.6}{2.491} \approx 1{,}398$$

$$1{,}823.0 - 91.2 - 472.3 + 216.3 = 1{,}475.8 \approx 1{,}475.9$$

最後一行與表上的 1,475.9 只差四捨五入。資金成本每年不同，所以折現要用累積的折現因子，不能直接用 1.0913^{10}。

老師課堂說明

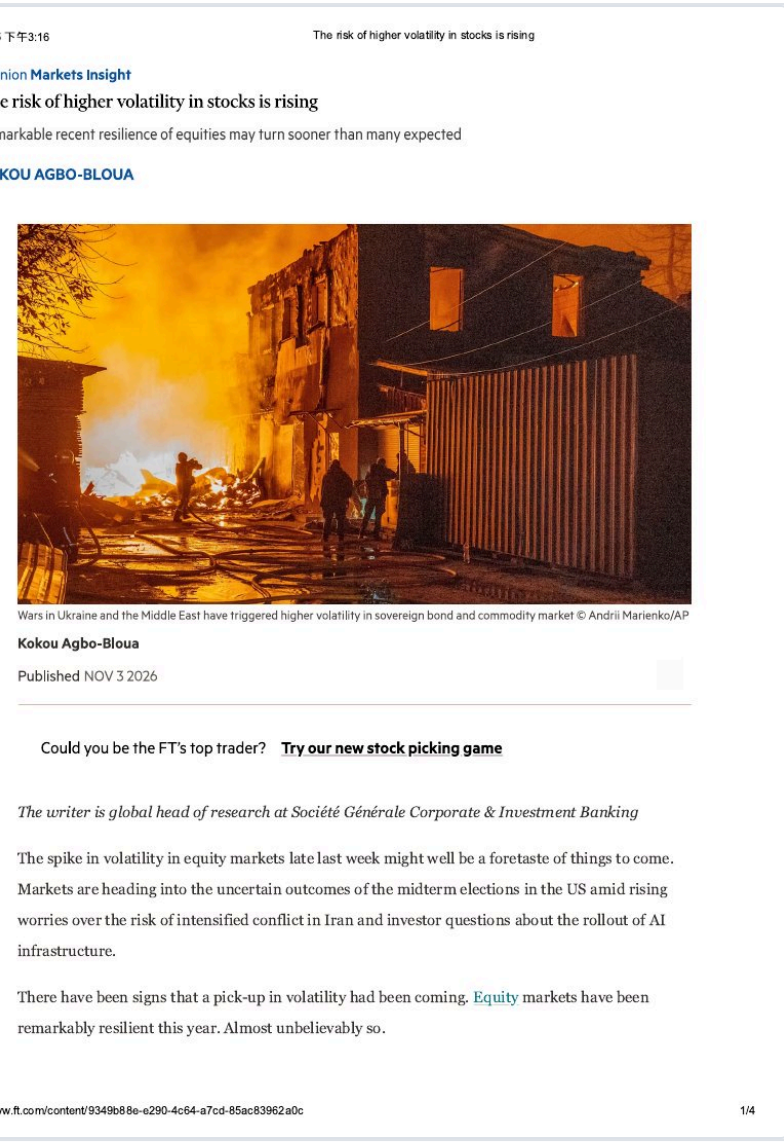
- 老師說 Anthropic 真正的歷史數字很少，2026 年也還差一季，所以 Damodaran 用一些保守的假設，套兩段式成長模型，做十年的 DCF，算出終值再走到股權價值，得到約 1.5 兆（1.475 兆）。
- 老師提醒這個數字也可能錯得很離譜，做財務的人因為處理的是數字，通常比行銷端保守。他的 1.475 兆和市場常講的 2 兆之間有很大的差距，要等上市之後才知道真正的價格。因為整個估值建立在假設上，他也做了敏感度分析。
- 這一頁幾乎用到前幾週的每一個工具：終值期的再投資率就是穩定期再投資率 g/ROC，終值用永續成長模型，再投資用銷售對投入資本比，最後的加減就是第九週正課要講的從營運資產到每股價值。
- 「債務與資本化承諾」把長期算力合約當成債，邏輯和營業租賃當成債務相同。選擇權只扣 5 億美元，用的是正課的選擇權價值扣除法。
- **即：同一套 DCF 流程，從營收成長、利潤率、再投資到資金成本一路推下來，故事講得多樂觀，最後就反映在這些數字有多大。**

易錯提醒

- 終值占營運資產現值的 76.7%，代表估值大部分取決於第 10 年以後的假設。看到這種結構，要特別檢查穩定期的成長率與 ROC 是否合理。

股市波動風險新聞｜PDF 第 1/4 頁

股市波動的風險正在升高：異常平靜的股市可能比預期更早反轉



2026/11/5 下午3:16 The risk of higher volatility in stocks is rising

Opinion **Markets Insight**

The risk of higher volatility in stocks is rising

Remarkable recent resilience of equities may turn sooner than many expected

KOKOU AGBO-BLOUA

Wars in Ukraine and the Middle East have triggered higher volatility in sovereign bond and commodity market © Andrii Marienko/AP

Kokou Agbo-Bloua

Published NOV 3 2026

Could you be the FT's top trader? **Try our new stock picking game**

The writer is global head of research at Société Générale Corporate & Investment Banking

The spike in volatility in equity markets late last week might well be a foretaste of things to come. Markets are heading into the uncertain outcomes of the midterm elections in the US amid rising worries over the risk of intensified conflict in Iran and investor questions about the rollout of AI infrastructure.

There have been signs that a pick-up in volatility had been coming. Equity markets have been remarkably resilient this year. Almost unbelievably so.

https://www.ft.com/content/9349b88e-e290-4c64-a7cd-85ac83962a0c 1/4

本頁中文重點

- **這篇在問什麼**：戰爭、利率、選舉一堆壞消息，美股指數卻異常平靜。作者認為平靜底下藏著風險，而答案是股票之間的相關係數（correlation）。這和本課的關係是：相關係數會影響股權風險溢酬（ERP），ERP 會影響折現率與估值。
- 文章出處：Financial Times 專欄 Markets Insight，作者 Kokou Agbo-Bloua（法國興業銀行 Société Générale 企業與投資銀行部門的全球研究主管），發布日 2026/11/3。頁首的 2026/11/5 是列印時間。
- 導言：股市近來驚人的韌性，可能比許多人預期的更早轉向。
- 上週後半股市波動突然升高，可能只是接下來情況的預告。市場正面對美國期中選舉結果的不確定，同時擔心伊朗衝突加劇，投資人也對 AI 基礎建設的推進有疑問。
- 已經有跡象顯示波動要回來了。股市今年的韌性非常驚人，幾乎令人難以置信。

老師課堂說明

- 老師先講投資人的角度：估值是 CEO、CFO 要向市場籌資時在意的事，投資人在意的是預期報酬，至少要拿到和風險相稱的要求報酬率。
- 1950、60 年代的華爾街相信聰明加上資訊就能賺錢。現代投資組合理論（Modern Portfolio Theory）在 1960、70 年代出現之後，大家知道風險和報酬一樣重要，所以專業投資人（共同基金、避險基金）都在做投資組合，享受分散（diversification）的好處。
- 兩檔股票的投資組合：預期報酬就是加權平均，例如 100 元放 40 在第一檔、60 在第二檔。風險（變異數）就複雜一點，除了兩檔各自的變異數，還多一項相關係數，也就是兩檔股票一起動的程度。兩檔同漲同跌是正相關，一漲一跌是負相關，沒關係就接近 0。老師說這條公式不用背，重點是投資組合的風險裡有相關係數這一項。
- 專業投資人的績效看夏普比率（Sharpe ratio）：報酬除以風險。比率越高，代表承擔同樣的風險賺得越多。
- 老師說他常建議朋友買指數，朋友說指數報酬比較低。老師的回答是：組成投資組合後預期報酬確實會下降，因為混進了報酬低的股票，但風險降得更多，所以夏普比率反而上升。報酬少一點，但更常、更有把握賺到錢。

先用數字看相關係數的威力

- 兩檔股票的標準差都是 30%，各放一半。只改變相關係數 ρ，投資組合的標準差就差很多。
- $\rho = 1$（完全同漲同跌）：30%，等於沒有分散。
- $\rho = 0$（互不相關）：21.2%。
- $\rho = -0.5$（常常一漲一跌）：15%，風險只剩一半。

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\rho\,\sigma_1\sigma_2$$

$$\sigma_p^2 = 0.5^2(0.3^2) + 0.5^2(0.3^2) + 2(0.5)(0.5)\,\rho\,(0.3)(0.3) = 0.045 + 0.045\rho$$

代入 $\rho = 1, 0, -0.5$，σ_p^2 分別是 0.09、0.045、0.0225，開根號得 30%、21.2%、15%。

- 即：兩檔股票越常反方向動，加在一起就越平穩，變異數裡的相關係數那一項可以把整體風險壓下來。

股市波動風險新聞｜PDF 第 2/4 頁

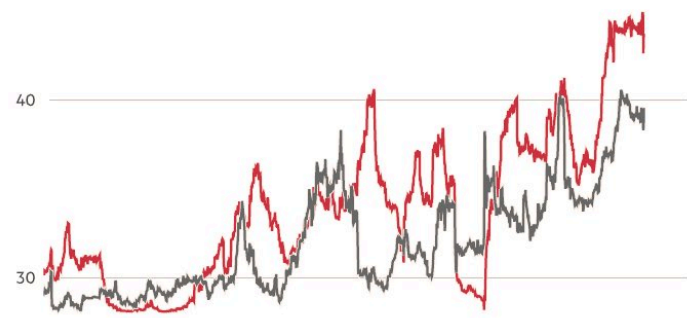
選擇權隱含的波動率很平靜：VIX 約 16、VSTOXX 約 18

Wars in Ukraine and the Middle East have triggered higher volatility in sovereign bond and commodity markets. But indices of investor expectations of equity volatility implied by options trading — the Vix index in the US and the VSTOXX index in Europe — have been relatively calm with an average level of 16.3 and 17.7 per cent respectively over the past three months, in line with long-term averages. The so-called fear gauge indices appeared "fearless" until midway through last week.

The most common explanation is the strong earnings growth driven by significant capital expenditure on the AI value chain, from hyperscale data centres and semiconductors to memory chips and energy infrastructure. Fiscal deficits with large tax cuts in the US and defence spending in Europe have also contributed to rising corporate profits.

Stock markets are masking unusually divergent moves beneath the surface

Weighted share of stock pairs with a negative three-month correlation (%)



The recent resilience is despite expectations that the wars in Ukraine and Iran increasingly appear to have little prospect of a resolution anytime soon. The risk of an escalation where both the straits of Hormuz and of Bab al-Mandeb can be shut in the context of depleted global energy reserves should have led to higher equity volatility risk premia. It did not.

本頁中文重點

- 烏克蘭與中東的戰爭讓主權債券與大宗商品市場的波動升高。但由選擇權交易推算出來、代表投資人預期股市波動的指數，也就是美國的 VIX 與歐洲的 VSTOXX，過去三個月平均分別只有 16.3% 與 17.7%，和長期平均差不多。這些號稱「恐慌指標」的指數，直到上週中都顯得「毫無恐懼」。
- 最常見的解釋是盈餘成長強勁，來自 AI 價值鏈的大量資本支出：超大規模資料中心、半導體、記憶體晶片到能源基礎建設。美國大幅減稅造成的財政赤字與歐洲的國防支出，也推高了企業獲利。
- 圖：「股市底下藏著異常分歧的走勢」，兩兩股票之間三個月相關係數為負的比例（加權），近年明顯上升。
- 儘管烏克蘭與伊朗的戰爭看起來越來越難以結束，股市還是很有韌性。在全球能源存量偏低的情況下，荷莫茲海峽與曼德海峽都可能被封鎖，這種升級的風險原本應該推高股市的波動風險溢酬（volatility risk premia），結果沒有。

老師課堂說明

- 老師說一般人會以為最近壞事很多，波動應該很大，但實際的波動卻比預期低，這篇文章就在分析原因，答案是相關係數。
- 文章提到的事件：上週美國期中選舉造成波動、伊朗戰爭、對 AI 的悲觀看法。戰爭推高通膨與能源價格，也影響主權債券，美國公債殖利率也曾經急升。
- VIX 是美國股市的整體波動指標，歐洲是 VSTOXX。數字大約 16 到 18，算相當小。發生了這麼多壞事，股市的不確定性卻很低，這很奇怪。
- 有人說是因為 AI 與科技帶來很好的基本面成長，以及龐大的財政赤字替企業分擔了負擔，推高了企業獲利。

股市波動風險新聞｜PDF 第 3/4 頁 升息與債市波動照理會傳到股市，股市卻沒有反應

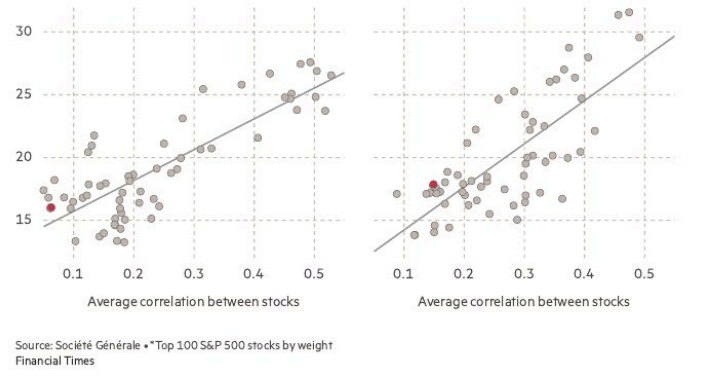
And having lost the argument that inflation would be "transitory" in 2022, central banks are not taking any chances and have embarked on a new tightening cycle with more hikes expected by the Federal Reserve, the European Central Bank, the Bank of England and the Bank of Japan over the next six to 12 months.

The subsequent rise in global bond yields will eventually force fiscal consolidation as debt-servicing costs rise, increasing concerns around sustainability. Higher bond yield volatility should have increased equity volatility, as the stream of future earnings that underpins stock valuations is discounted by the combination of bond yields and equity risk premia. Even corporate credit pricing has come under pressure in both investment-grade and high yields sectors.

So why did equity index volatility remain subdued? Is equity sentiment somehow immune to rising interest rates, geopolitical risks and higher credit spreads?

Index volatility tends to be higher when stocks move together

How closely stocks move together vs expected index volatility, Nov 2021 to Nov 2026



Source: Société Générale • *Top 100 S&P 500 stocks by weight
Financial Times

本頁中文重點

- 在 2022 年輸掉「通膨只是暫時」的爭論之後，各國央行不再冒險，已經展開新一輪緊縮。預期聯準會、歐洲央行、英格蘭銀行與日本央行在未來 6 到 12 個月還會再升息。
- 全球公債殖利率上升，償債成本升高，最終會迫使政府財政緊縮，也讓人更擔心財政能不能持續。公債殖利率的波動升高，照理應該帶動股市波動升高，因為支撐股票價值的未來盈餘，是用公債殖利率加上股權風險溢酬折現的。連公司債的定價，投資等級與高收益債都承受壓力。
- 那為什麼股市指數的波動還這麼低？股市情緒難道對利率上升、地緣政治風險與信用利差擴大免疫嗎？
- 圖：「股票越一起動，指數的波動就越高」，2021 年 11 月到 2026 年 11 月，美國（S&P 前 100 大與 VIX）和歐洲（Euro Stoxx 50 與 VSTOXX）的股票平均相關係數，對三個月平均的預期波動。兩張圖都是相關係數越高，預期波動越高。

老師課堂說明

- 老師說債市波動照理應該帶動股市波動，因為股票的現金流比債券更不確定：股東是剩餘請求者，排在最後一順位才拿錢。但這次債市波動很大，股市卻看不到。
- 文章的「未來盈餘用公債殖利率加股權風險溢酬折現」，就是本課 DCF 的折現率：無風險利率加上 β 乘 ERP（CAPM）。

股市波動風險新聞｜PDF 第 4/4 頁

風險沒有消失，只是被存在負相關的股票裡

Well, the underlying picture for individual stocks has been a different story from the broad index. When the gains of a given sector or stocks are offset by the losses of another, the index moves very little. For example, oil and gas companies versus transportation companies, AI hardware versus AI software, companies that benefit from higher interest rates such as financials versus those that suffer, like utilities. Looking at equity index performance can be as misleading as looking at the tip of an iceberg, especially for volatility.

And we are now seeing high sector rotation by investors and the dispersion of stock returns. Take, for example, the Euro Stoxx 50 index. If you look at all the possible pairs of single stocks in the index, the proportion with a negative correlation for three-month periods over the past five years has risen to an unprecedented high of 30 per cent. It is 40 per cent for the S&P 500.

Apart from current investment themes such as the energy supply and AI demand shock, such "social distancing" dynamics might also be a symptom of a new market structure, reflecting the more aggressive deployment of systematic long-short investing strategies.

Equity volatility risk is being transformed and stored in negatively correlated pairs. A broad change in the correlation of these stock pairs could gradually, or suddenly, force more components of the index to move in the same direction. The result could be sharply higher index volatility if, for instance, the odds of a recession increase materially.

Markets have an equivalent to the law of conservation of energy in physics. Risk is never destroyed, only transformed. We can see how risk is transforming and the manifestation of it in equity markets is coming sooner than many think.

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本頁中文重點

- 個股的情況和大盤完全不同。當某個產業或某些股票的漲幅，被另一些的跌幅抵銷，指數就幾乎不動。例如石油天然氣公司對運輸公司、AI 硬體對 AI 軟體、受惠於高利率的金融股對受害的公用事業股。只看指數，就像只看冰山一角，對波動尤其如此。
- 現在投資人頻繁地在產業之間輪動，個股報酬也很分散。以 Euro Stoxx 50 為例，把指數裡所有股票兩兩配對，過去五年中三個月相關係數為負的配對比例升到前所未有的 30%，S&P 500 則是 40%。
- 除了能源供給與 AI 需求衝擊這些投資主題，這種「保持社交距離」的走勢，也可能反映新的市場結構：系統性的多空策略（long-short）用得更積極。
- 股市的波動風險正在轉換，被存放在負相關的股票配對裡。這些配對的相關係數一旦全面改變，不論是慢慢地還是突然地，都會迫使指數裡更多成分股往同一個方向動。例如經濟衰退的機率明顯上升時，結果可能是指數波動急劇升高。
- 市場也有類似物理學能量守恆的定律：風險不會被消滅，只會轉換形式。我們看得到風險正在轉換，它在股市爆發的時間會比很多人想的更早。

老師課堂說明

- 老師說文章的答案是相關係數：某些股票漲、另一些跌，就是負相關，加總起來互相抵銷，指數就動得很少。油價上漲對運輸業不好、對油氣公司好。升息對銀行好、對要借錢的公司不好。
- 大約四成的股票配對是負相關。所以即使有壞消息，一部分股票跌、另一部分漲，方向互相抵銷，整體波動看起來就低。
- 危險在於壞事累積到某個程度，負相關突然轉成正相關，市場波動就會一下子大幅上升，股價大跌，影響市場與經濟。
- 老師把它接回估值：市場的整體風險可以拆成平均的個股變異數加上平均相關係數。ERP 有兩個組成，一個是投資人整體的風險趨避程度（壞時機大家不願意冒險），一個就是市場的風險。相關係數由負轉正 → 市場波動上升 → ERP 上升 → 折現率上升 → 估值下降、股價下跌。
- 老師說他希望衝擊是慢慢來的，但戰爭、通膨、各國政府的財政赤字，加上可能的 AI 泡沫，也可能帶來衰退。重點是估值、公司理財和股市投資人的行為與報酬波動，彼此非常緊密地連在一起。

從相關係數到估值：一條因果鏈

- 很多檔股票各放一樣多時，投資組合的變異數大約是「平均個股變異數除以股票數」加上「平均相關係數乘平均個股變異數」。股票數很多時，第一項幾乎消失，剩下的市場風險主要由平均相關係數決定。
- 數字例：100 檔股票，每檔標準差 30%。平均相關係數 0.1 時，指數標準差約 9.9%。升到 0.3，約 16.6%。升到 0.5，約 21.3%。個股本身沒有變得更危險，只是一起動了，指數的波動就翻倍。

$$\sigma_m^2 \approx \frac{\bar{\sigma}^2}{n} + \left(1 - \frac{1}{n}\right)\bar{\rho}\,\bar{\sigma}^2$$

$$n = 100,\ \bar{\sigma} = 30\%: \quad \bar{\rho} = 0.1 \Rightarrow \sigma_m \approx 9.9\%$$

$$\bar{\rho} = 0.3 \Rightarrow \sigma_m \approx 16.6\%$$

步驟	發生什麼	對估值的影響
1	負相關的股票配對轉成正相關	個股開始一起漲跌
2	市場整體波動 σ_m 上升	分散的效果變小
3	投資人要求更高的風險補償，ERP 上升	股權成本 $r_f + \beta \times \text{ERP}$ 上升
4	折現率上升	同樣的現金流折現回來的價值變低
5	估值下降	股價下跌

- 即：平常一漲一跌互相抵銷，大盤看起來很穩。一旦大家一起跌，市場風險跳升，投資人要更多補償，股價就被往下壓。

本單元術語

中英術語對照

英文取自課堂教材，中文譯名以本筆記內文使用的說法為準

中文	English	一句話說明
美國國庫券	Treasury bills	美國政府發行、一年內到期的短期債券，課堂上當成無風險利率的參考。
殖利率	yield	以目前價格買進並持有到期，可以拿到的年化報酬率。
基點	basis point（bp）	利率的最小計量單位，1 個基點等於 0.01 個百分點。
期限溢酬	term premium	投資人因為把錢鎖得比較久而要求的額外報酬。
再融資	refinancing	舊債到期時再借新債把它還掉，利率上升時這件事會變貴。
財政主導	fiscal dominance	政府債務壓力大到讓貨幣政策被迫配合財政需要。
金融壓抑	financial repression	用政策手段把利率壓在通膨之下，讓債務的實質負擔慢慢變小。
其他收入	other income	損益表上本業以外的收入，大型科技公司的股權投資評價利得就列在這一行。
稅前盈餘	earnings before tax（EBT）	扣掉利息與所有費用、但還沒扣所得稅的獲利。
超大規模業者	hyperscalers	自建大型資料中心、提供雲端運算的少數幾家科技公司。
特殊目的載體	special purpose vehicle（SPV）	為單一專案另設的公司，用來把該專案的資產與負債和母公司隔開。
表外融資	off-balance-sheet financing	不出現在資產負債表上的資金安排，估值時仍然要算進負債。
殘值保證	residual value guarantee	承諾資產在期末至少值多少錢，不足的差額由保證人補。
主經紀商業務	prime brokerage	銀行替避險基金提供融資、借券與結算的業務。
自營交易	proprietary trading	銀行拿自己的錢下注，金融海嘯後受到監管限制。
內部化	internalisation	把一個客戶的多頭和另一個客戶的空頭對起來，不必到市場上調券。
造市	market making	同時掛出買價與賣價，賺取價差並提供流動性。
點心債	dim sum bond	在香港發行的人民幣計價債券。
熊貓債	panda bond	外國機構在中國境內發行的人民幣計價債券。
公開說明書	prospectus	公司上市前給外部投資人看的完整揭露文件，美國申報用的是 S-1。
首次公開發行	initial public offering（IPO）	公司第一次把股票賣給公開市場的投資人。
公益公司	public benefit corporation	依法必須在股東利益之外，同時兼顧公共利益與公司宗旨的公司型態。
長期利益信託	Long-Term Benefit Trust（LTBT）	Anthropic 治理結構的頂層，有權選任董事會七席中的四席。
雙重股權	dual-class shares	同時發行一般股與超級投票權股，讓少數人用少量股份掌握多數投票權。
風險因素	risk factors	公開說明書裡揭露公司可能遭遇風險的章節。
可轉換票據	convertible notes	到期前是債、可以依約轉換成股份的證券。
基石投資人	anchor investor	在公開發行時承諾認購一大筆股份、替發行定錨的大型投資人。
淨收入留存率	net revenue retention	既有客戶今年花的錢除以去年花的錢，高於 100% 代表老客戶越花越多。
調整後獲利	adjusted profit	公司自行加回某些費用（例如股票薪酬）後報出的獲利，不是會計準則的數字。
總體可及市場	total addressable market（TAM）	產品理論上可以觸及的全部市場規模，估營收時的起點。